Pathology

FMGE

Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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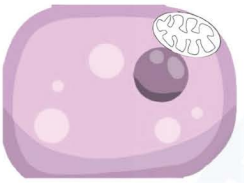
CELL INJURY

INTRODUCTION OF CELL INJURY



Balanced State of cell.

↓

"Homeostasis"

❑ CAUSES OF CELL INJURY

MC → ..... → ..... [.....]

($\downarrow O_2$)

Hypoxic
Anemic
Cytotoxic

Severity :-

..... >> ..... >> .....

(2-4min) (20-30min) (last)

Stress

↓

Stress → (N) Cell ⟷ Adaptation ⟷ Reversible cell Injury

↑↑ ↓ ↑Stress

Stress → Irreversible injury

(Cell Death)

❑ OUTCOMES OF CELL INJURY

1. Adaptation
2. Reversible Injury
3. Irreversible Injury
4. Intracellular Accumulations
5. Cellular Ageing

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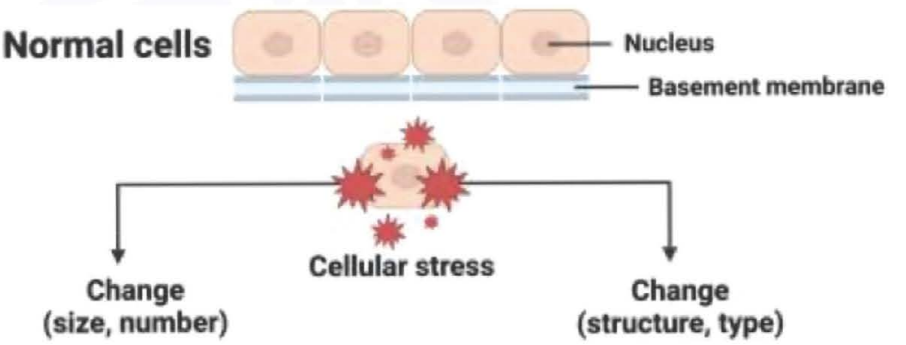
Adaptation

- Reversible changes

eg:- a) Hypertrophy
b) Hyperplasia
c) Metaplasia
d) Atrophy



(A)

.....

(B)

.....

(C)

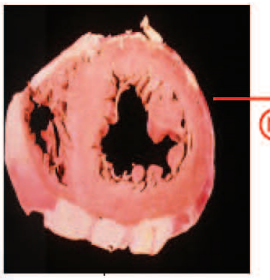
.....

(D)

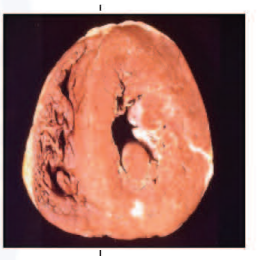
.....

Active space

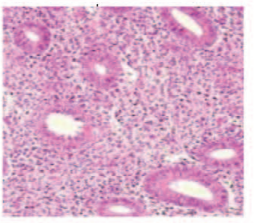
	Hypertrophy	Hyperplasia
No. of Cell →		
Size of Cell →		
Nature →		
Ability to replicate →		
DNA/Nuclear content		
Reason →		
Mechanism →		
E.g • Uterus - Pregnancy • Breast - Puberty - Pregnancy - Lactation • Cardiac Muscle • Endometrium		



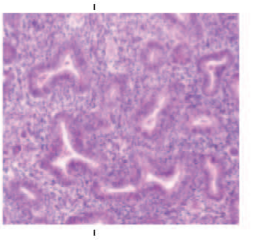
(A) CUT SECTION OF HEART



(B) LVH



(A) N ENDOMETRIUM



(B) ENDOMETRIAL HYPERPLASIA

✦ c) Metaplasia: - Change in ..... into another .....

Mechanism: .....

Type - 2

Epithelial | Mesenchymal

....

....

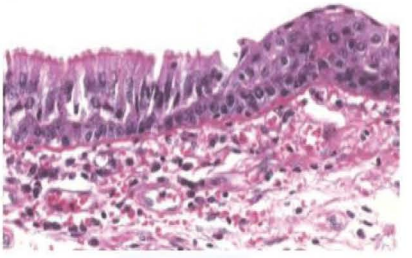
....

....

....

eg:- • Resp epi:- Smoking

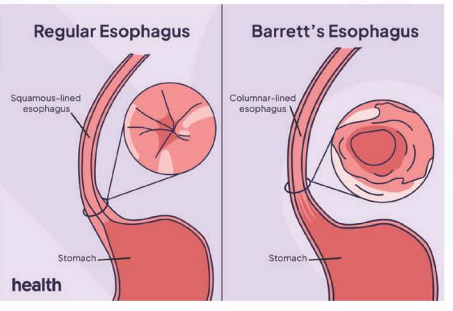
• GIT epi:- Stones

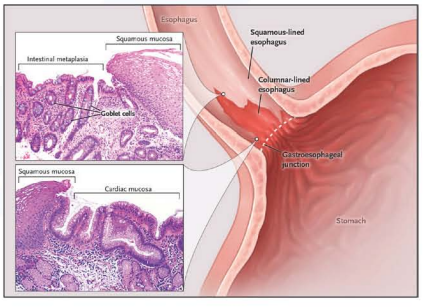


COLUMNAR METAPLASIA

.....

.....





- Hallmark of Barret's Esophagus → .....
- Function → .....
- Stain → .....
- Location → .....
- BE is a premalignant cond^{n}.: .....

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✦ d) Atrophy: .....

	Physiological	Pathological
eg:	• Menstrual cycle	• Disuse
	• Embryogenesis	• Denervation
		• Neurodegenerative d/s
		• Senile.

- Atrophy $\xrightarrow[UPP]{}$ Autophagy (Self eating)
(Starvation)

(UPP: Ubiquitine Proteosome Pathway)

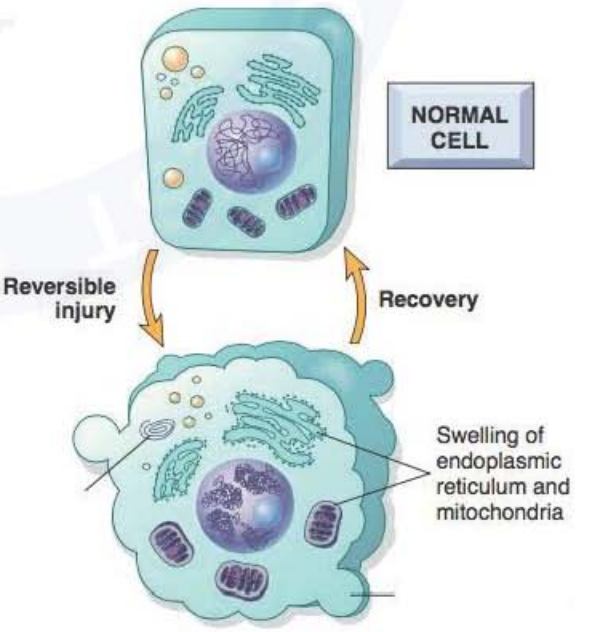
Marker: LC-3

REVERSIBLE INJURY

✦ Injuryto ..... pump → ..... inside cell → ..... insidecell

✦ Earliest Morphological Change -
(i) .....

✦ Earliest Change Overall
(ii) .....
(iii) RER → Ribosome detachment
↓
..... Protein Synthesis
↓
..... Lipase
↓
TG's (Cytoplasmic Blebs)
↓



IRREVERSIBLE INJURY

Extreme Stress → Permanent damage to pumps

↓

.....

Deposit inside mitochondria

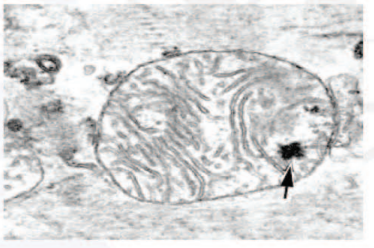
↓

.....

.....

.....

.....



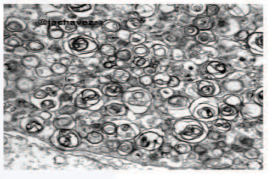
Activate Enzymes

↓

a) ATP ase → .....

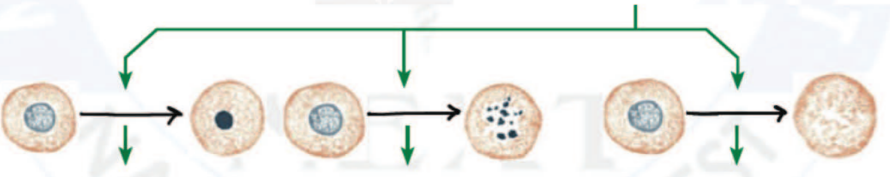
b) Protease

c) Phospholipase → .....



d) Endonuclease or → .....

DNA ase



Chromatin Condensation	Chromatin Fragmentation	Chromatin Dissolution
.....	.....	.....

eg: a) Necrosis
b) Apoptosis
c) Necroptosis
d) Pyroptosis
e) Ferroptosis

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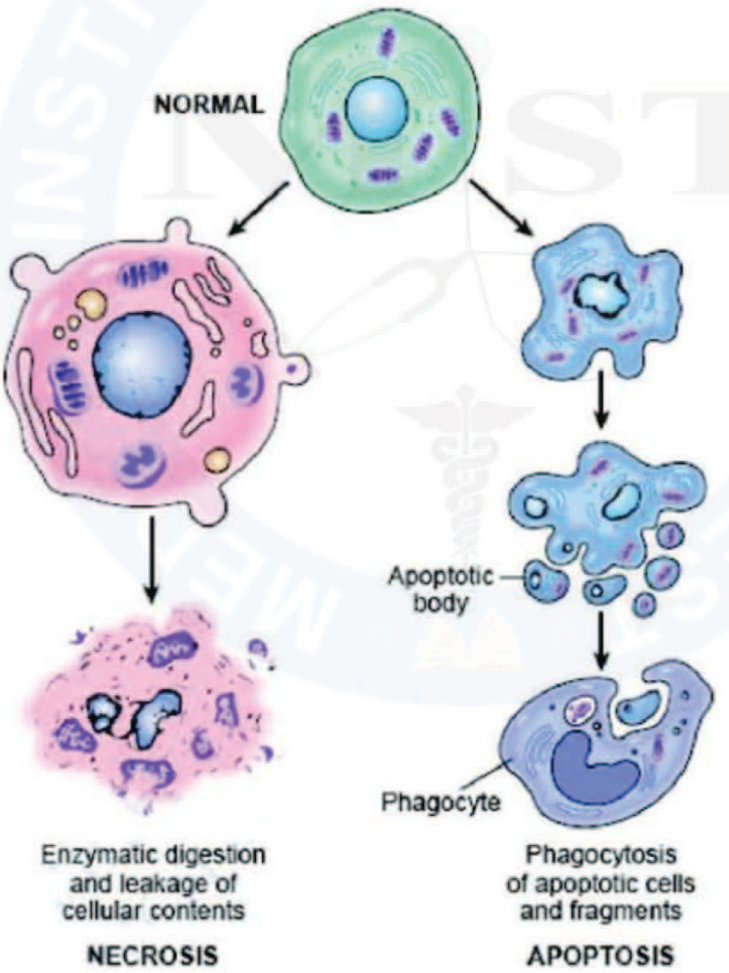
NECROSIS	APOPTOSIS
•	.....
•	.....
• cell size:	.....
• PM:	.....
• Infl^{n}:	.....
• Nuclear Changes:	.....



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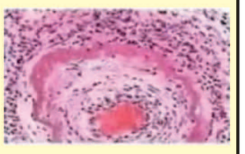
NECROSIS

Type of necrosis	Seen in	Image
1. Coagulative necrosis • d/t ..... •	• Ischemia to all ..... (Except .....) • Thermal Injury • Zenkers Degeneration • Dry gangrene	
2. Liquefactive necrosis • d/t - ..... • Coagulative necrosis with superimposed bacterial infection	• Ischemia to ..... • Wetgangrene	
3. Caseous necrosis • CheesyDeposition • Granuloma (+)	• ..... • Syphilis • Histoplasmosis • Coccidomycosis	
4. Fat necrosis • Chalky white deposit	• Traumatic Injury - Breast - Buttock - Abdomen • Enzymatic - Pancreatitis ↑ Lipase TG → FFA + Ca	
5. Fibrinoid Necrosis	Seen in Vessels : • Vasculitis • Malignant Hypertention • Immune Complex deposition • Rheumatic Carditis (Aschoff Body)	

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APOPTOSIS

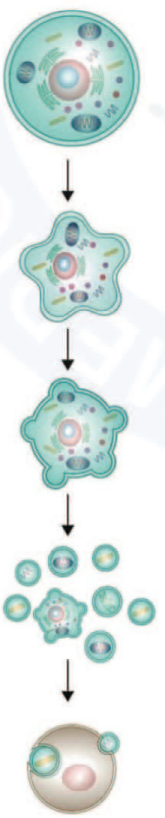
- Programmed cell death

  or

- Caspase dep. cell death

2 Phases:

- Initiation
  - Intrinsic → Mitochondrial pathway
  - Extrinsic → Death Receptor pathway
- Execution



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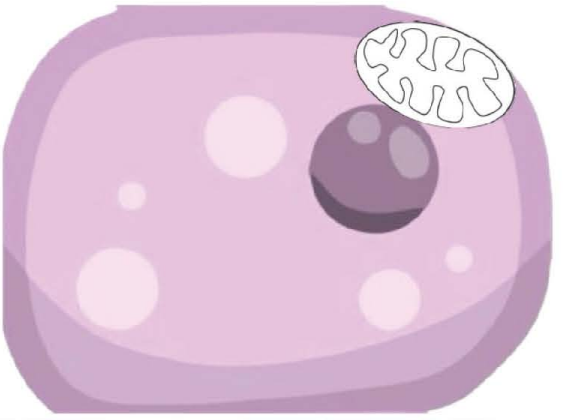
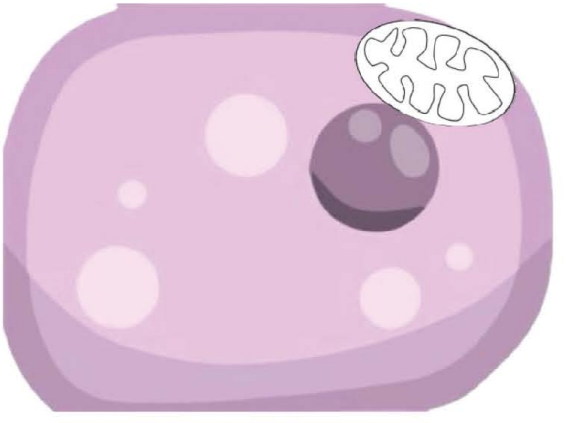
Intrinsic pathway



Sensors	Proapop	Antiapop
PUMA	BAK	BCL
NOXA	BAC	MCL
BIM	BAX	BCL-XL
BID	BCL-XS	BCL-XS
BAD	BAD	

Extrinsic Pathway

- Death Receptor Pathway



✦ Execution Phase: .....

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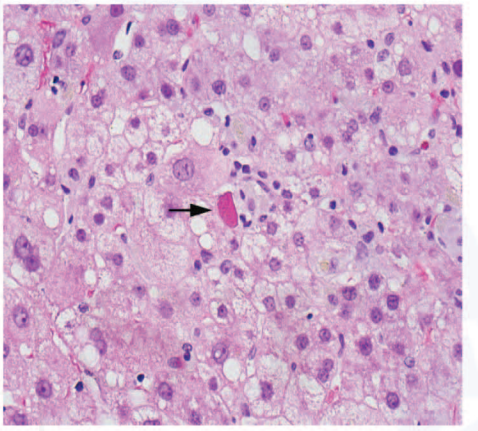
Stress → Initiation → Intrinsic → caspase 9
Or
Extrinsic → caspase 8
→ Execution (Activation of 3, 6, 7)

✦ Caspase: C + ASP + ASE

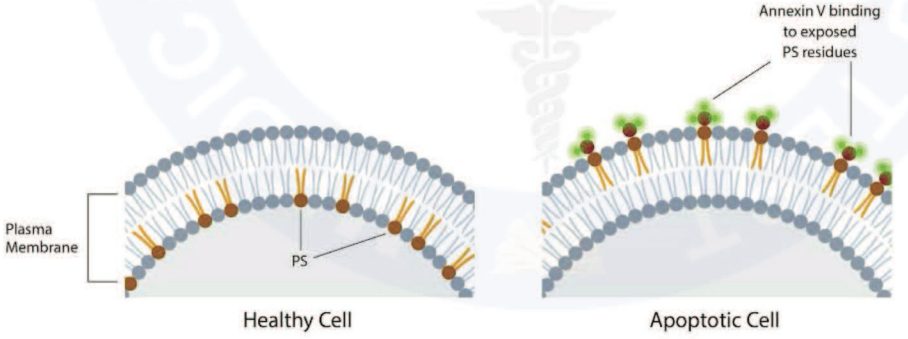
C → made up of cysteine
ASP → Aspartic acid
ASE → cut

	Initiator	Executioner
Intrinsic →	.....	.....
.....		.....
Extrinsic →	.....	.....
.....		



✦ Diagnosis:

i) Marker:



ii) Stain .....

(Terminal UDP neck end labelling)

- Inhibitor of Apoptosis ↓ .....

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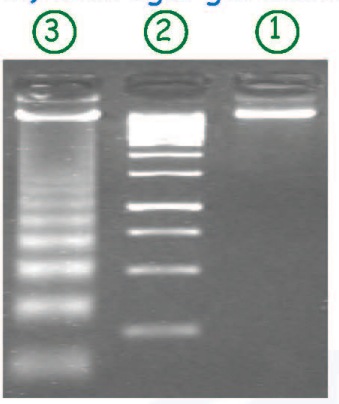
iii) DNA agar gel electrophoresis

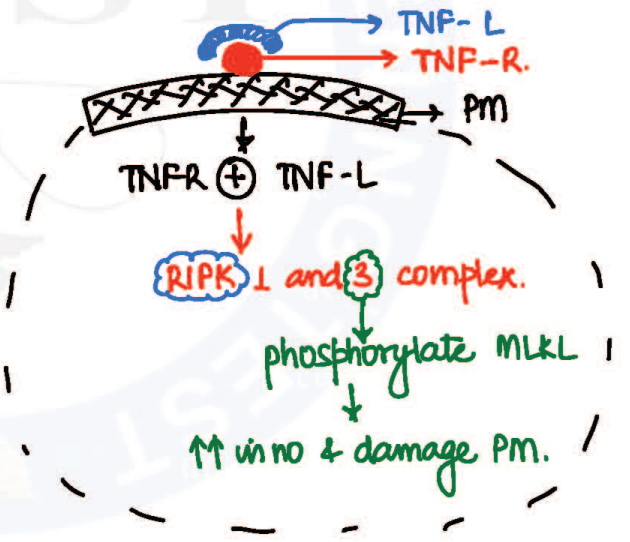


1. .....
2. ..... →
→
3. ..... →

NECROPTOSIS

- Programmed Necrosis
- Caspase independent cell death

Receptor Interacting Protein Kinase



eg: Physiological → Bone growth plate
Pathological → • Neurodeg d/s
• Pancreatitis

Necroptosis is useful in .....

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PYROPTOSIS

↓ Apoptosis a/w Fever

Microbial toxin ↑

↓

NOD receptor Secrete .....

↓ ↑

..... → Activates .....

FERROPTOSIS

↑↑Fe/ROS

↓

↓↓protectⁿ from glutathione dependent antioxidant

↓

↑↑Lipid peroxidation

↓

↑↑PM permeability

↓

cell death

M/ch feature: Mitochondrial cristae damage

Outer mitochondrial layer damage

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INTRACELLULAR ACCUMULATIONS

Glycogen :

- Glycogen storage d/s
- PAS ⊕
- DM

Protein: PAS ⊕

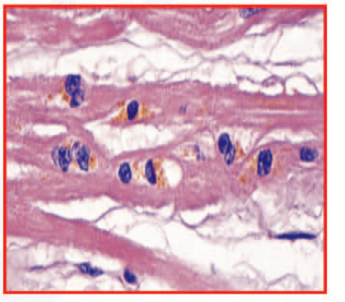
Fat:

- TG's: Fatty Liver
- Cholesterol: Xanthoma, Atheroma
- PL: Myelin figures
- Sudan Black B
- Oil Red O

A) Pigments

i) Lipofuscin - aka ↓ Perinuclear pigment

- Lipochrome
- Wear & Tear
- Senile
- Ageing



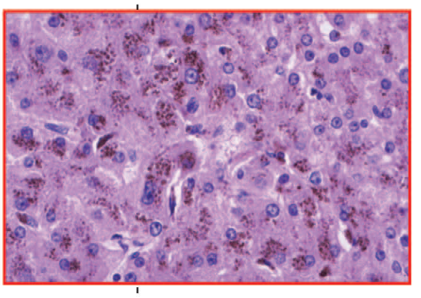
ii) Hemosiderin:

- Hb derived
- Dark Red/brownish color

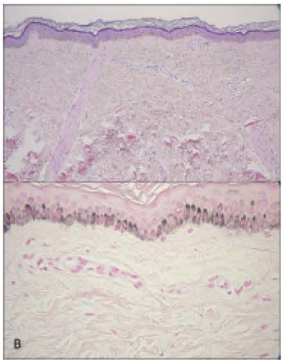
Cause:

- Repeated blood Transfusion
- Hemolytic anemia

Stain: Prussian Blue/Perl's stain



iii) Melanin:
- Black in color
- Masson Fontana stain



B) Calcium: Stain: Von Kossa, Alzarin Red

↓

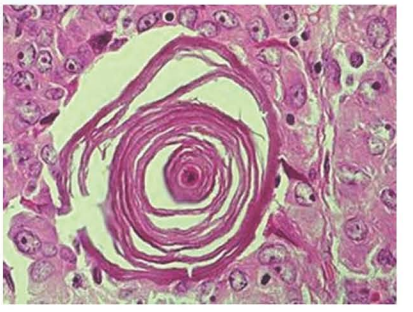
Pathological Calcification

Dystrophic	Metastatic
•	.....
• S. cal levels: -	.....
eg: A →	• .....
R →	• .....
T →	• .....
↓	• .....
.....	

P : .....

S : .....

M : .....



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CELLULAR AGEING

Telomere → (End part of ch)

- Biological clocks
- c̄ each division

↓

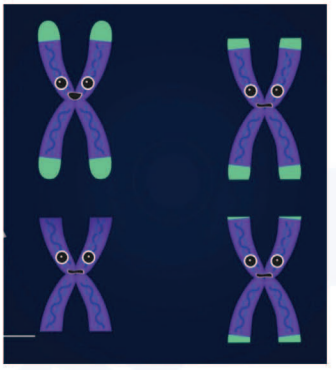
length of ch/Telomere shortens

↓

reaches to a point of no further division.

↓

.....



Telomerase: .....

↓ ↓

Present in: .....

G : .....

S : .....

T : .....

Delay in cellular Ageing

- Calorie Restriction
- Sirtuins
  - present in Red wine
  - ↓Apoptosis
  - ↓Cell proliferatⁿ

Active space

FREE RADICAL INJURY

aka → • Reactive O_2 species
• highly charged particle having unpaired electron in outermost orbit

Fⁿ of FR: • PM damage (Lipid peroxidatⁿ)
• Protein damage
• DNA damage

eg;
- OH^- → Hydroxyl .....
- ONOO → Peroxynitrite
- OCl → Hypochlorite .....
- H_2O_2 → Hydrogen peroxide
- O^2 → Superoxide

✦ Formation of FR

MIST

i) Heber Weis Rxn

.....

ii) Fenton Rxn

.....

✦ Removal of FR

i) Antioxidants: a) Vitamins .....

.....

b) Transferrin

c) Lactoferrin

Active space

ii) Enzymes:

- Catalase
- Peroxidase
- Glutathione Peroxidase

- Superoxide Dismutase (SOD) → .....

SOD 1 ↓
- Cytosol
- Cu-Zn

SOD 2 ↓
- Mitochondria
- Mn

- SOD not a true scavenger

other eg: of enzymes forming FR.

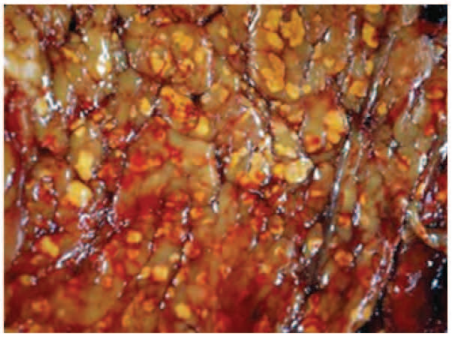
- Xanthine Oxidase
- NADPH oxidase / Phagocytic Oxidase

EXERCISE

1. True about dystrophic calcification is
   A. Calcium level is elevated
   B. Occur in normal cells
   C. Mitochondria involved earliest
   D. None of the above

2. A surgeon was operating on a case of acute pain abdomen. On exploration of abdomen, mesenteries shows the following. What is your diagnosis?

Active space



A. Nutmeg Liver
B. Necrosis in fat tissue leading to Ca accumulation in tisues
C. Putrefaction
D. Gall stones

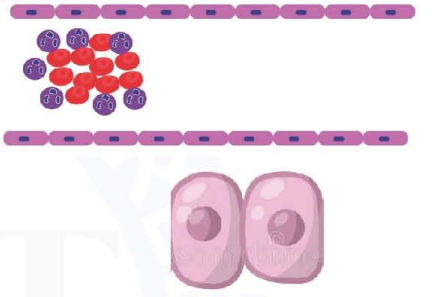
3. After binding to fas ligand (cd95l), fas (cd95) self-associates and activates fas-associated death domain protein (fadd), which in turn induces apoptosis by stimulating
   A. Bcl-2
   B. Caspase 8
   C. Cytochrome a3
   D. Cytochrome p450
4. All are true about apoptosis except?
   A. Apoptotic cells appear as round mass of the intensely eosinophilic cytoplasm with dense nuclear chromatin fragments
   B. Apoptosis of cells induce inflammatory reaction
   C. Condensation of chromatin
   D. Considerable apoptosis may occur in a tissue before it becomes apparent in histology
5. On autopsy of a 75 yrs old man who died during abdominal surgery; the heart was found to be atrophied and brownish in colour. Deposition of the pigment is due to:
   A. Abnormal calcium homeostasis
   B. Lipid peroxidation
   C. Elastic lamina breakdown
   D. Abormal high rise of iron

Active space

INFLAMMATION

INTRODUCTION

- Protective Responce
- Mediated by vascularized tissue so as to bring molecule of host defence system (WBC) from circulation to site of injury to kill the pathogen.



CLASSN OF INFLAMMATN :

1. Based on fluid type
   - Serous
   - Purulent
   - Fibrinoid
   - Catarrhal

2. Based on duration

Acute				Chronic
Sudden	←	Onset	→	Delayed
.....	←	WBC cells	→	.....
Mild	←	Tissue Inj	→	Severe
Prominent	←	Sign & Symp	→	Milder
.....	←	Outcome	→	.....

Active space

SIGNS & SYMP OF INFL^{N}

1. RUBOR → Redness
2. CALOR → Heat/ ↑temp
3. DALOR → Pain
4. TUMOR → Swelling

→ Celsus

5. FUNCTIO LESSA → Loss of f^{n} → Virchow

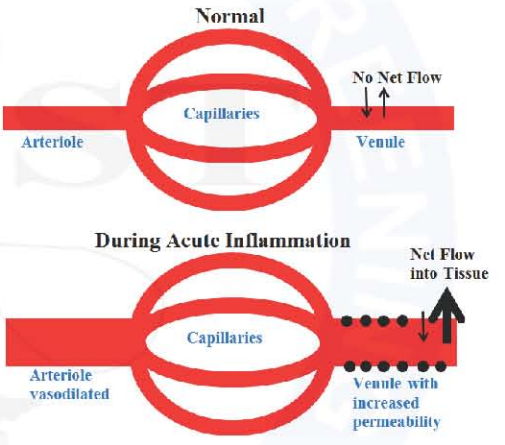
ACUTE INFLAMMATION

2 Events → 1. Vascular (.... Steps)
→ 2. Cellular (.... Steps)

✦ (I) VASCULAR:

i) .....

- d/t .....
- Mainly Seen in arterioles
- Cause: • Rubor
  • Calor



ii) .....

- d/t .....
- Mainly Seen in .....
- Cause: • Dalor
  • Tumor

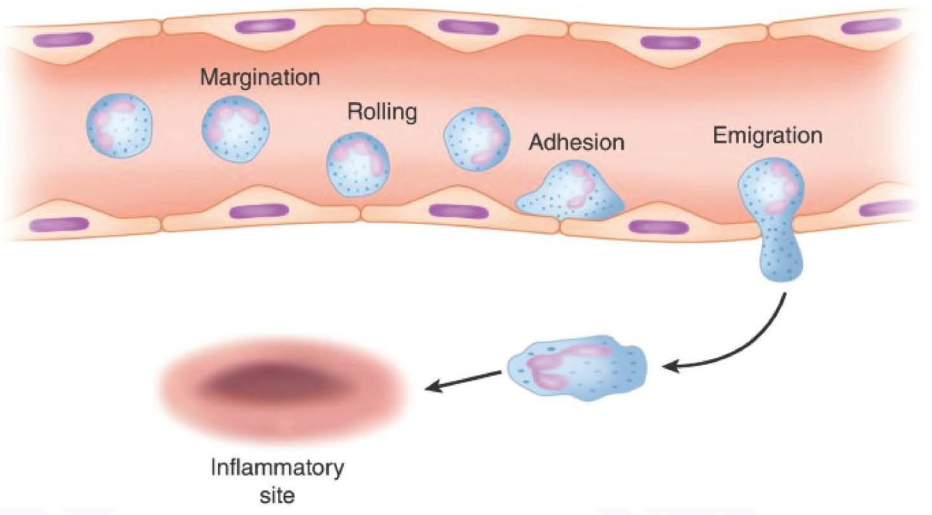
(i) .....
(ii) Endothelial retraction
(iii) Direct Endothelial Injury
(iv) Transcytosis

iii) STASIS: d/t ↑VP, more fluid outside vessel & more cells inside vessel → make blood viscous → STASIS

MIST

Active space

✦ (II) Cellular Events: 7



(i) Margination: movt of WBC towards endothelium

(ii) Rolling:
- ..... Attachment of WBC c endothelium
- Mediated by

Selection / CD62 receptors	Ligands
a) L (Leukocyte) b) E (Endothelial) c) P (Platlet /Endothelium)	....

(iii) Adhesion:
- Attachment of WBC c endothelium
- mediated by .....

Integrin	Ligands
ß 1 / VLA - 4 ß 2 / LFA - 1/CD 11/18	V Cam I Cam

Active space

LAD (Leukocyte Adhesion defect) -

LAD 1 : - d/t deficiency of .....
- delayed Seperation of .....

LAD 2 : - d/t deficiency of .....
- .....

LAD 3 : - d/t ferm T3 gene defect.
- Bleeding.

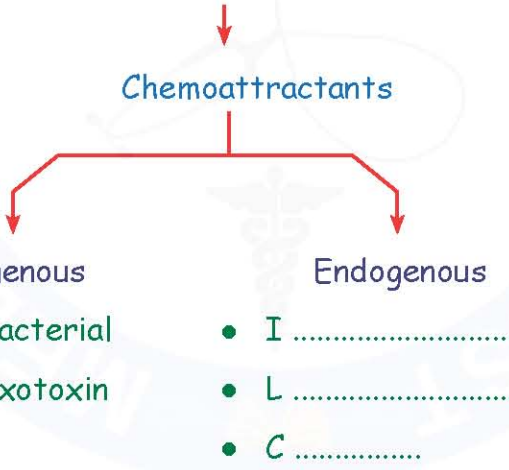
(iv) Pavementing: Whole endothetial lining is covered by WBC.

(v) Diapedesis/Transmigration: Movt of WBC from vessel into tissue.

- Mediated by ..... (Marker of Endothelial cell)

(Plt. endothelial cell adhesion molecule)

(vi) Chemotaxis: ..... movement of WBC towards pathogen under chemical agent gradient.



(vii) Phagocytosis: (3)

A) Recognition and attachment

- Scavenger receptor
- Mannose receptor
- Opsonin receptor → • .....
  • .....

Active space

B) Engulfment: Formation of phagolysosome

↓

Defect → .....

C	-	CNS inv.
HE	-	Hemorrhage
DI	-	Decrease Immunity
A	-	Albinism
K	-	Coarse Cytoplasmic granules.

- .....
  Recurment Infection
- M > F
- ..... gene defect
- Albinism

Occulocutaneous albinism

C) Killing: 2 ways

O_2 dependent

O_2 independent

↓

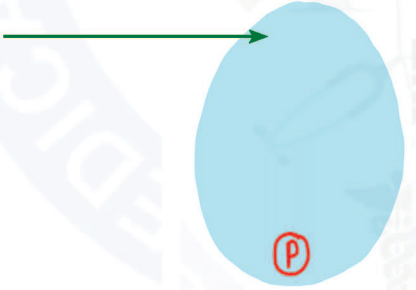
Via Lysosomal granules

↓ 1°

Azurophilic

↓ 2°

Specific



➢ Ch. Granulomatous d/s (O2 dependent)

- .....
- Recurrent infection with Catalase +ve
- Male > female
- Defect in Synthesis of .....
- ..... gene defect → .....

CGD

Screning Test - .....

Confirmatory Test - .....

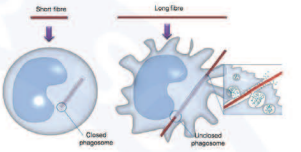
MIST

Active space

➢ Bruton's Hypogammaglobulinemia
- XLR
- Male > female
- ..... defect
- defect in .....

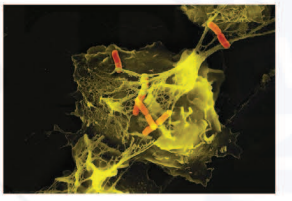
MISCLLANEOUS POINTS

1. Frustated Phagocytosis :-
   - Phagocyte are unable to engulf a target particle d/t its large & Irregular str. leads to partial engulfment.

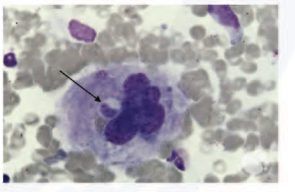


Frustated Phagocytosis

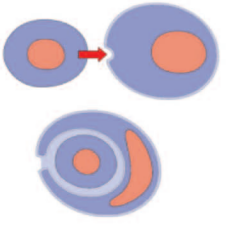
2. NETS : (Beneficial Suicide)
   - Neutrophilic extracellular traps.
   - Suicidal pathway : .....
   - Chromatin material of neutrophils Comes out & traps the antigen
   - ..... is required.
   - Seen in .....
   - ↑ Risk of autoimmune disorder


NETS

3. Emperipolesis :
   - Cell .....with in cell - No Killing
   - E.g. Non-Hodgkin Lymphoma
     Rosai Dorfman
     AIH
     CLL


Megakaryocytic Emperipolesis

4. Entosis:
   - Cell ..... with in cell - No Killing


Entosis

Active space

SYSTEMIC EFFECT OF INFLAMMATION : -

- ✦ Fever - IL -1, IL-6 (Pyrogenic cytokiness)
- ✦ Leukocytosis

Acute phase reactants :

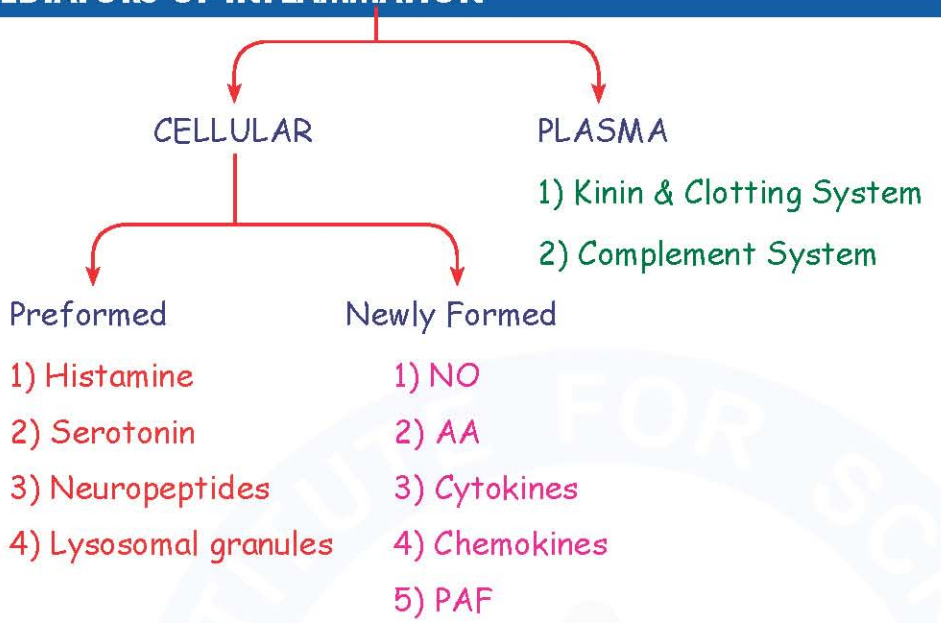
Acute Inflammation Response.

1. (+) respnse : - hight at the time of inflammation
   - Factor VIII
   - Hepcidin
   - IL-6
   - Ceruoplasmin
   - CRP
   - Fibrinogen
   - VWF
   - Haptoglobin

2. (-) respnse : - Low at the time of inflammation
   - FIlbumin
   - Transcortin
   - Transferrin

Active space

MEDIATORS OF INFLAMMATION



Preformed:

1. Histamine:
   - .....
   - Richest source: .....
     ↓
     Toluidine Blue

   Act^{n}:
   - .....
   - .....
   - .....

2. Serotonin:
   - Richest source: .....
   - Major role → NT in GIT
   - Act^{n} is same as Histamine (weak)

3. Neuropeptide:
   - e.g.: substance P

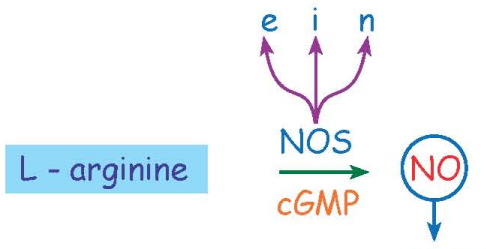
   Act^{n}:
   - Pain

4. Lysosomal Granules: 1° 2°

Active space

Newly Formed:

1) NO:



L - arginine

e-endothelial

i- inducible

n - neuronal

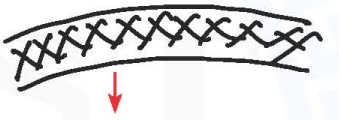
Act^{n}:

i) .....

ii) .....

iii) .....

2) Arachdonic Acid (AA) .....



AA

(COX) Cyclooxygenase | Lipooxygenase (LOX)

(i) Prostaglandins	5-Lox	12-Lox
PGD2	(ii) Leukotrines	(iii) Lipoxins
E2	LTA4	Lx A4
F2	B4	B4
G2	C4	
H2	D4	
I2	E4	

Act^{n}: Same as histamine ⊕ Plat antiagg

PGE_2 → .....

PGF_2 → .....

Thromboxane A_2: • .....

• .....

Active space

3) Cytokines: (IL/IFN/TNF) (Highly Specific)

- Made up of peptides

eg:
- IL-1 .....
- IL-4, 6 .....
- IL-5 .....
- IL-8 .....
- IL-10, 13, TGF-β .....
- TGF β .....
- VEGF .....
- IFN (α) .....
- TNF (α)
- IFN - γ .....

4) Chemokine:

- Chemoattractant
- made up of cysteine
- 4 types:

a) C-X-C (α)	..... Recruitment	IL - 8
b) C-C (β)	..... Except Neutrophil	Rantes, Eotaxin,
c) C (γ)	..... recruitment	MCP-1 Lymphotactin
d) CX3C (δ)	- Mononuclear cells	Fractalkine

5) PAF: Platelet Activating Factor

- .....

Fn:
- agg of platalets
- VD, ↑VP, CSM

Active space

Plasma Mediators

i) Kinin and Clotting System

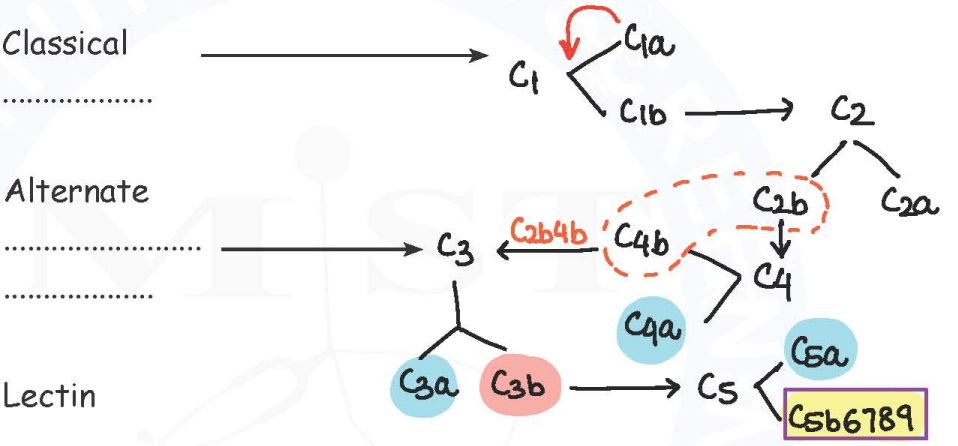
Prekallikrein $\xrightarrow{FXII}$ Kallekrein $\xrightarrow{HMWK}$ Bradykinin

Actⁿ: Pain + Same as histamine (weak)

ii) Complement System

- > 20 complement products (C_1-C_9)
- Present in inactive state

Complement Actⁿ : 3 pathways



Fⁿ of CS

- Opsonization: .....
- Chemoattractⁿ: .....
- Cell lysis: .....

↓

.....

CHRONIC INFLAMMATION

❑ Features

i) Infiltration by mononuclear cells (mϕ & Lymphocytes)

ii) Tissue destruction - .....

iii) Repair

Active space

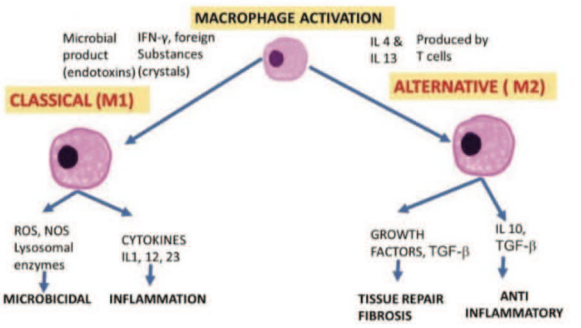
❑ Cells of CI:

i) Mϕ: .....

- mϕ → Tissue

↓

monocyte → Blood



❑ Activatn of mϕ

eg:

Organ	Macrophages
Brain	Microglia
Lungs	Dust Cell
Liver	Kupffer Cell
Kidney	Mesangial Cell
Spleen	Littoral Cell
Placenta	Hoffbeaur Cell
Bone	Osteoclast
Synovium	Type A Synoviocyte

ii)Lymphocyte

iii) Eosinophil:

- 1-6% of total WBC
- Main cells against.....
- IL-5 Eo
- Eotaxin (chemokine)

iv) Basophil:

- 0-1% of total WBC
- Basophil → Blood

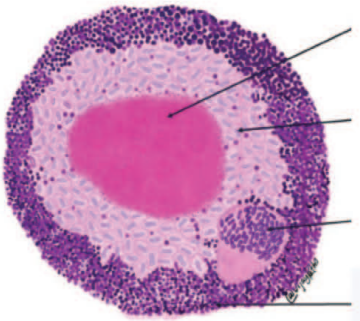
↓

Mast cell → Tissue

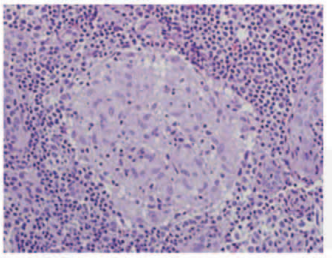
Active space

GRANULOMATOUS INFLAMMATION

→ Characterized by Granuloma Formatⁿ



Ag → CD4TH$_1$ → IFN-γ → mϕ Activatⁿ

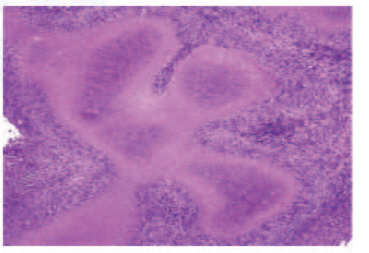


✦ Types of Granuloma:

- Immune → Granuloma (+) .....
- Non immune → Granuloma (-) .....
- Caseating → Granuloma (+) ..... necrosis → .....
- Non-caseating → Granuloma (-) ..... necrosis → .....

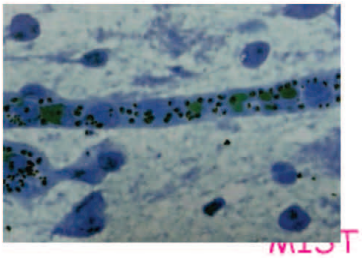
✦ STELLATE:

- ..... Shaped
- ..... infiltration
- eg : • Cat scratch disease
  - LGV



✦ DURCK'S:

- seen c̄ cerebral malaria (P Falciparum)



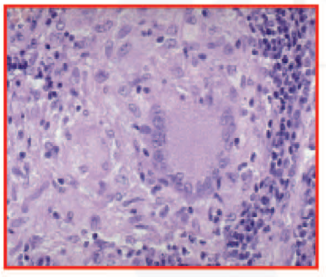
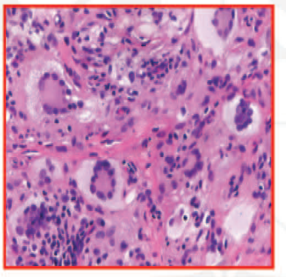
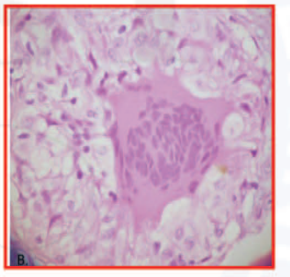
Active space

- DOUGHNUT :
- EOSINOPHILIC:
  - Central eosinophilic infiltrate
  - seen in .....
- Types of Giant Cells: 3 on Basis of nuclear arrangement

Peripheral → .....

Central → .....

Haphazard → .....

REPAIR

REGENERATION → damaged cells replaced by ..... of cells → depends upon proliferatⁿ capacity → (3)

SCAR FORMATION → damaged cells replaced by .....

Active space

i) Stable Cells
- divide only on stimulus
- aka QUISCENT CELLS

eg: • Liver Go → G_1
GD

ii) LABILE CELLS
- continuosly dividing
- G2- M

eg: • Skin epi
• epi of vagina, cervix
• HSC

iii) PERMANENT CELLS:
- Do not divide at all (Go)

eg: ♡, Neurons

STEPS OF SCAR FORM
- Angiogenesis
- Depositn of granulation tissue
- Remodelling .....
.....
.....

- Granulation tissue
  - .....
  - .....
  - .....
  - .....

Small, clean, uninfected

deep, unclean
infected damaged margins
Myofibroblast
(contraction)

MIST

Active space

Events in Scar Formation

- Immediately: Platelet form clot / Neutrophils↑
- Day 1: Neutrophils↑↑ / Lymphocytes↑
- Day 3: Lymphocytes↑↑ (Granulation tissue ⊕)
- Day 5:
  - Max granulation tissue
  - Collagen appears
  - Complete epidermal thickness recovery
- Day 7-14: Max collagen
- Day 28:
  - Scarring completes
  - Collagen III replaced by I

Pathological Aspecs of Repair

DEFICIENT → Ulceration
eg: Diabetic foot

EXCESS
- Proud Flesh → excess granulatⁿ tissue
- Hypertrophic Scar
- Keloid

MIST

HYPERTROPHIC SCAR

KELOID

Active space

Hypertrophic scar		Keloid
Raised scar ..... wound boundary	Character	Raised scar even ..... wound boundary
	Race	
F = M	Sex	F > M
Children	Age	10-30 y
Spontaneous (18-24 months)	Duration	(-)
(-)	Site	..... > Earlobe > umbilicus
	Genetic predisposition	
Itch	Symptoms	itch, pain
	Reoccurence	
	Rx	Intrakeloidal Inj. of Triamcinolone
Thin well-arranged collagen	Histological pattern	Thick haphazard bindle collagen

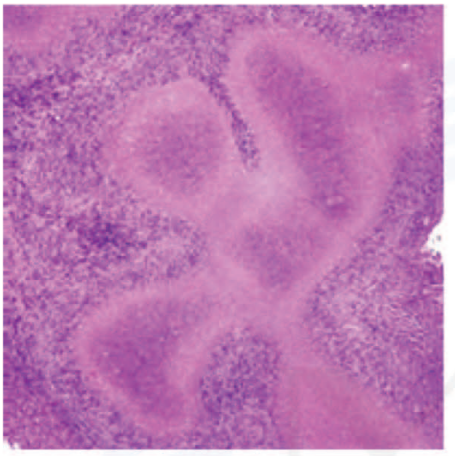
WOUND STRENGTH:

- Wound never attains 100% original strength
- Maximum wound strength → ..... of original (3rd mth)

Active space

EXERCISE

1. 18 year old girl presents with fever from 5 days and a cervical lymph node swelling. A lymph node biopsy was done and granuloma with central neutrophilic infiltrate was found. Whats the probable diagnosis?



A. Malaria

B. Tb

C. Lympho granulo venerum

D. Leprosy

2. Granulation tissue in wound healing is seen at which stage of healing

A. 1st week

B. 2nd week

C. 3rd week

D. End of 2 months

Active space

3. A 3-year-old boy presents with recurrent bacterial and fungal infectionsprimarily involving his skin and respiratory tract. Physical examinationreveals the presence of oculocutaneous albinism. Examination of aperipheral blood smear reveals large granules within neutrophils, lymphocytes,and monocytes. The total neutrophil count is found to be decreased.further workup reveals ineffective bactericidal capabilities of neutrophils due to defective fusion of phagosomes with lysosomes. Pathogenesis regading the following condition is ?

   A. Mutation in PHOX gene

   B. Mutation in HBX gene

   C. Mutation in LYST gene

   D. Mutation in RAS gene

4. A 47-year-old male presents with pain in the midportion of his chest. the pain is associated with eating and swallowing food. Endoscopic examinationreveals an ulcerated area in thelower portion of his esophagus. Histologicsections of tissue taken from this area reveal an ulceration of theesophageal mucosa that is filled with blood, fibrin, proliferating blood vessels,and proliferating fibroblasts. Mitoses are easily found, and most of thecells have prominent nucleoli. Which one of the following correctly describes this ulcerated area?

   A. Caseating granulomatous inflammation

   B. Dysplastic epithelium

   C. Granulation tissue

   D. Squamous cell carcinoma

Active space

5. All of the following responses are seen in acute inflammation except

A. Vasodilatation

B. Neutrophil migration

C. Granuloma formation

D. Increased vascular permeability

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

IMMUNITY

INTRODUCTION

Ability of body to fight against infections

	Native/Innate	Adaptive / Acquired
• Defence →	..... line	..... line
• Activation →	always	require ..... exposure
• Memory →	-	.....
• Specificity →	.....	.....
• Lag time →	.....	.....
• Duration →	.....	.....
• Components →	(i) Skin Epi barrier (ii) Dendritic cell-APC (iii) WBC (Neu, Mϕ) (iv) Complements C3b, C5a (v) ..... (vi) ..... Cells	(i) B-Cell → ..... (ii) T-Cell → .....

Active space

PATTERN RECOGNITION RECEPTORS

Pattern

Pathogen associated molecular pattern (PAMP)

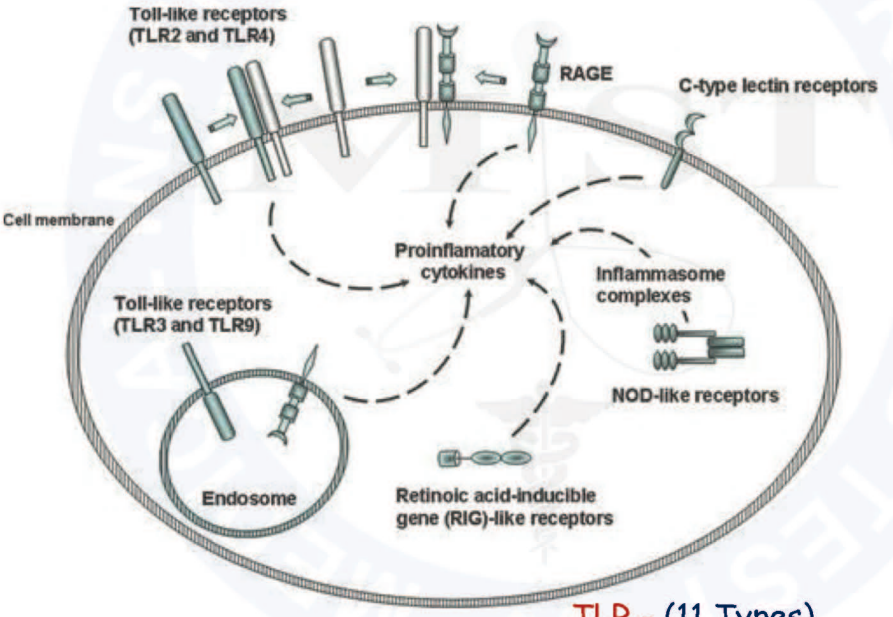
Damage associated molecular pattern (DAMP)

Types of PRR → 3 → an basis of locatⁿ

Plasma memb

Endosomal

Cytoplasmic



TLR - (11 Types)

2 .....

3 .....

4 .....

5 .....

6 .....

7 .....

8 .....

9 .....

TLR → Toll like receptors

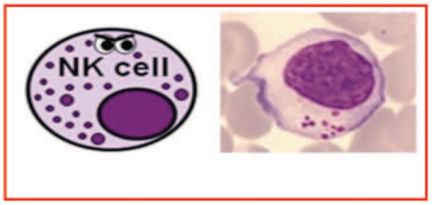
✦ MOA:

MIST

Active space

NK CELLS

- Natural killer cells
- Large granular Lymphocyte
- Null cell

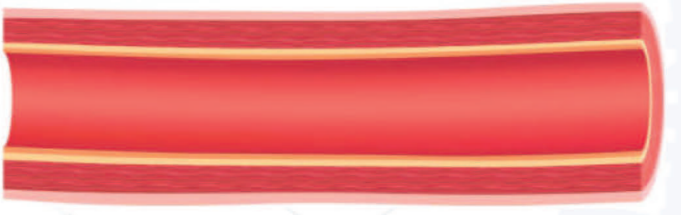


....

....

- active against →
  - .....
  - .....

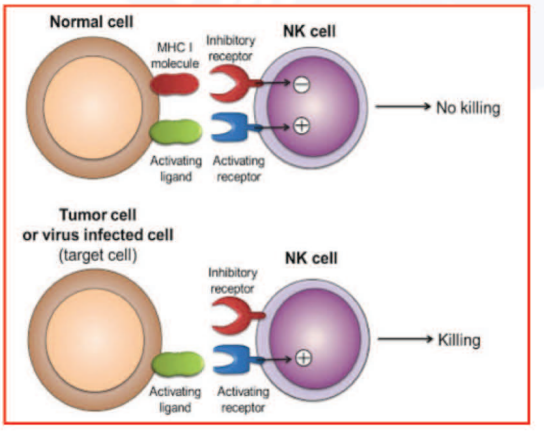
- NK cells cannot kill anything having .....
- Marker: .....
- MOA:



- IL-2
- IL-15

→ ↑es ..... of NK - Cells

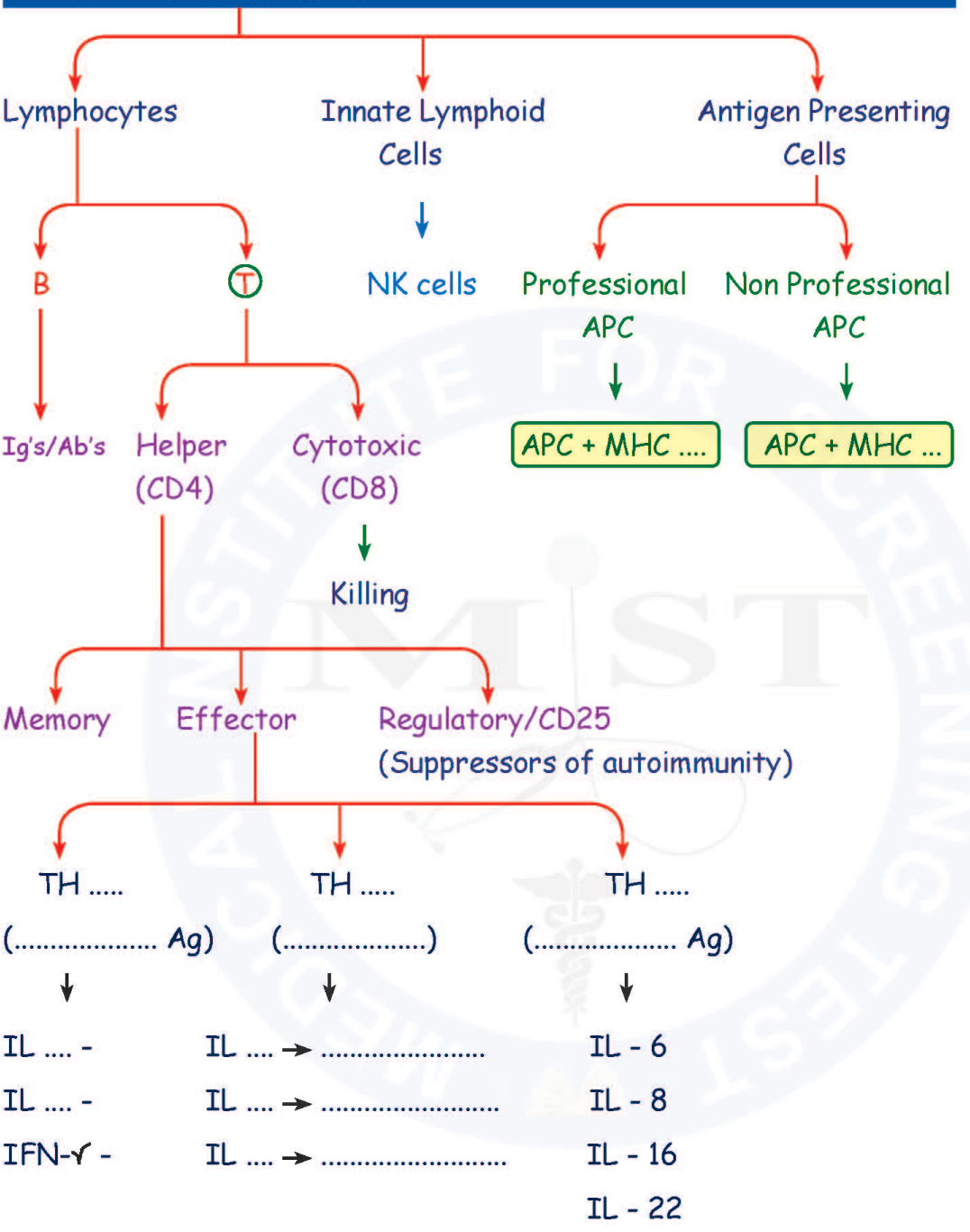
- IL-12 → ↑es ..... NK - Cells



NK Cell are .....Unrestricted

Active space

TYPES OF IMMUNE CELLS



Active space

❑ **Professional APC**

↓

- APC + ...

..... Cellular Antigen

eg
- Dendritic cells
- Mϕ
- Langerhan's cells
- B-cells

❑ **Non-Professional APC**

↓

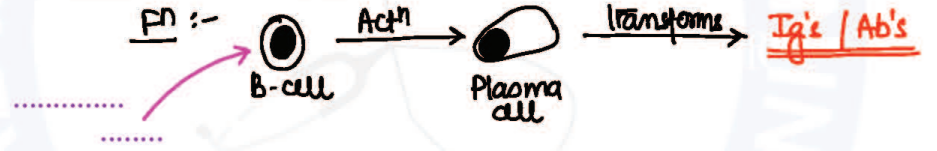
- APC + ...

..... Cellular Antigen

eg:
- Fibroblast
- Endothelial cells
- Thymic cells
- Thyroid epi cells

B-CELLS

- 10-30% of Total lymphocytes
- Origin → .....
- Maturatⁿ (central organ) → .....



✦ BCR: B-cell Receptor:

i) Ig M&D: Ag binding sites on BCR

ii) α and β: Signal transduction molecule
(CD79a/CD79b)

iii) CD21/CR-2: • help the BCR in formatn of Abs
[CD - cluster of differentiatn] • binds to EBV
[CR- compliment receptor]

Active space

iv) CD40:
- Co-stimulatory molecule
- CD40(-) → permanent inactivation of BCR → Anergy
- attaches to T-cell(CD40L) for B-cell activation

✦ B-cell activation

T cell independent	T-cell dependent
↓	↓
• Ag → LPS	• Ag → Proteinaceous
• .Only ..... Ab formation	

T-CELLS

- 60-70% of total lymphocytes

CD4 (70-80%) CD8 (20-30%)

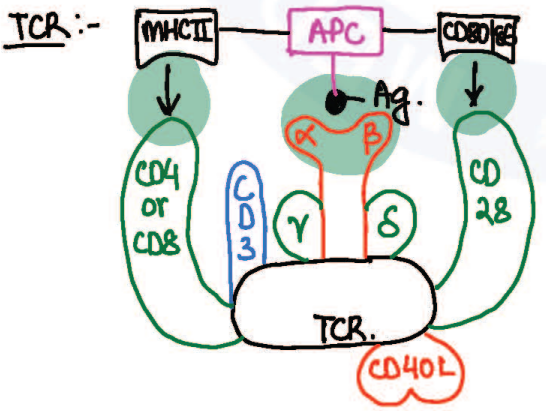
Origin → .....

Maturatn → .....

Normal - CD4 : CD8 = 2 : 1

↑ es - .....

↓ es - .....



i) α and β chains : Ag binding sites for professional APC

ii) γ and δ chains : " " " non-professional APC

MIST

Active space

iii) CD3 : Signal transduction molecule

iv) CD4/CD8: Co-receptors

- Rule of 8

  CD x MHC = 8

  .....

  .....

v) CD28 :
- Co-Stimulatory molecule
- CD28 ⊖ → ANERGY
- Binds to CD80/86 on APC

vi) CD40L :
- ligand for attachment to CD40 present on B cell.

HYPERSENSITIVITY REACTION

✦ 4 types:

I - A → Anaphylactic / immediate

II - C → Cytotoxic /Ab mediated

III - I → Immune Complex mediated

IV - D → Delayed /cell mediated

❑ **i) Type I HSN:**

Pollen (Allergen) → APC → .....

1st exposure

IL-4 IL-5 IL-13

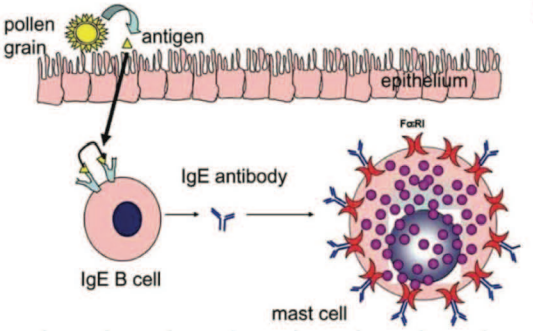
B cells

↓ class switching

.....

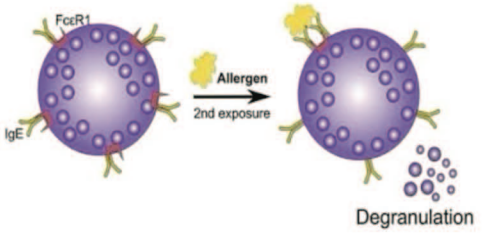
Attach to Fc-R on mast cell

Mast Cell .....



Active space

2nd exposure (<2 hrs)



IgE crosslinks c Ag

↓

Mast cell activatⁿ and .....

↓

.....

↑VP ↑VD CSM

>2 hrs:

↑IL-5 → ↑Eo

↑IL-13 → ↑Mucin

• Damage to resp epi cells → 3C's (BA)
  - Creola Bodies
  - Cirshman Spiral
  - Charcoat Leyden

• Destroy Allergen

eg;

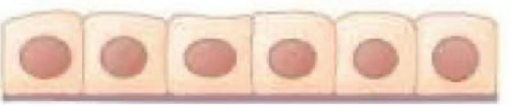
- Food allergy
- Allergic rhinitis
- .....
- Hay Fever
- Prausnitz Rxn
- Casoni Test
- Schultz Dale Phenomenon

❑ **ii) Type II HSN**

Cytotoxic/Ab mediated .....

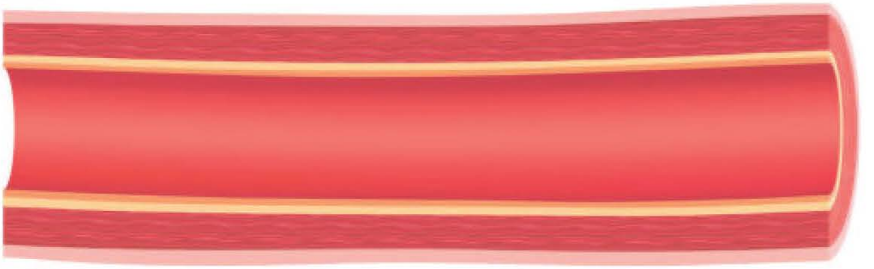
3 mechanism:

a) Inflammatory:



Active space

b) Phagocytic:



c) Receptor dystⁿ:

eg: My → Myasthenia Gravis
Blood → Blood transfusion rxn
Gp → Good Pasteur synd, Grave's d/s
Is → Immune hemolytic Anemia, ITP
Rh → Rh d/s of newborn.
Positive → Pemphigus vulgaris / Pernicious Anemia.

❑ **iii) Type III HSN:** Immune Complex mediated

(Ag + Ab)

.....

Steps:

a) IC formatn

b) IC deposition { arteries → endothetites, vasculitis
capillaries → joints (↑Filteration pressure)
kidney

c) Damage

Active space

MIST

eg: S → SLE, serum sickness Lepra Type II

H → HSP (Henoch Scholein Purpura)

A → Arthus Rxn

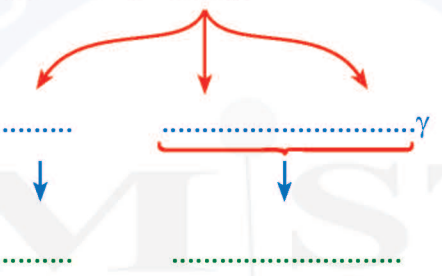
R → Reactive Arthritis

P → PSGN

❑ **iv) Type IV HSN:**

Delayed/cell mediated (..... → .....)

Ch. Ag → APC → .....



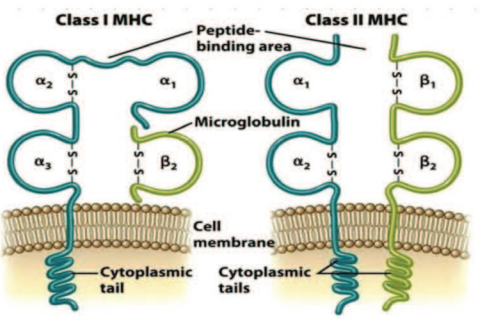
eg:
- Tuberculin Test (Montoux Test)
- Multiple Sclerosis
- Lepra Type I
- Contact dermatitis
- Type I DM

MHC

Major Histocompatibility Complex

aka - HLA - Human leucocytic Antigen

- Gene for MHC is located on .....



Active space

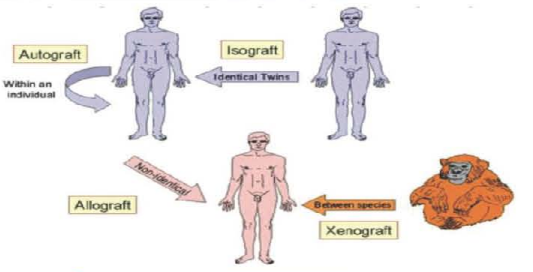
MHC I	MHC II
•	.....
.....	
+	
.....	
•	.....

Uses of MHC - a) Organ Transplant → Bone Marrow transplant 6/6
→ Visceral Organ 3/6
→ Cornea 0/6

b) Paternity Testing

GRAFT REJECTION

- ISO → .....
- AUTO → .....
- ALLO → .....
- XENO → .....



Graft Rejection → solid orga

Hyperacute	Acute	Chronic
•	.....	.....
•	.....	.....

↓ Acute Humoral reject^{n}

↓ Acute cellular rejection

Host → immuno competent → Graft

GVHD: GRAFT VS HOST DISEASE

Graft → Host

immunocompetent

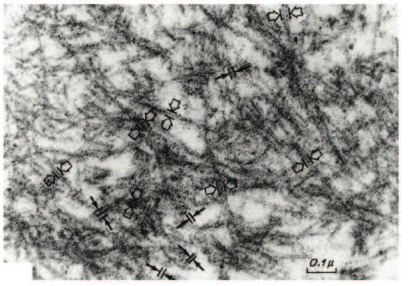
- After Bone Marrow Transplant
- Type IV HSN
- Organ: Skin, gut, liver

MIST

Active space

AMYLOIDOSIS

Abn deposition of Amyloid protein



Electron microscopy → x-ray crystallography

- diameter → ..... β-pleated sheats
- fibrillar st

Amyloidosis → 3 types

1) Localized
2) Systemic
3) Familial

✦ 1) Localized:

- Amyloid calcitonin (A cal) → .....
- A-β → .....
- Apr → .....

✦ 2) Systemic

1° → ..... → ..... → (K/λ)

2° → .....

- Present in plasma cell disorders

.....

Active space

✦ 3) Familial

A) Familial amyloidotic polyneuropathy

- AD
- ATTR → ➢ Mutant
  ➢ Amyloid Transthyrretin

B) Famililial Mediterranian Fever

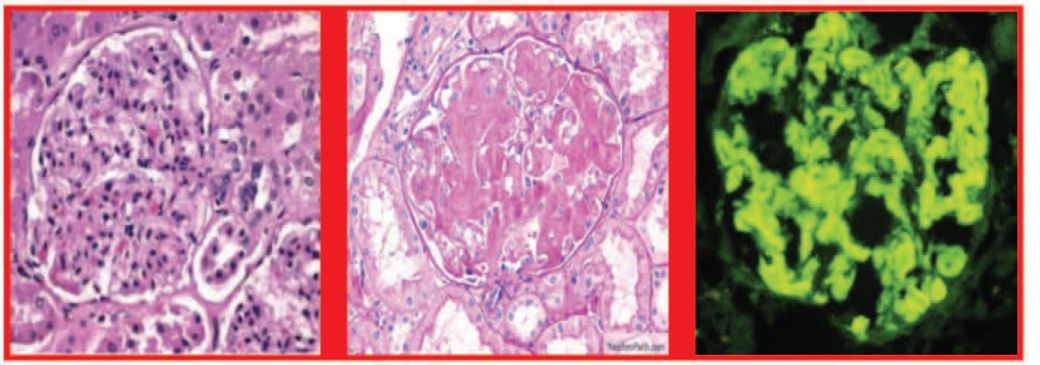
- AR
- AA depositⁿ (d/t Pyrin protein)

Δ: ➢ Stains:

i) H & E → .....

ii) Congo Red → .....

iii) Thioflavin → .....



(A) (B) (C)

Congo Red → LM → .....

Congo Red → Polarized → ..... microscope

- EM/X-ray crystallography
- Rectal biopsy
- Abdominal fat aspirate
- MC organ affected in Amyloidosis (mesangium)

Active space

- Other organs: Heart
  - arrythmias → MCC of death in Amyloidosis
- Spleen in Amyloidosis
  - (Lal) Red pulp (Sinusoid) → Lardaceous spleen
  - White pulp (Safed) (Follicles) → SAGO Spleen

POINTS TO REMEMBER :

AL	→ ..... Amyloidosis (MM)
AA	→ ..... Amyloidosis (SAA), Cancer
ATTR	→ .....
AB2M	→ .....
APr	→ .....
ACal	→ .....
AB	→ .....

IMMUNODEFICIENCY DISORDER :

- 1° → 1. Bruton → Lymphocyte maturation defect
- 2° → MCC - Malnutrition; Other - HIV/AIDS Cancer

2. Hyper IGE
3. Hyper IGM
→ Lymphocyte activation Defect

4. Wiscott Adrich Syn.
5. Alaxia Telangiectasia
→ Systemic disease

MIST

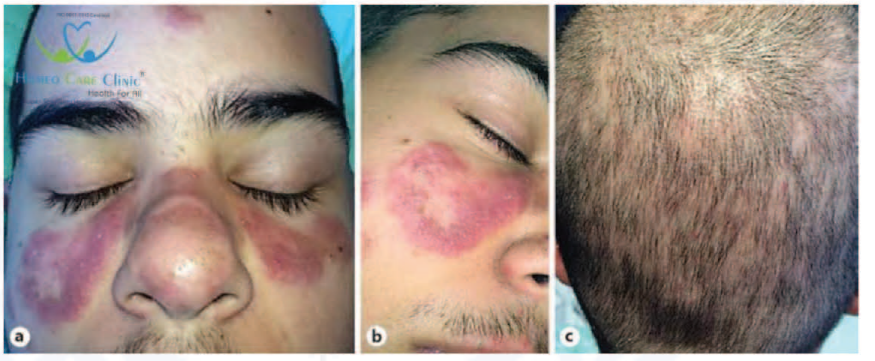
Active space

✦ Bruton : aka → X-linked Agammaglobulinemia

- Absence of ..... & ..... in peripheral blood & Lymphoid tissue
- d/t mutation in ..... responsible for B-cell development
- ..... IgM, G, A, E
- Case → Child with recurrent persistent, sino-pulmonary infection / severe bacterial infection flows yrs.of age pt. with strong family history of immunodeficiency relevant with x-linked inheritance.
- Rx → IvIg infusions

✦ Hyper IgM Syndrome

- Defect in CD40® & CD 40L → ..... IgM ; .... IgG, A, E
- C/F : ↑ Risk of Infections
- Dx : S. immunoglobulin profiling



✦ Hyper IgE Synd / Job Syndrome

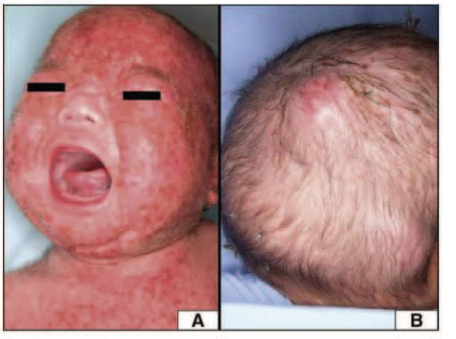
- ..... gene defect

F - Facies (Coarse)

A - Abscess (Cold)

T - Teeth

E - Eczema (↑ IgE)



Active space

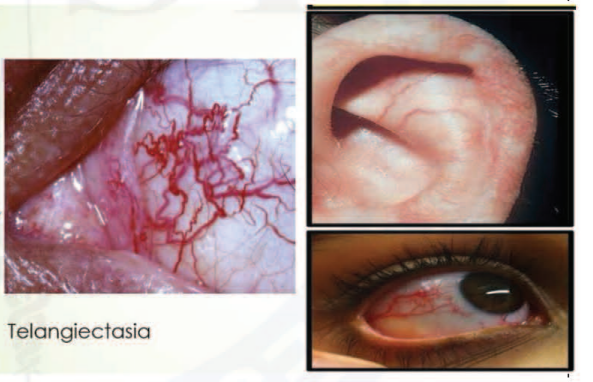
✦ Wiskott Aldrich Synd.

- XLR
- ..... gene defect
- Wiskot Aldrich (Triad) - Thrombocytopenia
  (WATER) - Eczema
  - Recurrent infection

- E.g. profile : ↓ IgM
  ↑ IgE
  ↑ IgA
  (N) IgG

✦ Ataxia Telangiectasia

- Autosomal recessive condⁿ
- ..... Gene defect (Ch 11q 22)
- Ataxia + VD + Immunodeficiency



Telangiectasia

Active space

EXERCISE

1. A 60 year old female is suffering from renal failure and is on hemodialysis since last 8 years. She developed carpel tunnel syndrome. Which of the following finding will be associated

   A. AL B. AA
   C. ATTR D. Beta 2 microglobulin

2. 12. Bee sting will activate following cell in human body in cases of abnormal high response

   A. Th2 cells B. Macrophage
   C. Mast cells D. Neutrophils

3. A 48 yr old male died suddenly due to cardiac arrest. On biopsy taken from the heart, pinkish hyaline material found to be amyloid seen deposited in cardiac myosites as shown in image. Correct statement?

   A. ATTR deposit
   B. B2 amyloid deposit
   C. Left ventricular hypertrophy seen
   D. Rhematic carditis depicting aschoff bodies

4. Hyperacute graft rejection is due to

   A. Preformed antibodies
   B. Cytotoxic t-lymphocyte mediated injury
   C. Circulating macrophage mediated injury
   D. Endothelitis caused by donor antibodies

5. Which of the following statements about graft versus host disease is least correct?

   A. Occurs when host is immunocompromized
   B. Occurs when donor cells are immunocompromized
   C. Is also called runt disease
   D. A common cause is stem cell transplantation

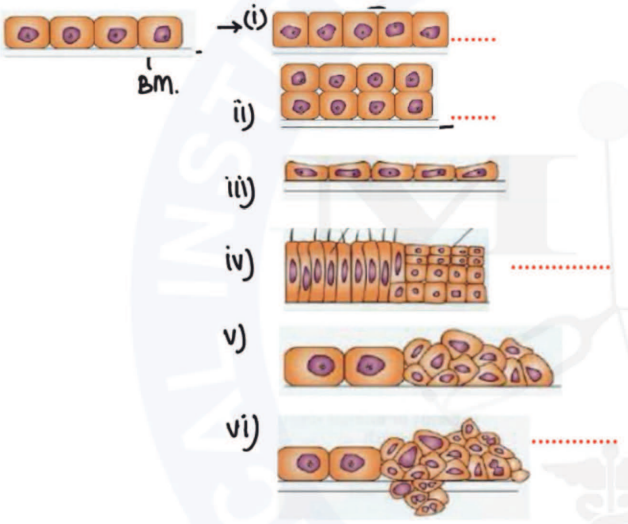
Active space

NEOPLASIA

INTRODUCTION OF NEOPLASIA

Neoplasia → New ⊕ process of growth

Neoplasm → New ⊕ growth



❑ **Differentiatⁿ:**

Well — Moderate — Poorly

❑ **Desmoplasia:** Abundant stroma around tumor cells.

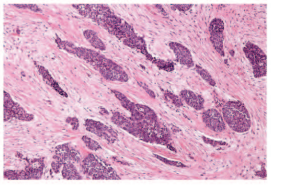
✦ Pleomorphism: → Variations in size & shape

✦ Choriostoma: (N) tissue on (Abn) locatⁿ

eg; Endometriosis

✦ Hamartoma: (Abn) tissue at (N) locatⁿ

eg; Pulm Hamartoma



◀FIG 4.1 Desmoplasia

MIST

Active space

NOMENCLATURE OF TUMOR

✦ i) Tissue of Origin:

Epithelial

Benign	Malignant
.....	.....
✦ Squamous	
Papilloma	Sq cell Carcinoma
✦ Columnar	
Adenoma	Adeno Carcinoma

Mesenchymal

Benign	Malignant	
.....	.....	
Osteoma	Osteosarcoma	Bone
Chondroma	Chondrosarcodma	Cartilage
Angioma	Angiosarcoma	Vessels
Myoma	Myosarcoma	M/S

Except^{n}:
- Lymphoma
- Seminoma
- Melanoma

} Malignant

✦ 2) Mixed Tumor:

Epi + Mes → Pleomorphic Adenoma

✦ 3)Germ layers (≥ 2) → Ectoderm, Mesoderm, Endoderm } Teratoma (Malignant)

a) Cystic: Dermoid cyst
b) Mature: Benign
c) Immature: Malignant

MIST

Active space

Features	Benign	Malignant
i) Rate of growth	.....	.....
ii) Capsule	.....	.....
iii) Mitotic figure	.....	.....
iv) Invasion	.....	.....
v) Metastasis	.....	.....

- Hallmark of Malignancy → .....
- Surest sign of malignancy → .....
- Malignant tumor never metastatize → .....

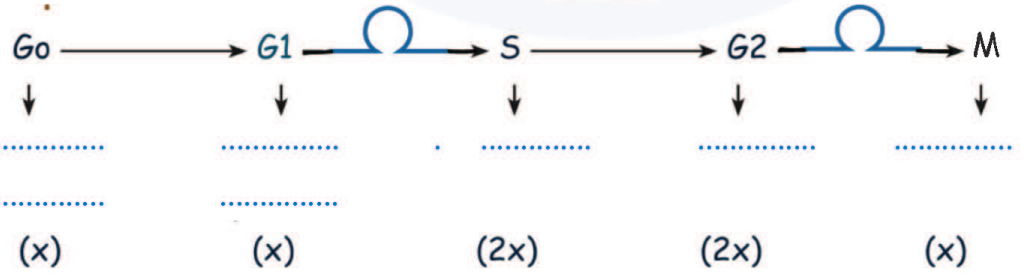
↓

.....

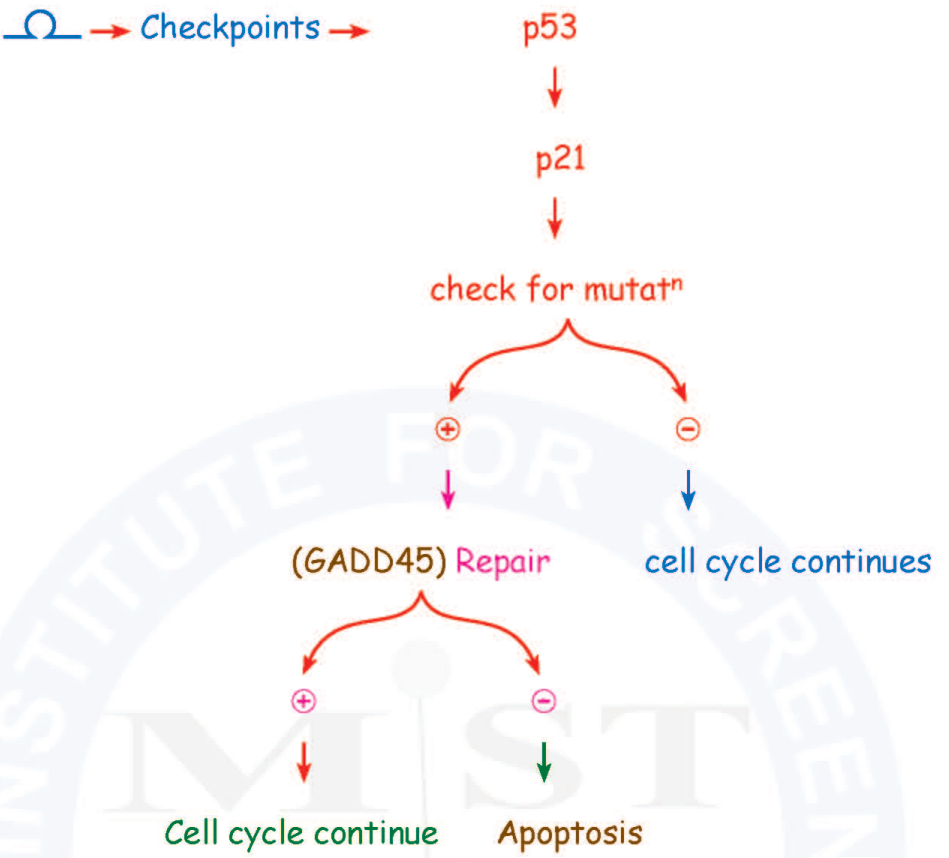
✦ Features:	Metaplasia	Dysplasia	Anaplasia
• Reversibility	.....	.....	.....
• N/C ratio	.....	.....	.....
• Polarity	.....	.....	.....
• Invasion	.....	.....	.....
• Differentⁿ	.....	.....	.....
	.....	.....	.....

CELL CYCLE

❑ **4 phases:**



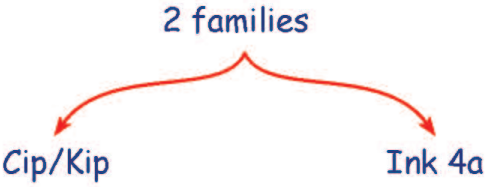
Active space



❑ Cell Cycle Proliferators

Cyclins — Cyclin Dep.Kinase (CDK)

❑ Cell Cycle Inhibitors (Brakes)



Active space

Max DNA Synthesis : .....

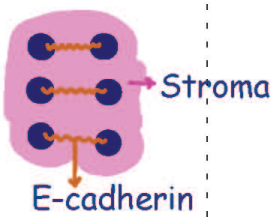
	MOST RR	MOST RS
Phase		
Cell		
Tissue		
Organ		

HALLMARK OF CANCER

M	Metastasis
E	Evasion of Apoptosis
A	Angiogenesis
T	Evasion of T cells
S	Self sufficiency in growth signals
T	Tumor suppressor gene inactivation
A	Altered Metabolism
L	Limittless Replication potential

MIST

Active space



✦ 1) M - Metastasis

- Spread of tumor cell away from original site.

Hematogenous	Lymphatic	Direct Spread.
.....	.....	- Ovarian ca. (mucinous) ↓ Peritoneal. Cavity (aka Pseudomyxoma Perotoni)
Except^{n}: - Renal cell Carcinoma - Hepatocellular carcinoma	Except^{n}: - Synovial Sarcoma - Alveolar Rhabdomyo Sarcoma	- Medulloblastoma ↓ drop metastasis. (Directly goes to CSF)

✦ 2) E- Evasion of Apoptosis

- d/t inactivation of TSG and overexpression of antiapop mol.

↓

↑↑BCl-2 → Follicular Lymphoma

↑↑BCL-6 → DLBCL

✦ 3) A - Angiogenesis

- Format^{n} of new vessels → ↑↑ VEGF, ↑↑ PDGF, ↑↑ FGF

✦ 4) T - Evasion of T-cells

CD4 CD8

Tumor cell -
- lacks Ag expression
- inhibition of CD 28.
- lack of MHCI

MIST

Active space

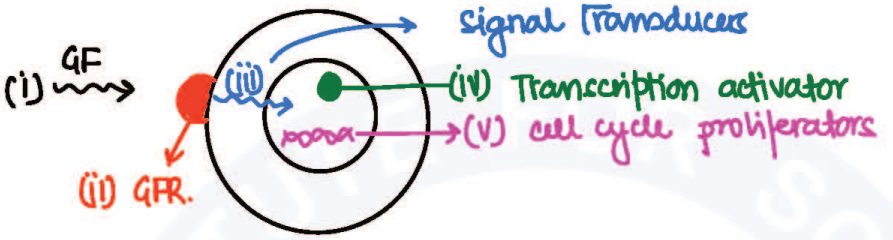
✦ 5) S - Self sufficiency of growth signals

(N) individual → Protooncogenes ((N) physiological genes)

↓ mutation

Oncogenes

5 families:



- PDGF .....
- FGF .....
- HGF .....
- PDGFR .....
- EGFR 1 .....
- EGFR 2 .....
- RET .....
- KIT .....
- ALK .....
- RAS - K .....
  - H .....
  - N .....

MC Oncogene mutated in humans

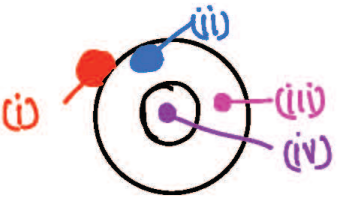
- ABL .....
- BRAF .....
- MYC - C .....
  - N .....
  - L .....
- CDK 4 .....
- Cyclin D_1 .....

MIST

Active space

✦ 6) T - Tumor Suppressor gene inactivation

4 types;



	Location	Tumor
• p53	.....	.....
• RB	.....	.....
• NF 1	.....	.....
• NF 2	.....	.....
• BRCA 1	.....	.....
• BRCA2	.....	.....
• WT1	.....	.....
• WT 2	.....	.....
• VHL	.....	.....

❑ **(i) p53:** • Gaurdian of Genome
• Molecular policeman of cell cycle

- locat^{n}: 17p
- mutat^{n}: Li-Fraumeni synd

↓

multiple tumors at young age

B → Brain, Breast, Bone

L → Lung, Liver

A → Adrenals

S → Salivary glands, stomach

T → Tumors

- .....

Active space

❑ ii) **RB gene:** Governor of cell cycle

location: 13q.

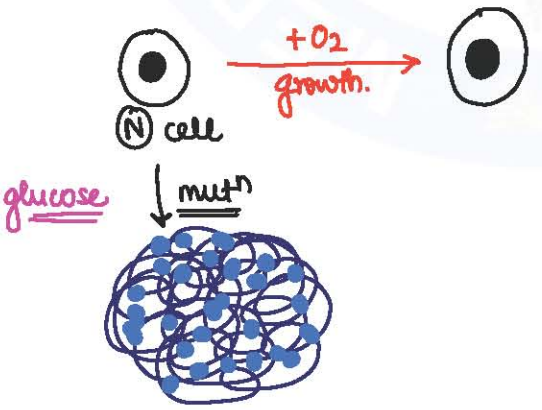
mut^{n}:
- B/L Retinoblastoma
- Pineoblastoma
- Osteosarcoma

→ T/L Retinoblastoma

☞ Knudson Hypothesis/LOH-Loss of Heterozygosity

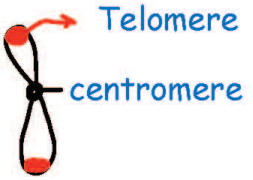
- Role of RB gene in cell cycle.

✦ 7) A - Alltered cellular metabolism (Warburg Hypothesis)



Active space

✦ 8) L - Limitless Replication potential



(Shortens c̄ each division
↓
when further shortening ⊖
↓
cell senescence)

Telomerase → Maintains Telomere length
↓
G S T → Unlimited Replicatn

ASPECTS RELATED TO TUMOR

❑ i) CANCER CACHEXIA → Loss of lean body mass
↓ .....
M/I cytokine → .....

❑ ii) PARANEOPLASTIC SYND

- Synd a/w cancers but can't be explained with anatomical locatn of cancer & spread.

eg: Lung Ca
a) Hypercalcemia
b) Cushing Syndrome
c) SIADH

Active space

❑ iii) TUMOR LYSIS SYND:

- HYPER .....
- HYPER .....
- HYPER .....
- ....

Ca xPO4 = constant

↓ ↑ = constant

❑ iv) STAIN

- Epithelial — Cytokeratin
- Mesenchymal — Vimentin
- M/S — Desmin
- Melanoma
- ChondroSa
- LipoSa

→ S100, HMB45

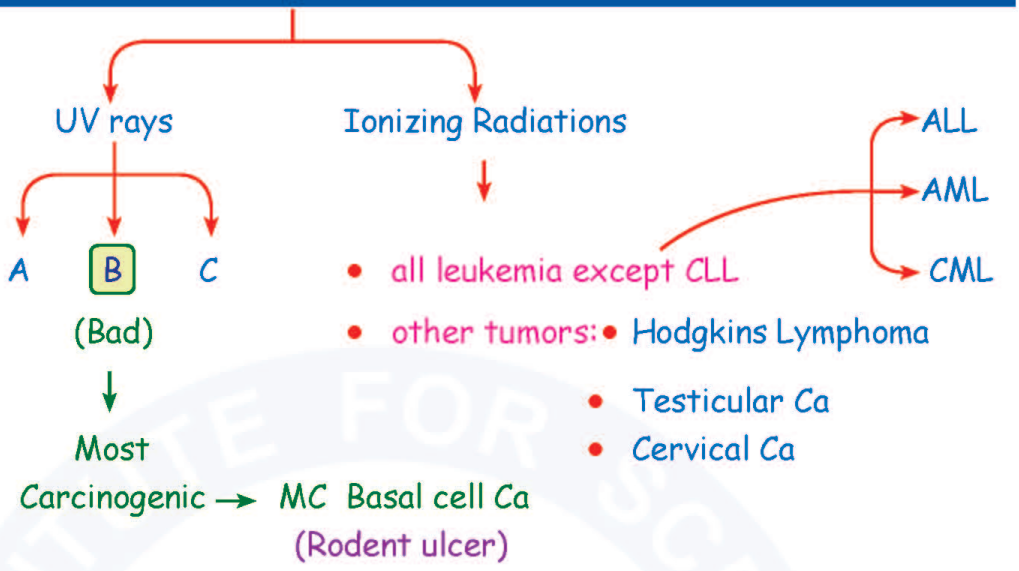
- Neuroendocine
  - .....
  - .....

Tumor Markers

- Calcitonin .....
- AFP .....
- βHCG .....
- PSA .....
- Catecholamine .....
- CEA .....
- Ca 72.4 .....
- Ca 19.9 .....
- Ca 15.3 .....
- Ca 125 .....

Active space

RADIATION CARCINOGENESIS



CHEMICAL CARCINOGENESIS

✦ AMES TEST

S. Typhimurium Chemical 1 → Histidine (-) → SAFE

Histidine (-) Chemical 2 → Histidine (+) → UNSAFE

Chemicals	Cancer
Polycyclic Aromatic Hydrocarbons	.....
Arsenic	• ..... • Hepatic angiosarcoma
Asbestos.	• ..... (MC) • ..... (Most Specific)
Aflatoxins	.....
Beta Naphthylamine / azodye	..... cancer
Benzene	.....
Polyvinyl chloride/Thorotrast	Hepatic Angiosarcoma
Cadmium.	..... cancer.

Active space

EXERCISE

1. Pancreatic cancer is having which of the following as an immunohistochemical marker

   A. Cytokeratin

   B. Calretin

   C. Synaptophysin

   D. Carcinoembryonic antigen

2. Phosphorylation of the Rb gene will have which of the following effect on cell cycle

   A. Inhibition of cell cycle at G1 phase

   B. Inhibition of cell cycle at G2 phase

   C. The cell cycle will progress as it is and the cell will divide

   D. There will be no effect on cell cycle as for Rb gene phosphorylation is not needed

3. Least Radioresistent Phase of cell cycle

   A. S

   B. G1

   C. G2-M

   D. G1-S

4. A 59-year-old male is found to have a 3.5-cm mass in the right upper lobe of his lung. A biopsy of this mass is diagnosed as a moderately differentiated squamous cell carcinoma.The paraneoplastic syndrome associated most commonly with this condition is

Active space

A. Cushing syndrome

B. Hypercalcemia

C. SIADH

D. Medullary Ca thyroid

5. Find the false matching

A. Cytokeratin- epithelial cells

B. Ca125- ovarian tumor

C. Calcitonin- papillary carcinoma thyroid

D. Afp- yolk sac tumor

Active space

GENETICS

INTRODUCTION OF GENETICS

Study of genetic material

- Genes
  - Genomics
    - = 25,000 genes
    - made up of
      - = 3.2 billion base pairs
        - CODONS
    - 2 parts
      - ..... → (Coding Region) (≤ 1.5%)
      - ..... → (Non - Coding Region) (≥ 98.5%)
        - Eg. - mi - RNA
        - Centromere
        - Telomere
        - Transposon /Jumping gene
- Chromosome
  - Karyotyping

Phenotype

- .....
- .....
- .....
- .....
- .....

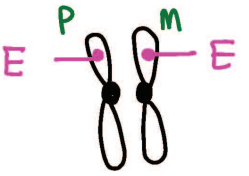
Genotype

- .....
- .....

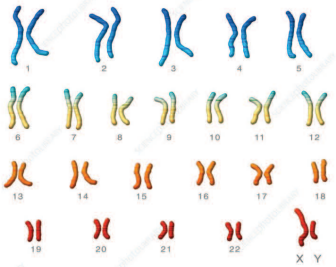
MIST

Active space

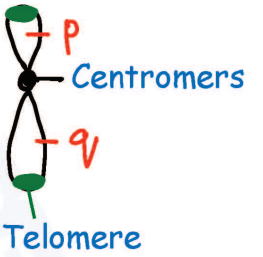
✦ Allele: Alternate form of a gene on another arm of chromosome.



✦ CHROMOSOME:



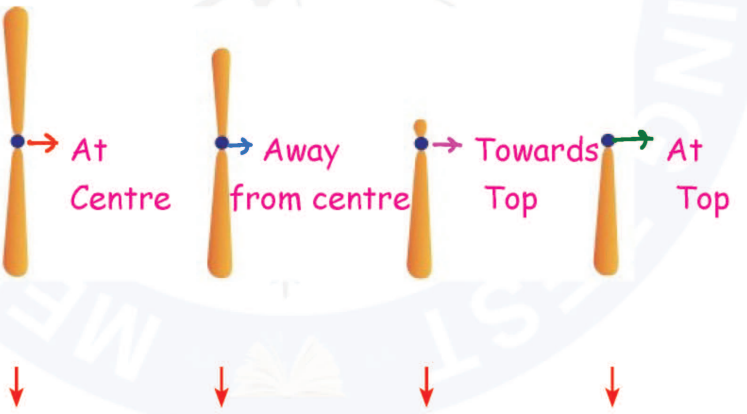
- n = 23 pairs
- 2n = 46 total ch
- 2Types: (i) Sex Chromosomes → X, Y → 1 Pair



(ii) Autosomes → 22 pair

- Largest Ch: .....
- Smallest Ch: .....

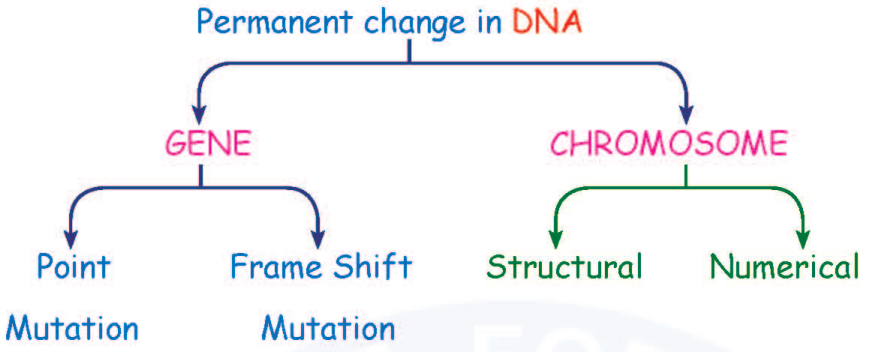
✦ Types of Ch on basis of centromere



....

Active space

MUTATIONS



✦ GENE MUTATION:

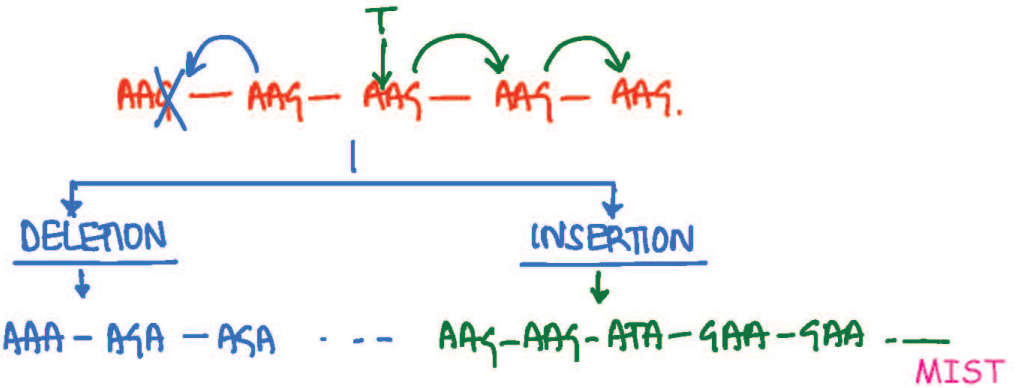
i) Point Mutatⁿ: → replacement of nucleotide

A) TAG → A amino acid.
mutatⁿ
TAA → A amino acid.

B) TAG → A amino Acid.
mutatⁿ
TAU → B amino acid.

C) TAG → A amino acid
mutatⁿ
UAG → X (No amino acid)

ii) FS mutⁿ:



MIST

Active space

CHROMOSOMAL MUTATIONS:

i) NUMERICAL:

EUPLOIDY

Total no of Ch after mutatⁿ are in exact multiplication of n = 23

eg: n = 23
2n = 46 } N

3n = 69 Triploidy

4n = 92 Tetraploidy

5n = Pentaploidy

6n =

ANEUPLOIDY

Total no of Ch after mutatⁿ are not exact multiple of n = 23

Trisomy

eg: (2n + 1)

47

SC → KFS

A → Trisomy 13
16
18
21

Monosomy

(2n-1)

45

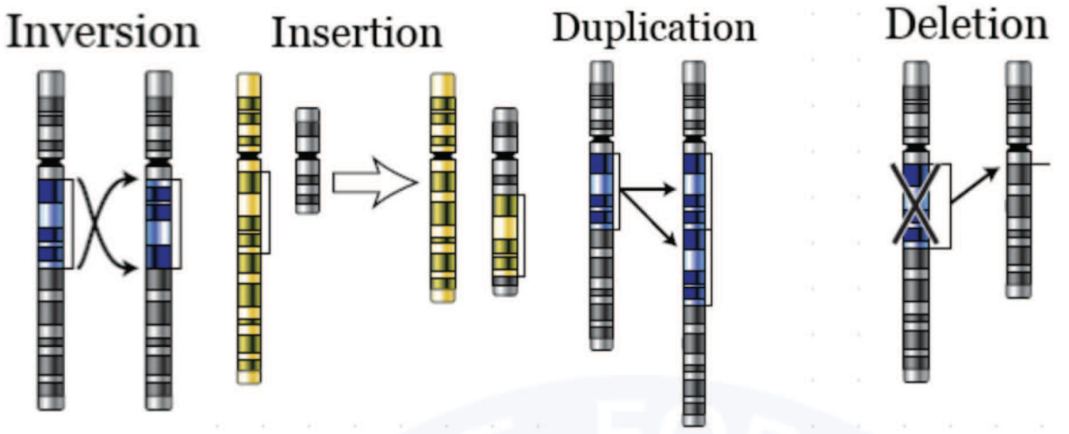
SC → TS

A → Not viable for life

ii) STRUCTURAL

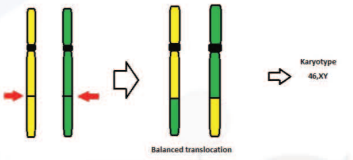
- Inversion
- Insertion
- Duplicatⁿ
- Deletion

Active space

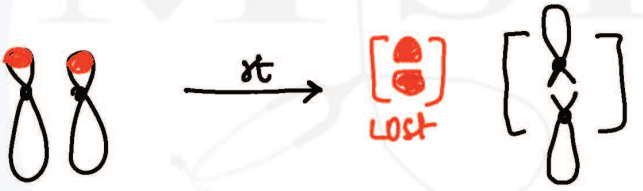


- Translocation (t) → Exchange of arms

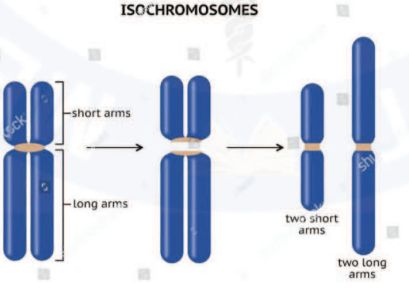
  a) Balanced:



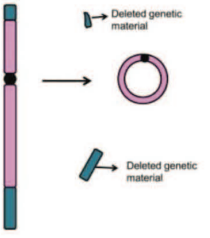
  b) Robertsonian: (rt) (ch 13, 14, 15, 21, 22)



- Isochromosome



- Ring chromosome: (r)



Active space

✦ Causes of Mutations

i) Anaphase lag

ii) Non- dysjunction during mitosis → .....

.....

iii) Non - " " meiosis → .....

.....

Pattern of Inheritence of Genetic Disease

- Single gene
  - Classical (Mendelian)
    - (i) AD
    - (ii) AR
    - (iii) XLD
    - (iv) XLR
  - Non-Classical (Non-Mendelian)
    - (i) Genomic imprinting
    - (ii) Gonadal Mosaicism
    - (iii) Mitochondrial Inheritence
    - (iv) TNR Mutations
- Multigenetic
  - Enviournmental
    - (i) CL & CP
    - (ii) DM
    - (iii) HTN

NON CLASSICAL/NON-MENDELIAN PATTERN

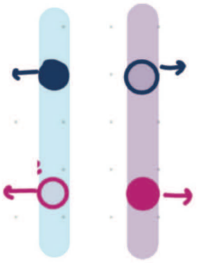
❑ **(i) Genomic Imprinting:** .....

↓

Mechanism: ..... → DNA methylation

Mechanism: ..... → Histone acetylation

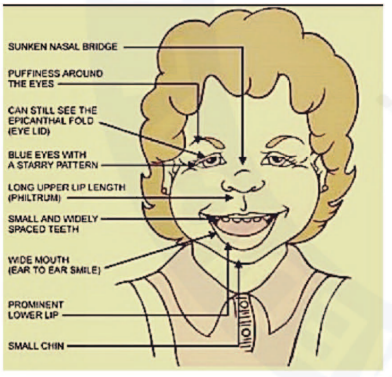
Active space

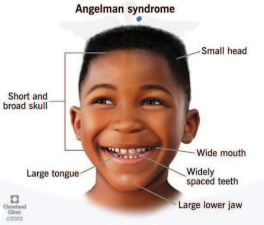


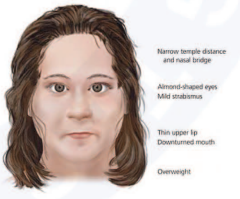
Cause of D/S
↓
MC → .....
↓
.....

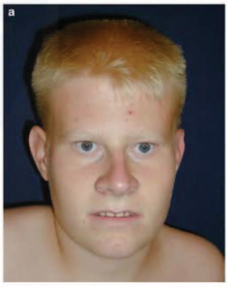
✦ Maternal deletion → • .....
↓
or
..... Duplication → UBE3A gene
or Disomy • Happy puppet's → Mental retardation
→ inapp. laughter

✦ Paternal deletion → .....
↓
or
.....Duplication or Disomy → SNORP gene
Overweight d/t ghrelin gene mutation
↓
↑↑Appetite ↓↓Satiety



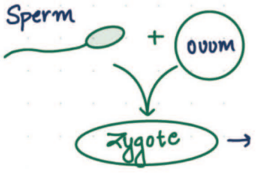






❑ **ii) Gonadal Mosaicism:**

✦ Parents (N) → offspring affected
↓
post zygotic mutations



eg: • Tuberous sclerosis
• Ospeogenesis Imperfecta
• Achondroplasia

MIST

Active space

❑ iii) Mitochondrial Interitence

aka: .....

eg:
- K → Kayrn's sayre syndrome
- L → Leber's Heroptic Neuropathy
- M → MELAS
- N → Narp Syndrome
- O → Ophthalmoplegia
- P → Pearson Syndrome

MC mit. int. d/s: .....

- M, E → Mitochondrial Encaphalopathy
- L, A → Lactic Acidosis
- S → Stroke like symp

NARP

- N → Neurological manifestatⁿ
- A → Ataxia
- R, P → Retinitis pigmentosa

Active space

❑ iv) TNR Mutations

EXONS

- Hungtington's chorea (CAG)
- Spinocerbellar ataxia (CAG)

INTRONS

- Fragile X - synd (CGG)
- Fredrich Ataxia (GAA)
- .....

d/to mutation of .....

(familial mental retardation)

CGG

5-55	56-199	> 200
.....	.....	.....

With Successive → .....

Generation → .....

↑

Anticipation

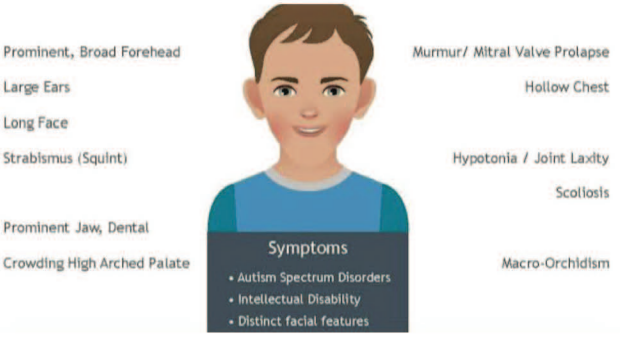
↑

FXS - ..... genetic cause of mental retardation

(MC genetic cause of MR - .....)

- MC Inherited cause of mental retardation.

✦ C/F:



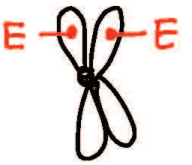
Active space

CLASSICAL / MENDELIAN PATTERN

❑ (i) Autosomal Dominant → Mutation

↓

Ch-involved



E → .....

E^m → .....

EE → .....

EE^m → .....

E^mE^m → .....

- AD disorders mainfest themselves
- .....
- .....
- .....
- .....

✦ D/S transmission rate:

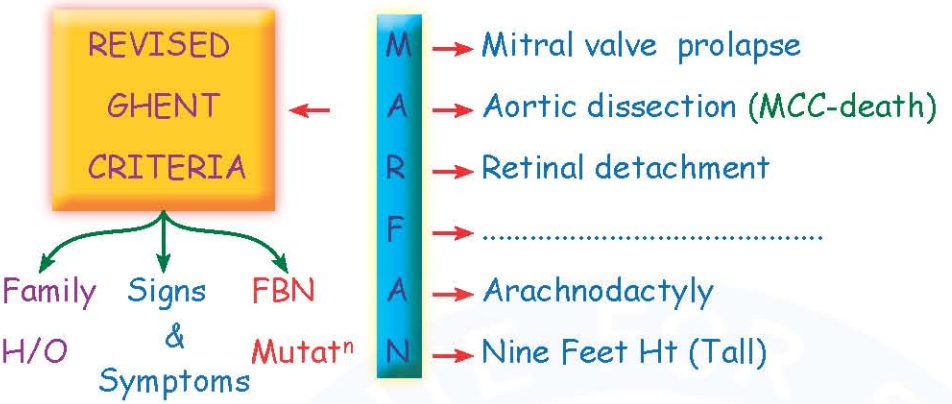
- No skipping of generation
- Incomplete Penetrance
- Variable expressivity

Example :

D → Dystrophia Myotonia
O → Osteogenesis Imperfecta
N → Neurofibromatosis
T → Tuberous Sclerosis
H → Her. Spherocytosis
A → Achonodroplasia
V → Von-Villibrand D/S
E → Ehler Dalnos Synd
A → APCKD
I → Intermittent Porphyria
M → Marfan's Synd (MC)

Active space

Marfan Syndrome :



❑ (ii) Autosomal Recessive → Weak mutatⁿ

- E → .....
- E^m → .....
- EE → .....
- EE^m → .....
- E^mE^m → .....

- AR mutations/disorders they manifest themselves in .....
- .....
- .....
- .....

✦ D/S Transmission rate:

Examples : A - Alkaptonuria
B - ß & α Thalassemia
C - Cystic fibrosis
D - Deafness (SNHL)
E - Emphysema (α-1 AT)
F - Fredrick Ataxia
H - Homocystinuria
G - Gaucher's d/s

Active space

❑ (iii) X - linked Recessive:

X → ..... X^m = ...

Y → .....

XX → ..... XY → .....

XX^m → ..... X^mY → .....

X^mX^m → ....

- Criss cross inheritence

Examples :
- AKSHAY : Agammaglobulinemia
- CAN : Ch granulomators d/s
- FIGHT : Fragile X-Synd (Better choice)
- DANCE : Duschenne Muscular dystrophy
- Give : G-6-P D def. anemia
- HIM : Hemophilia A & B
- COLD : Color Blindness
- WATER : Wisckott Aldrich Synd.

❑ (iv) X - linked Dominant:

X → ..... X^m → .....

Y → .....

XX → ..... XY → .....

X^mX → ..... X^mY → .....

X^mX^m → ....

.....

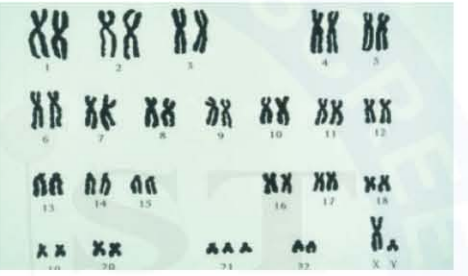
Active space

Examples :
- FAST → Fragile X-Synd
- C → Charcoat marie tooth D/S
- A → Alport Synd.
- R → Ricket's, RETT Synd.

GENETIC DISORDERS

AUTOSOMAL CYTOGENETIC D/S

✦ i) DOWN'S SYND: Trisomy 21



- MCC of DS: .....

  Other: rt (4%)

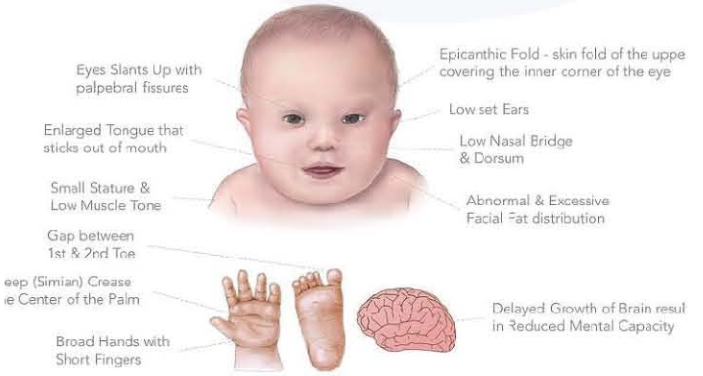
  Mosaicism (1%)
- MC genetic cause of

  .....

  (MC Trisomy overall → .....)
- All pts of DS → Alzheimer's d/s

  (> 40 yrs)
- MC leukemia in DS → .....

  .....
- MC Cardiac defect → • .....
  - VSD



◀WN'S SYNDROME

MIST

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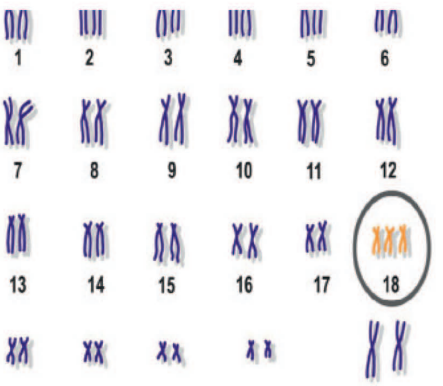
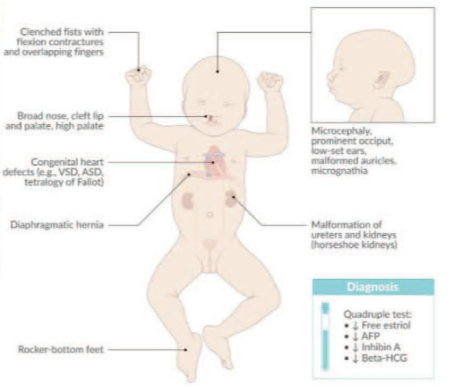
✦ ii) EDWARD SYND:

- Trisomy 18
- MCC: NDDM

⇨ C/F:

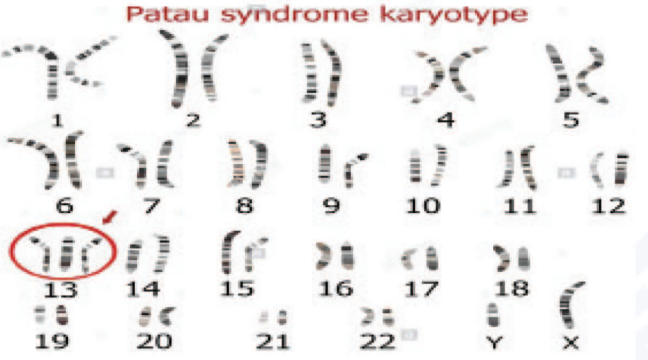


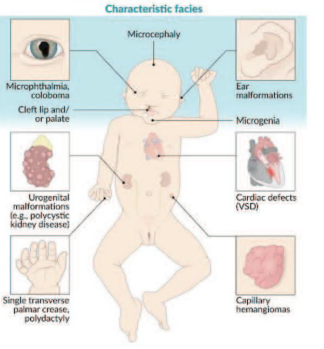


✦ iii) TRISOMY 13: PATAU SYND

MCC: NDDM

⇨ C/F





✦ iv) Cri-du chot d/s

- Ch5p -, cat like cry (mewing cry)

✦ v) Di-George Syndrome

C → Cleft lip palate

A → Abn facies

T → Thymic hypoplasia

C → Cardiac Abn

H → Hypocalcemia -- Trossaeu Sign, Chvostek Sign

22 → del 22q 11.2

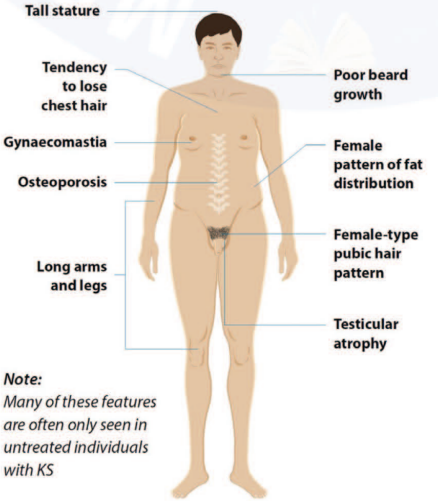
✦ vi) Trisomy 22

- aka - Cat Eye synd
- C/F: Vertical colobamas in eye

SEX CHROMOSOMAL DISEASE

❑ 1) KFS: KLEINFILTER'S SYN:

- MC-genotype: .....
- MCC .....
- MCC of KFS: .....
- More x Ch → .....
- ↑paternal/ ↑maternal age → ↑KFS.
- MC tumor c̄ KFS → .....
- MC CVS abⁿ c̄ KFS → .....



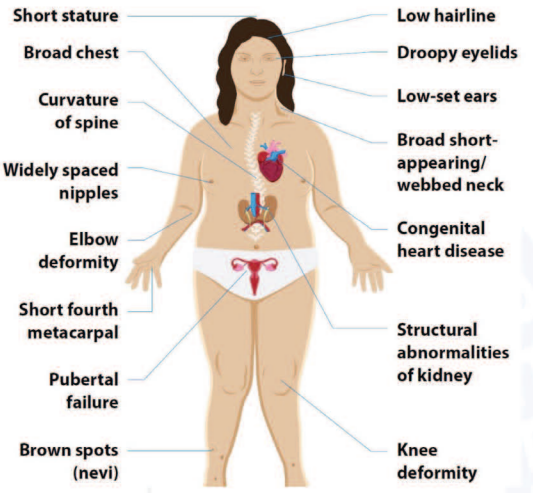
MIST

Active space

❑ 2) TURNER SYND:

- MC genotype - .....
- MCC .....
- MCC of TS → .....

✦ C/F:



❑ 3) NOONAN SYND:

- Karyotype - .....
- C/F → .....
- Mut^{n} → .....

DIAGNOSIS

❑ 1) FISH: FLOUROSCENSE IN SITU HYBRIDIZATION

- Color Based technique
- .....
- .....

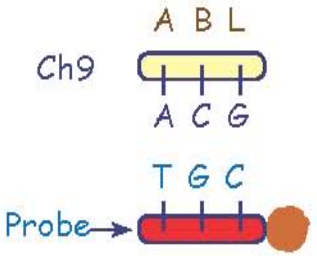
eg: Philadelphia ch → t (9:22)

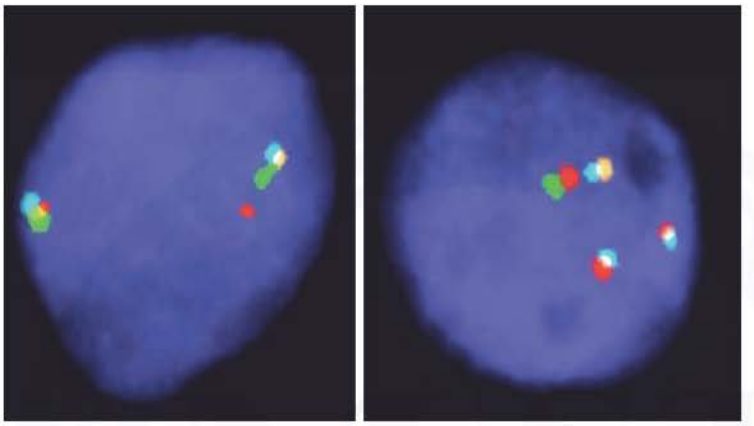
↓ ↓

ABL BCR

MIST

Active space



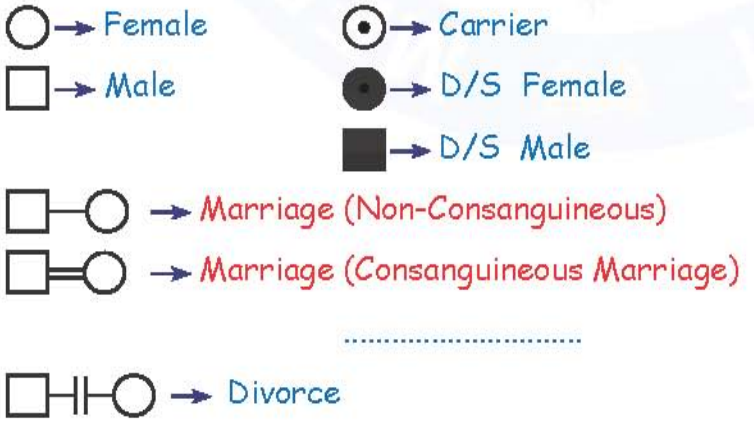


- **2) Antenatal Testing:** Chorionic Villi (≈ 9 wk)
- **3) Postnatal:** Fibroblasts, BM, Blood

MIST

PEDIGREE ANALYSIS

- **SYMBOLS:**



Active space

❑ STEPS OF PEDIGREE SOLVING

i) Rule out non-mendelian patterns if in options

↓

- Mitochondrial inheritence
- Gonadal mosaicism

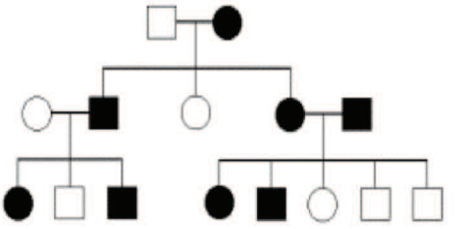
ii) Then if non-mendelian are not present

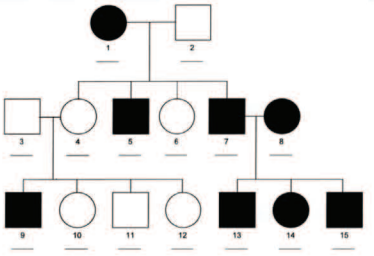
↓

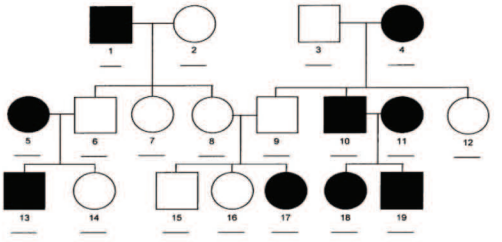
First rule out
- dominant → No-skipping of generation
- recessive → At-least 1 parent will be affected of D/S child.

iii) D/d B/w X-linked & Autosomal

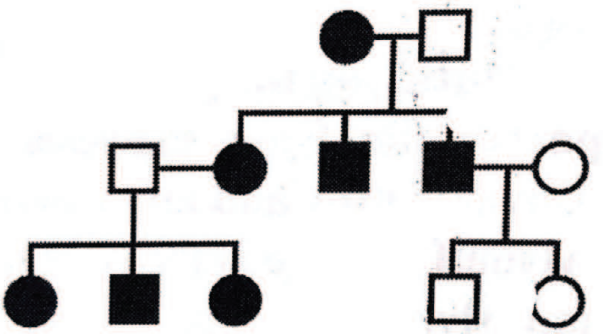
(on the basis of criss cross inheritence)

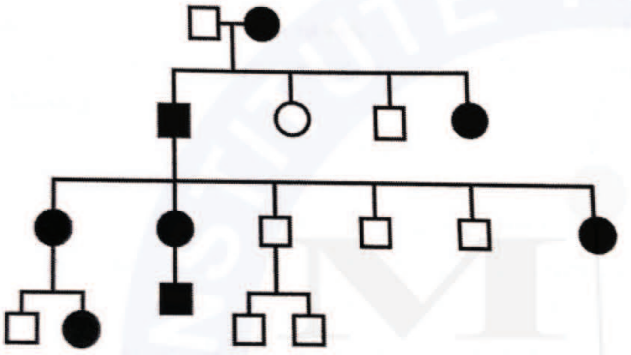


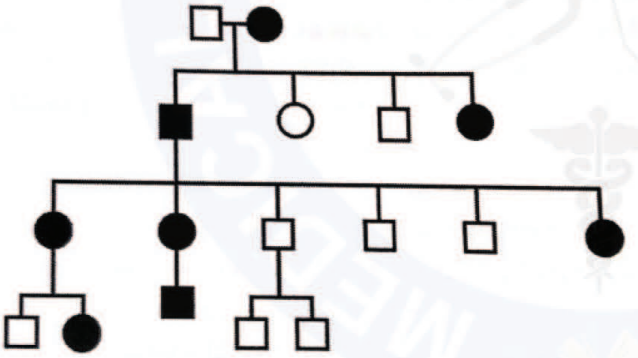


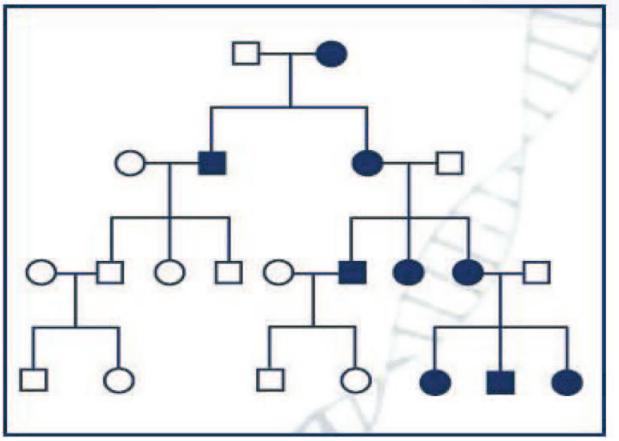


Active space





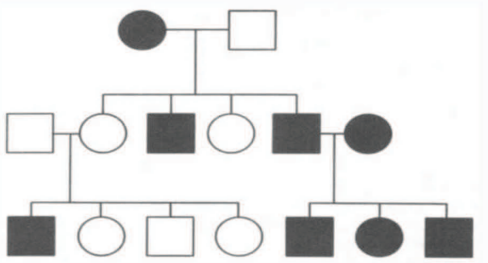


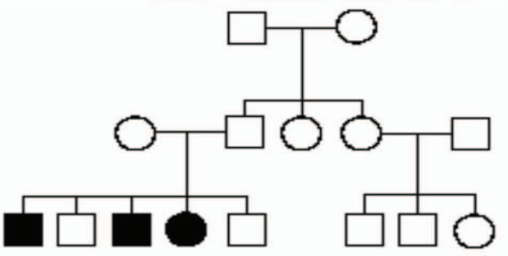


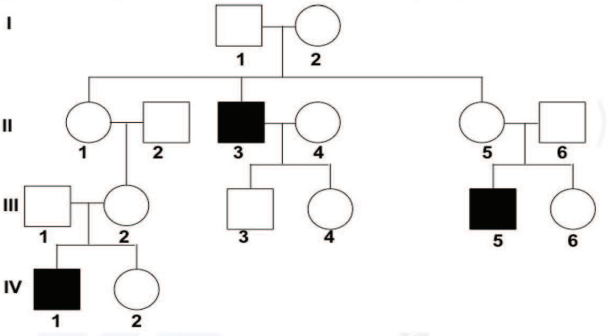
Active space

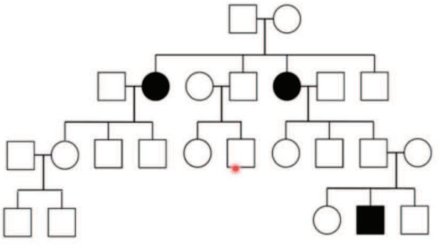
MIST

Read the pedigree chart and identify the pattern of transmission:



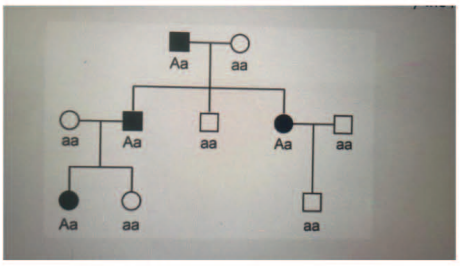


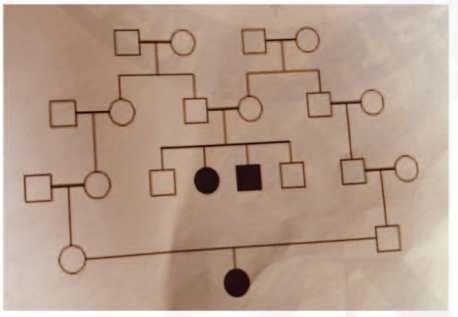


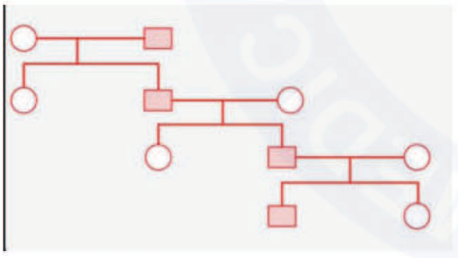


Active space

MIST







Active space

MIST

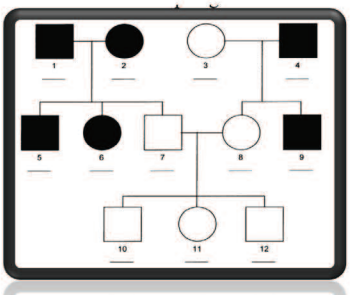
EXERCISE

1. Which one of the following is an autosomal dominant disorder
   A. Duchenn's muscular dystrophy
   B. Frangile x syndrome
   C. Fanconi's anemia
   D. Marfan syndrome

2. The clinical features of turner syndrome in girls include the following except:
   A. Normal iq
   B. Webbing of the neck
   C. Delayed puberty
   D. Gynaecomastia

3. Pattern of inheritance when disease gets transmitted from grandfather to father and then to daughter and son both is
   A. AR
   B, AD
   C. XLR
   D. XLD

4. Comment on pedigree



Active space

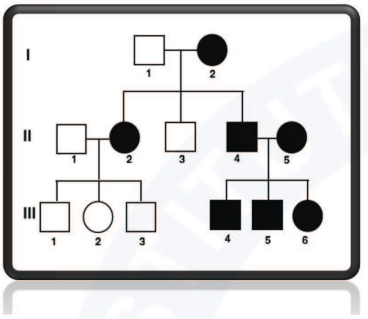
A. Autosomal recessive

B. X-linked recessive

C. X-linked dominant

D. Autosomal dominant

5. 32. Comment on Pedigree:



A. Autosomal recessive

B. X-linked recessive

C. X-linked dominant

D. Autosomal dominant

Active space

BASICS OF HAEMATOLOGY

Hematology : Study of (N) & pathologic aspect of blood & Blood cells.

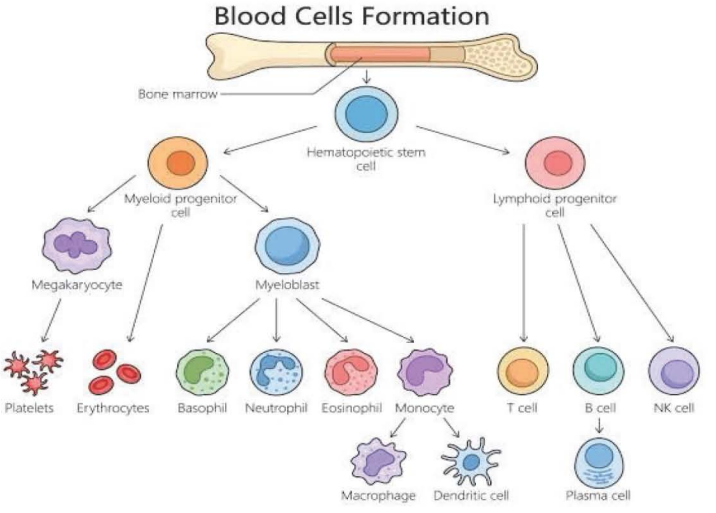
Hematopoiesis : Formation of blood cells.

↓

Inside BM (Intramedullary) | Outside BM (Extramedullary) Liver & Spleen

RBC Contain→Hb (Gower 1→Gower2→Portland→HbF→HbA2→HbA)

Age	Organ
3rd wk of Gestation	.....
3rd month of IUL	.....
4th month of IUL	.....
Just before birth	.....
After birth	.....



Active space

Bone Marrow :

	Child	Adult	Old
Cells	75%	50%	25%
Fat	25%	50%	75%

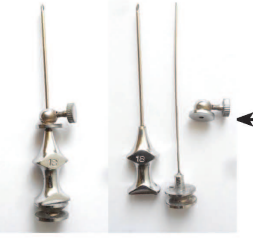
- % cellularity of the BM : 100- age of the Pt e. g 100-25 = 75%

➢ Cellularity dec↓ses as age↑ses.

✦ Bones involved in hematopoiesis :
Long bones & axial skeleton ⟶ Needles used

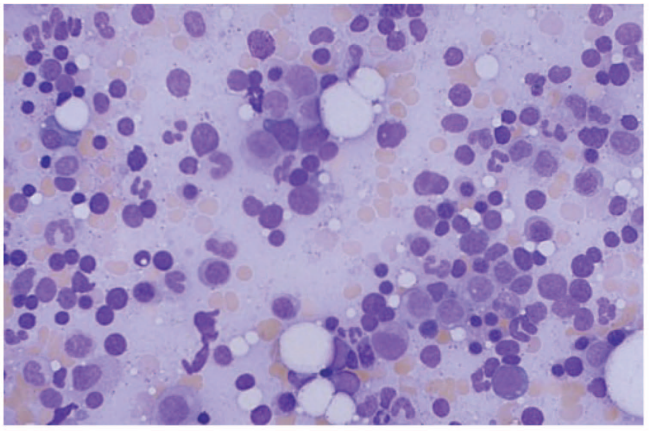
← Jamshedi

✦ Site of BM aspiration / Biopsy :
Child : Shin of Tibia
Adult : Iliac crest in PSIS (MC)
Sternum (Best)



←Salah

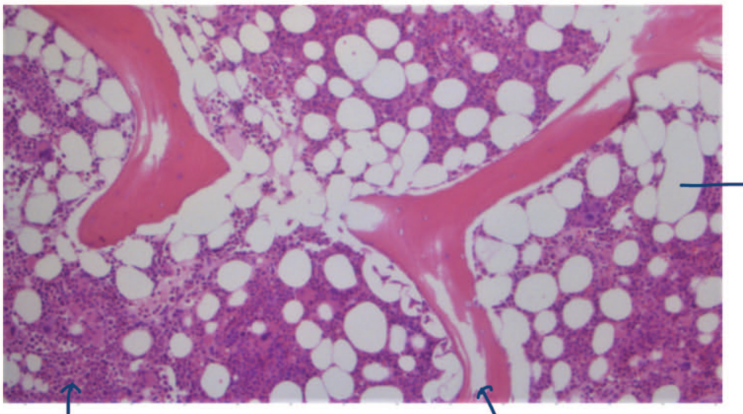
✦ Interpretation of a BM aspirate



Contain Mature & Immature cells

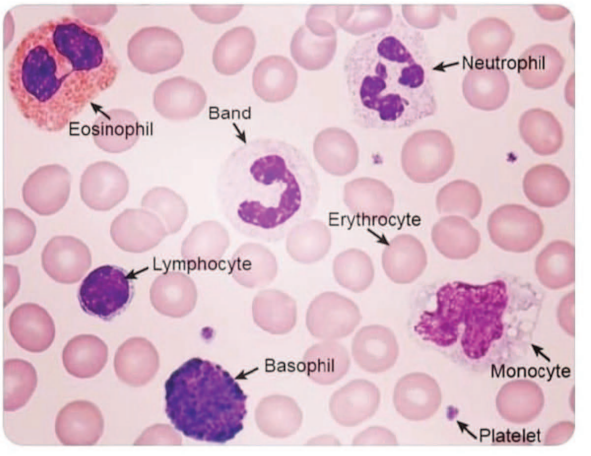
Active space

✦ Interpretation of a BM Biopsy



✦ Interpretation of a peripheral Smear.

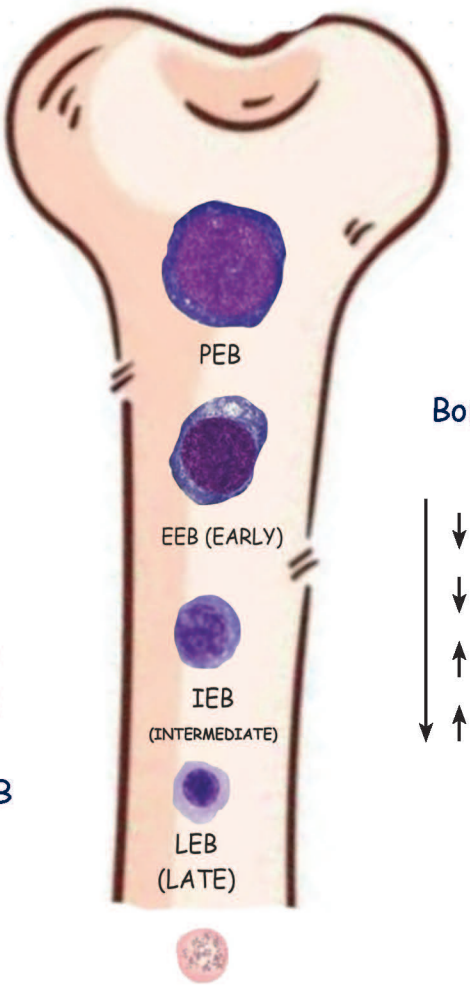
Contains only mature cells.



Active space

RBC DISORDERS

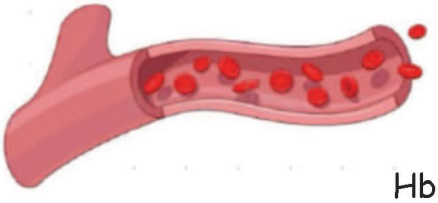
✦ RBC Development



Proerythroblast
(Hb Production
.....)
↓
Basophilic EB
↓
Polychromatic EB
(Hb ..... appears)
↓
Orthochromatic EB
↓
Reticulocyte
1-3 days ↓
RBC

Bone Marrow

↓ Cell Size
↓ Nuclear Size
↑ Maturation
↑ Number



Blood Vessel

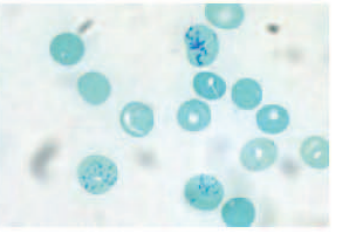
Hb Seen on EM → .....
Hb Seen on LM → .....

Note : Nucleus in RBC develovpment is only seen inside the cells present in BM.

Active space

✦ **Reticulocyte :**

- Precursor of RBC (Immature RBC)
- Contain .....



Reticulocytes

Stain : Supravital Stains

(Stains living structure of a cell In Vitro)

- (Do not contain alc)

e. g.- Brilliant Cresyl Blue

- .....

Ⓝ **Reticulocyte count** (Retic count)

No. of Reticulocytes / 100 RBC	Ⓝ Retic count ✦ Child- ..... ✦ Adult- .....

↑	↓
-	- .....
- Rx	- BM Suppression
• IDA	- Pure red cell aplasia.
• Meg. A.	

✦ RBC

(N) Count : 4. 0 - 5.5 million/Cmm

Size → : .....

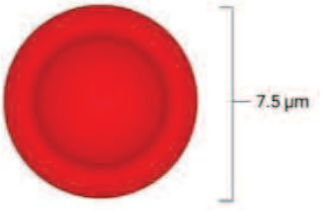
Volume → : .....

Shape → : Biconcave/dumble shaped

Property → : - Non-Nucleated

- Highly flexible
- Central pallor

Side view (cut)



Top view

Active Space

✦RBC Indices

1. Mean corpuscular Volume [MCV]	.....	(N) → 80-100 fl (femtolitres) < 80fL → ..... > 100fL → .....
2. Mean corpuscular Hemoglobin [MCH]	.....	(N) → 27-33 pg (Picograms) < 27pg → ..... > 33pg → .....
3. Mean corpuscular Hb Concentratⁿ [MCHC]	....	(N) → 32-37 g/dl
4. Red cell distribution width [RDW]	Degree of anisocytosis ↓ diff In size	(N) → 11.5-14.5 %
5. Packed cell Volⁿ/Hematocrit [PCV]	- .... (PCV = 3 X Hb)	(N) Male→15X3= 45% (N) Female→14x3= 42% Hb

Active space

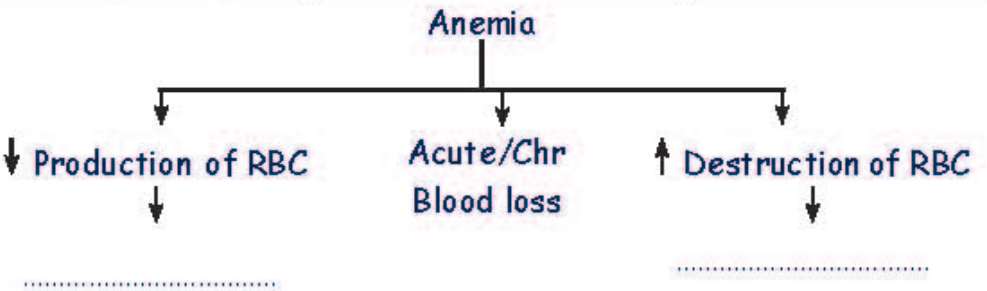
ANEMIA

✦ ..... in RBC mass / ..... in Hb content based on : Age, Gender
Geographical location.

✦ Anemia's based on MCV :

Microcytic	Normocytic	Macrocytic

<7u	7- 8u	<9u
< 80fl	80-100fl	> 100fl
S -Sideroblastic A I -Iron def. A T -Thalassemia A -Anemia of chr d/s L -Lead poisoning.	- Kidney d/s → ↓ EPO - ACD	M-Megaloblastic. A A- Alcohol L- Liver disease



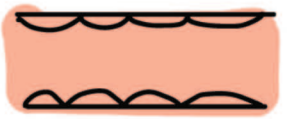
Active space

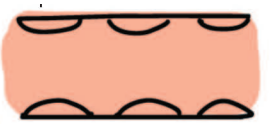
HEMOLYTIC ANEMIA

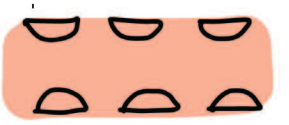
✦ Anemia d/t ..... destructⁿ of RBC.

➨ Based on site of defect :

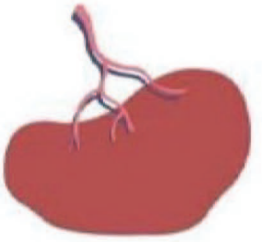
- **Intrinsic defect** (c̄ In RBC)
  - **RBC Membrane defect**
    - Hereditary Spherocytosis.
    - PNH
  - **Hb defect**
    - SCA
    - Thalesemia.
  - **Enzyme defect**
    - G6PD deficiency.
- **Extrinsic defect** (Outside RBC's)
  - **Immune mediated**
    - AIHA
    - MAHA





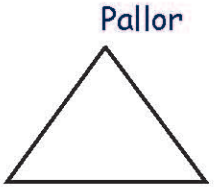


➨ Based on site of Hemolysis :

Intravascular Hemolysis (IVH)	Extravascular Hemolysis (EVH)
	

Active space

C/F :



Haptoglobin
protein
that binds to free
hemoglobin released
from .....

Lab finding :

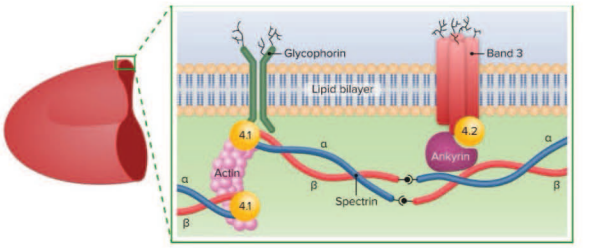
CBC :

- Hb : .....
- TLC : .....
- PLC : .....
- Retic : .....
- ..... Unconjugated Bilirubin.
- S. Haptoglobin .....
- S. LDH ↑

Active space

HEREDITARY SPHEROCYTOSIS

✦ AD > AR

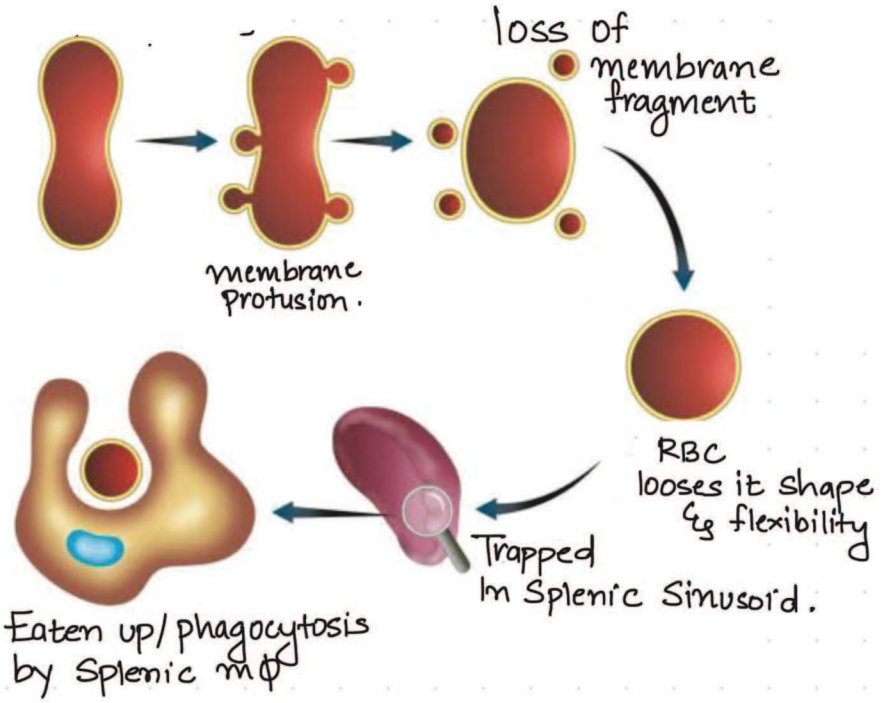


✦ Spherical RBC → Spherocytes → ...O2 binding / delivering capacity
→ .... Life span →mild →..... days
→severe →..... days

✦ RBC memb (N) contain 4 proteins

SPECTRIN > ANKYRIN > BAND 3 > PALLADIN
(Protein 4 . 2)

- Most abundant — M/C
- ↓ Causes m/Severe form of HS — deficient In HS .



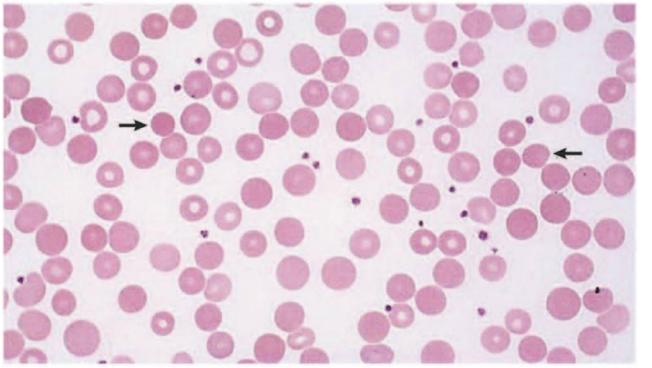
↓

Extravascular Hemolysis

Active space

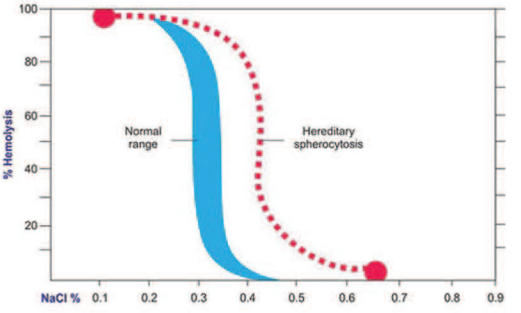
CBC :

RBC	.....
Hb	.....
WBC	.....
PLC	.....
MCV	.....
MCH	.....
MCHC	.....

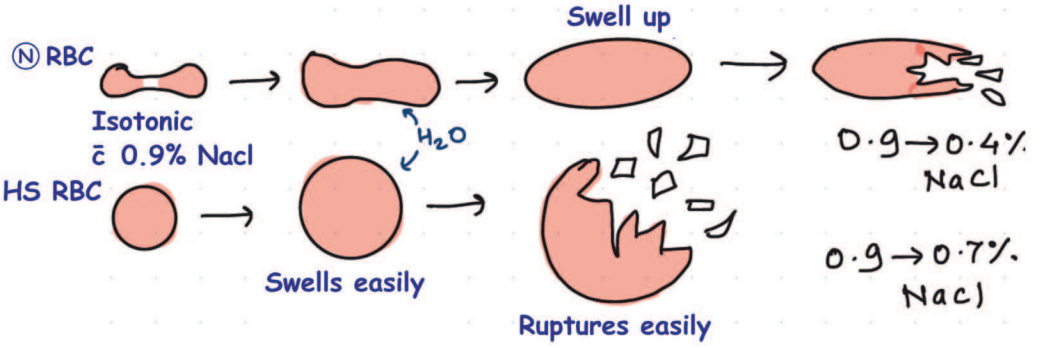


Spherocytes - Small, Spherical
- Central pallor

Ⓝ RBC → ..... central pallor



- Screening Test :
  (Osmotic Fragility Test):



Confirmatory Test : → .....(Eosin malemide test)

Treatment : → • Blood Tranfusion

• Elective Splenectomy → Complication → Infection

↓

Capsulated Organism → Meningococcus (always administered Meningoccal Vaccine)

Rxoc :- • BMT

Active space

PAROXYSMAL NOCTURNAL HEMOGLOBINURIA

- Acquired d/S (.....Hemolysis)
- Stem cell defect
- Ⓝ RBC membrane contains 2 GPI Anchors - d/t .....gene

DAF / ..... [Decay accelerating Factor]

MIRL / ..... [Membrane Inhitor of Reactive lysis]

- PIGA gene (Phosphatidyl Inositol glycan A) → Synthesis of GPI → Protect against complement. System

Pathogenesis : Mutⁿ In PIGA gene

↓

↓↓ DAF or ↓↓ MIRL

↓

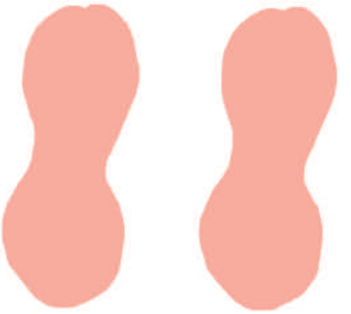
No protective covering against CS.

↓

Lysis of RBC Via CS (MAC)

↓

..... Hemolysis.



Sleep → Passive hypoxia → ↓PH → Activates complement. S

↓

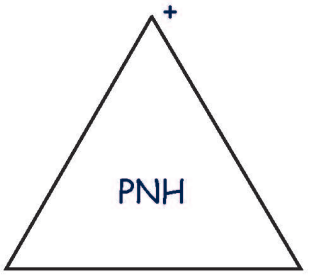
I/V hemolysis. ← MAC

Active space

PNH → RBC, WBC, Plt (Pancytopenia c̄ HS megaly)

PNH → - AML
- Aplastic Anemia
- MDS

Pancytopenia.

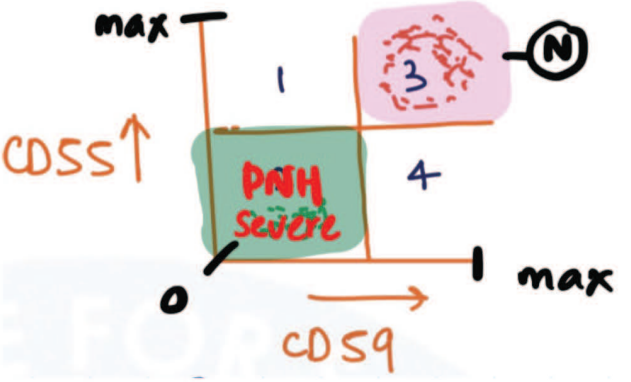


..... (I/V)

(MCC of .....) Hemolytic. A

✹ Screening Test : → • .....
• Acidified Serum test/HAM'S Test

✹ Confirmatory Test : Flaer Flow Cytometry



Rx : • BT
• Eculizumab [C5 (-)]

Rxoc : BMT

Active space

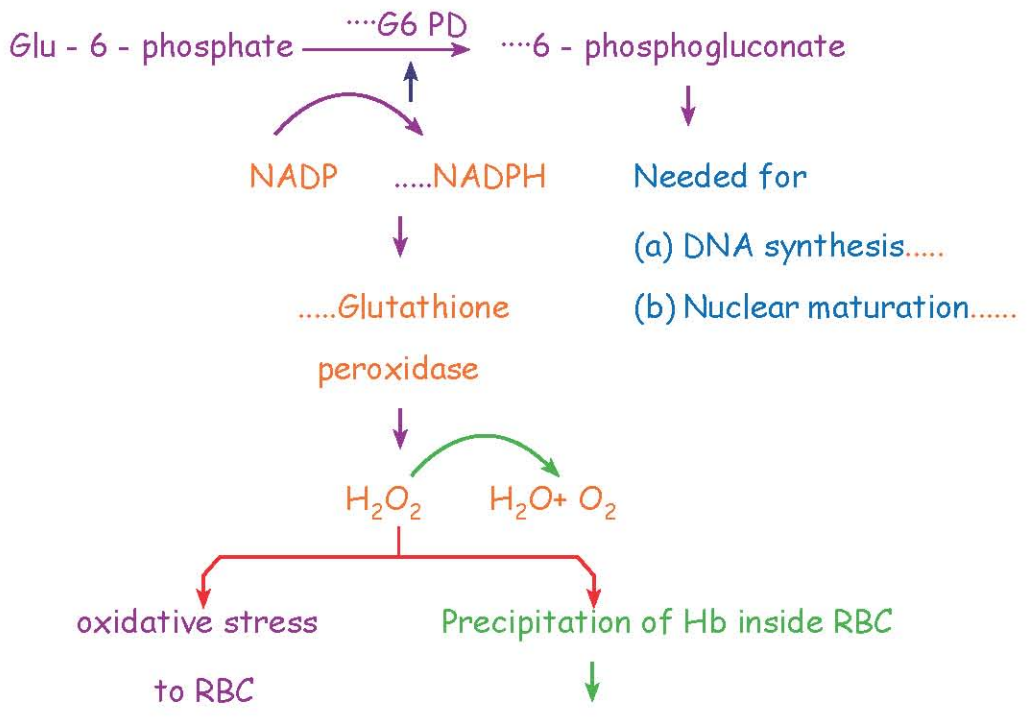
G6PD DEFICIENCY ANEMIA

- X-Linked recessive disorder
- M > F

Etiology : Enzyme G6PD (Glucose-6-phosphate dehydrogenase)

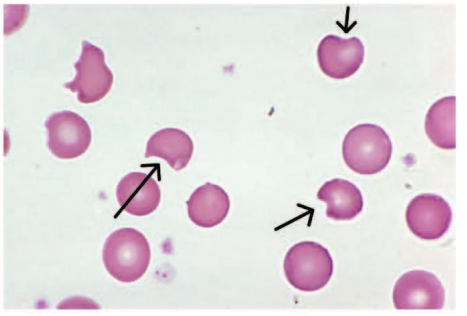
- Fava beans
- Antimalaria drugs, Sulfonamides.
- Infectⁿ (Pneumonia, Hepatitis)

Variants :
G6PD A-B
G6PD Mediterranean.

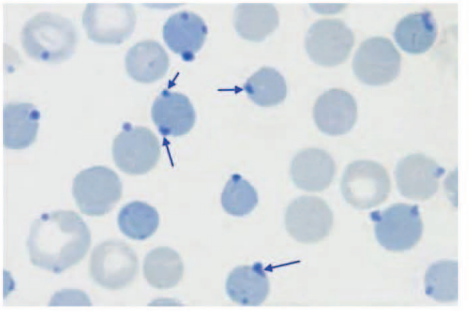


Active space

PBS :



Bite Cells



Heinz Bodies

CBC :

- Hb .....
- WBC
- PLC
- MCV
- MCH
- MCHC

← .....

- ..... Retic count
- ..... unconjugated bilirubin
- ..... S. haptoglobin.

✹ Protectⁿ against P. Falciparum. ←

- G6PD def A.
- Sickle Cell A.
- Thalassemia

IOC : G6PD enzyme assay

Screening : 1. Meth Hb reductⁿ Assay. 2. Fluroscence test.

Active space

Rx : Self-limiting → Avoid oxidative stress
→ Antioxidant.

IMMUNE HEMOLYTIC ANEMIA

- Type II Hypersensitivity reaction
- Ab → RBC destruction → Anemia

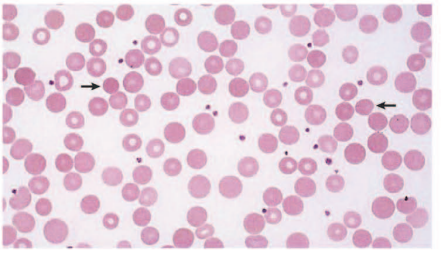
IHA

- Auto Immune
  - WARM AIHA
  - COLD AIHA
- Alloimmune
  - Rh d/s of Newborn
  - ABO Incompatibility.

Auto Immune HA

WARM AIHA	COLD AIHA
M/C ..... : Active at 37°C ↓ RBC coated with IgG ↓ Destroyed In Spleen ↓ E/V Hemolysis.	..... : Active at 0-4°C ↓ Activates CS ↓ Complement Mediated Hemolysis

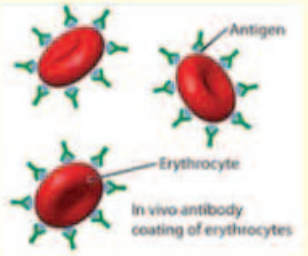
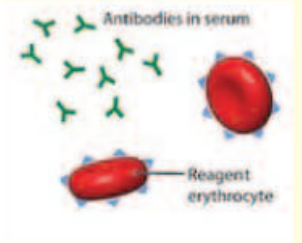
Spherical RBC
↓
Spherocytes
↓
D/D → HS
→ AIHA
→ Blood transfusion uxn.



Agglutin Test / Coomb's Test → To detect Ab's in blood

- ○ IHA
- ○ HS

Active space

DIRECT COOMB's	INDIRECT COOMB's
AKA Direct Agglutination test [DAT] ↓ Ab's directly attached to RBC ↓ Autoimmune 	AKA in Direct Agglutination test [IAT] ↓ Ab's +nt, But not attached to RBC ↓ Alloimmune 

RH INCOMPATIBILITY

Rh Antigens :

Short arm of chromosome

Types : C, C, D, E, e

↓

Most Immunogenic.

If Rh (Antigen D)

→ +nt ⟶ Rh + Ve

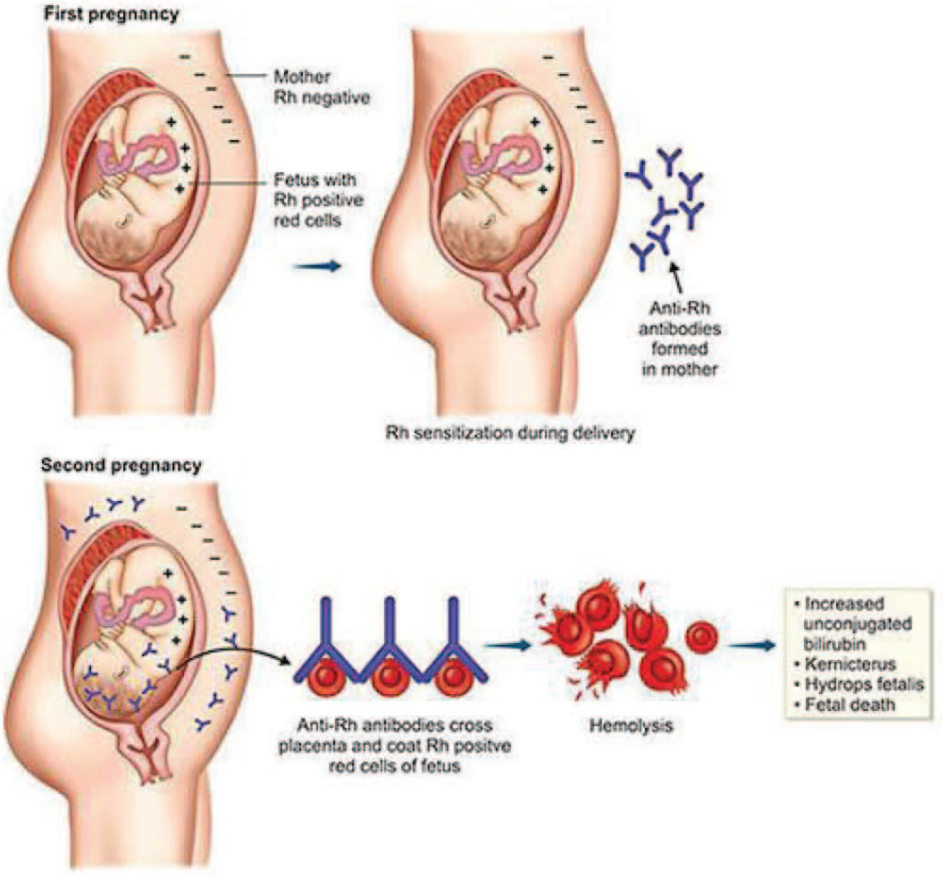
→ - nt ⟶ Rh - Ve

Eg.

If Rh-ve exposed ⟶ Rh +ve

Active space

In Rh-ve Pregnancy :

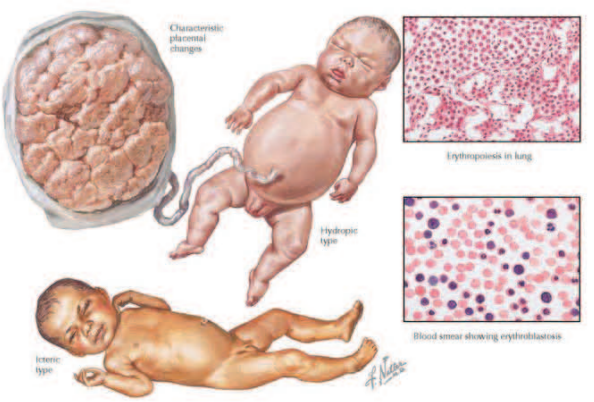


Coomb's test : .....

Prevention : Anti-D I/M Inj.

[Stop maternal Immune. S from producing Ab's]

Dose : 300 mg

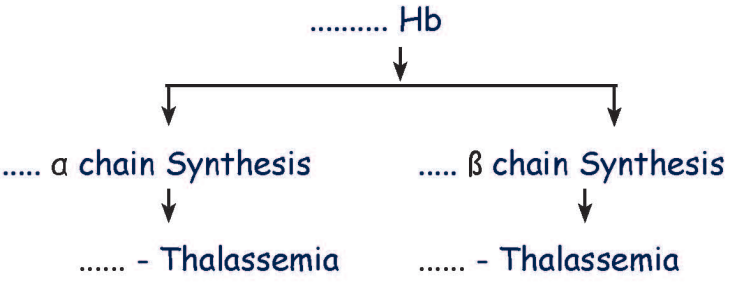


Hydrops Fetalis.

Active space

THALASSEMIA

- Autosomal Recessive
- .....

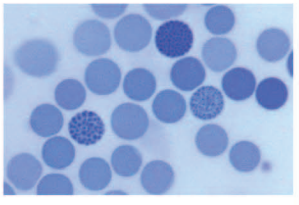


Ⓝ (Hb → Electrophoresis)

Types	ADULT	CHILD	COMPOSITION
Hb A	.....	.....	.....
Hb A_2	.....	.....	.....
Hb F	.....	.....	.....

α - Thalassemia

- Pathogenesis :



Type	Genotype	Findings
1 Gene Deletion		
2 Gene Deletion		
3 Gene Deletion		
4 Gene Deletion		

ß - Thalassemia

Pathogenesis : .....

ß - Major ..... ß - Minor ..... ß - Intermediate

✦ Defect :

ß : (N) ß-chain

ßo : Complete deficiency of B-chain.

ß+ : Partial deficiency of ß-chain

ß-Thal → ... ß Chain Synthesis

↓

.....

↓ → (↑α_2, ... & ↑α_2 .....)

↓

..... (.....)

(MCHC Anemia)

↓

Ineffective Erythropoeisis → Spleen (E/V Hemolysis)

↓

Anemia → ..... Iron absorption

(Tissuehypoxia ↑EPO) Blood transfusion (Rx)

↓

Extramedullary erythropoiesis

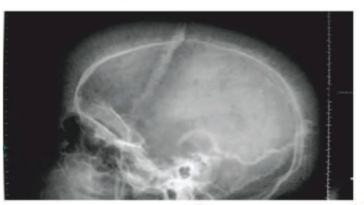
↓

Systemic Iron Overload (2° hemachromatosis)

↑

MCC of ..... in B-Major

X-ray

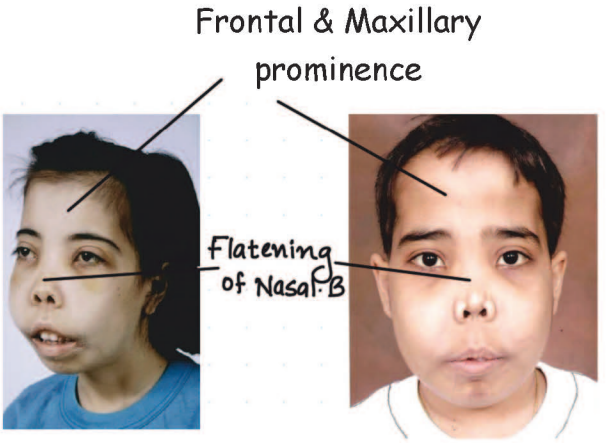


Crew cut app hair on end app.

Active space

Clinical Features :

- ✦ Chipmunk facies (Sev. Cases)
- ✦ Anemia, Pallor, Fatigue
- ✦ Hepatosplenomegaly.
- ✦ Jaundice.



Investigatⁿ :

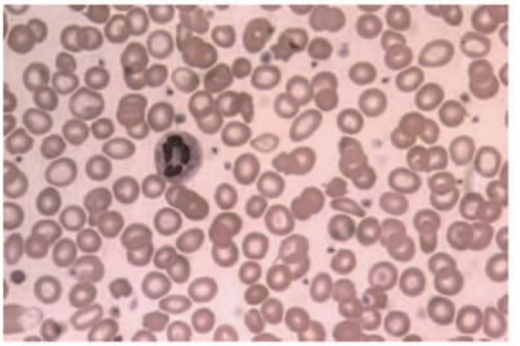
1. CBC : HB → ..... gm% (major)
→ ..... gm% (major)

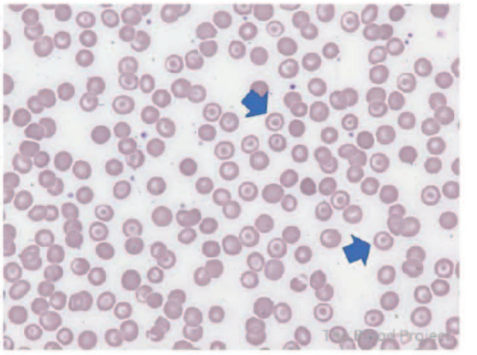
Hb	....	MCV	....
PLC	....	MCH	....
WBC	....	MCHC	....
Retic	....	RDW	....

Iron profile

S. Iron ..... (Major)
..... (Minor)

2. PBS :





Active space

3 : Screening :

✦ Nestroft :
(Naked eye Single tube Red cell Osmotic fragility test):

Ⓝ RBC → 0.9 → 0.4 % Nacl
[Flexible]

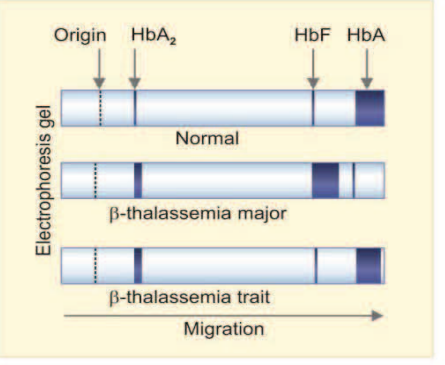
Spherocyte → 0.9 → 0.7 % Nacl
(Rigid)

Thalassemia → ..... show osmotic fragility.
[.....]

4. Hb Electrophoresis :

a) ß-major
↓
.....
↓
.....
↓
.....

b) ß-minor
↓
.....
↓
.....
↓
.....



c) 3α - gene deletion : .....
d) 4α - gene deletion : .....

5. Globin gene Sequencing > HPLC > Hb electrophoresis.
(.....)

Treatment :

- Repeated blood Transfusions :
  ↑↑ Felevels → Fe toxicity → Fe chelators (Desferroxamine)

RxOC :

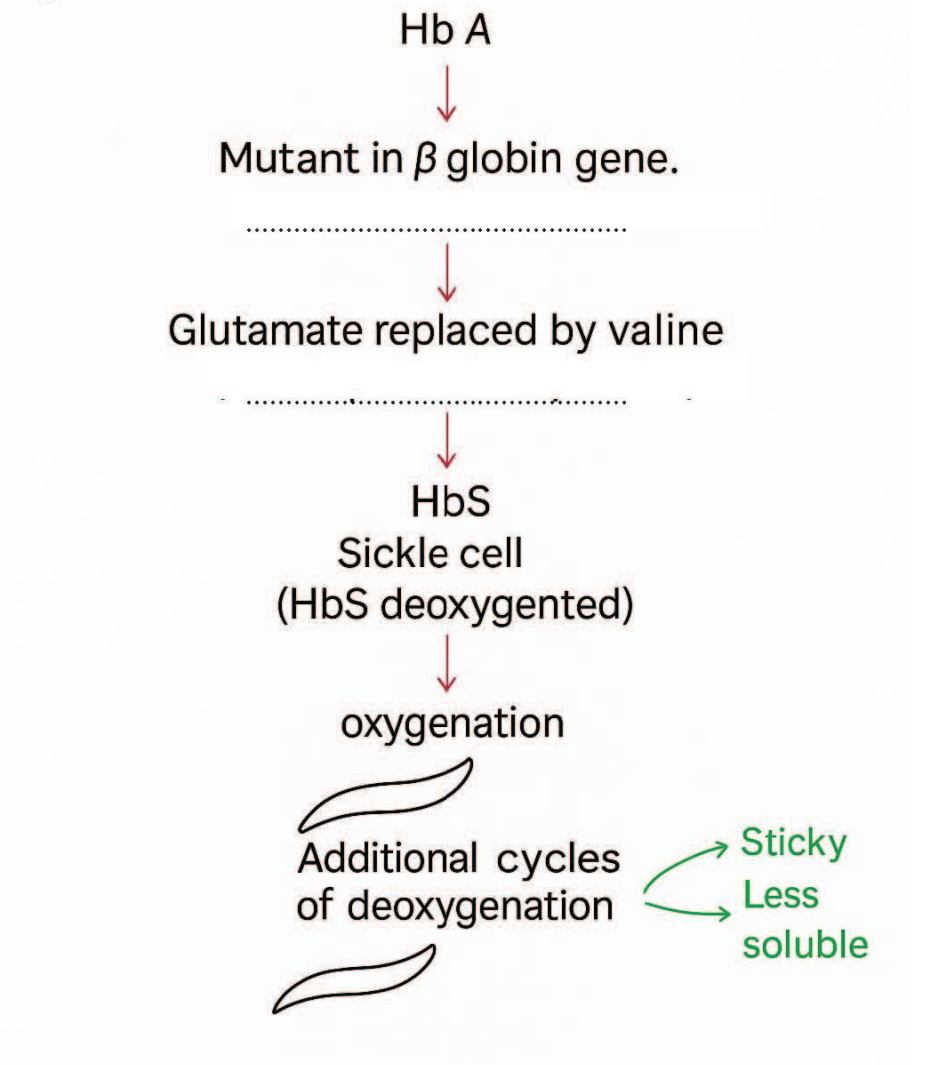
- Bone Marrow Transplantation.

Active space

SICKLE CELL DISEASE

- Autosomal Recessive
- M= F
- Common In africans

Pathogenesis :



Active space

Sickle cells

Microvascular occlusion

↓

vasocclusive crisis

MC (Complication) in .....

Goes to Spleen

↓

phagocytosed by

↓

E/V hemolysis.

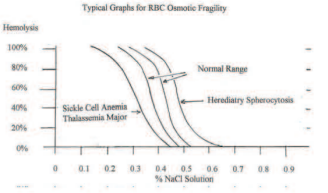
↑

(Splenomegaly)

Osmotic Fragility Test (OFT)

↓

Curve shifts to left



D/D In HS.

↓

Curve shifts to Rt

↑ Sickling	↓ Sickling
• Acidosis • Hypoxia • Dehydration	• HbF

Sickle Cell Disease

SICKLE CELL TRAIT

↓

HbA ..... HbS .....

SICKLE CELL ANEMIA

HbS .....

Active space

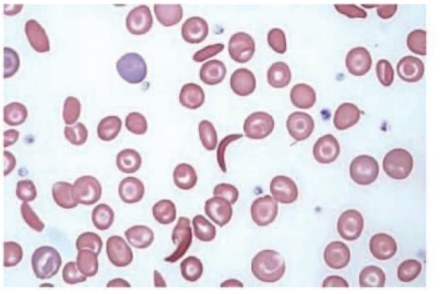
Complications : 3 'CRISIS'

Vaso occlusive Crisis	Aplastic Crisis	Sequestration Crisis
✦ Hand foot syndrome (small bones)	Parvovirus B19 Infection	Pooling of Blood in Spleen ↓ Hypovolemia ↓ ↑ Mortality
✦ AVN (Longbones)		
✦ Dyspnea, pain in chest, PAH		
✦ Papillary Necrosis		
✦ Ischemia (Acute coronany Syndrome)		
✦ Spleen → Shrunken → Auto Splenectomy Spleen		

Investigation :

CBC

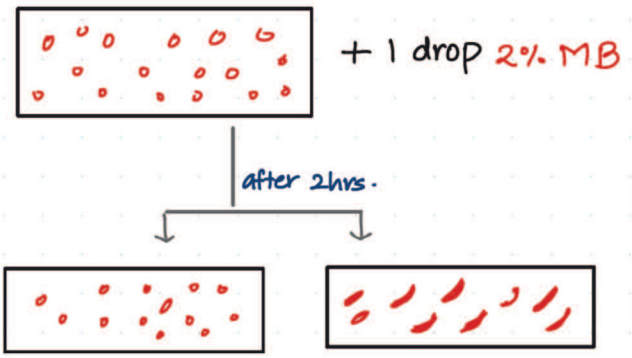
- ✦ Hb
- ✦ WBC
- ✦ PLC
- ✦ ESR
- ✦ Retic count
- ✦ UC bilirubin

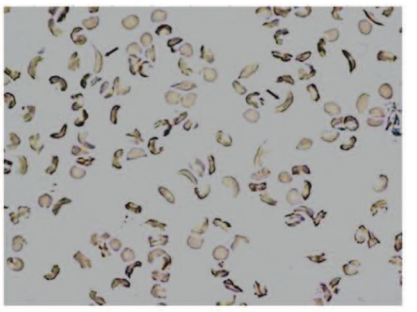


X - ray : Fish month vertebrae
Hair on end App,
Crew Cut App.
↓
(d/t Extramedullary Haematopoiesis)

Active space

Screening test : (1) Sickling test (2% Metabisulphite / Na dithionite)
(O_2 Consuming agents)





(2) Hb Electrophoresis :

(3) High performance liquid chromatography (HPLC)
- (Gives % of diff types of Hb)

(4) IOC : .....

- Rx : Hydroxyurea (↑ HbF), Pain killers, O_2 Therapy, BT
- Rxoc → BMT (Bone marrow transplantat^n)

Active space

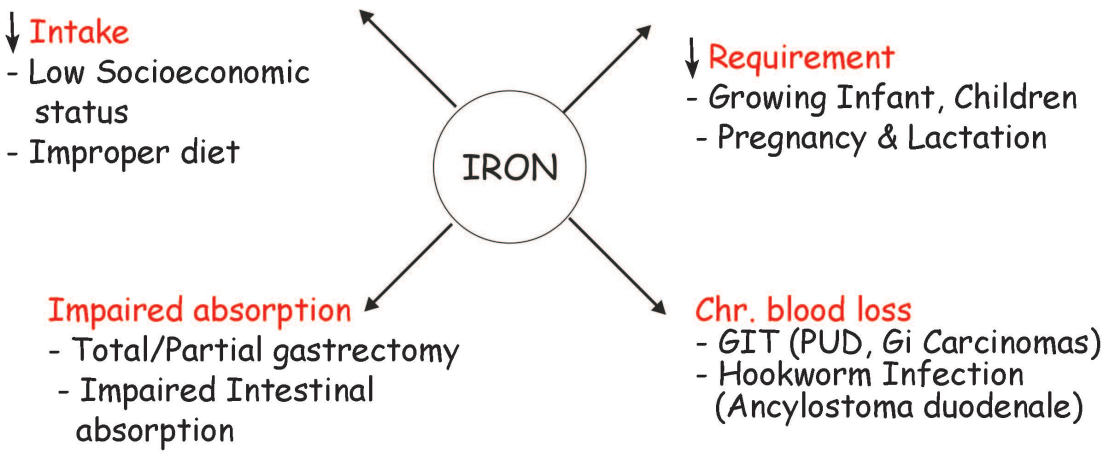
IRON DEFICIENCY ANEMIA

MC anemia overall

Daily requirement
10 - 20 mg/day.

Types of Iron → Fe^{2+} → Ferrous → ● .....
Site : ● .....
→ Fe^{3+} → Ferric → ● .....
● .....

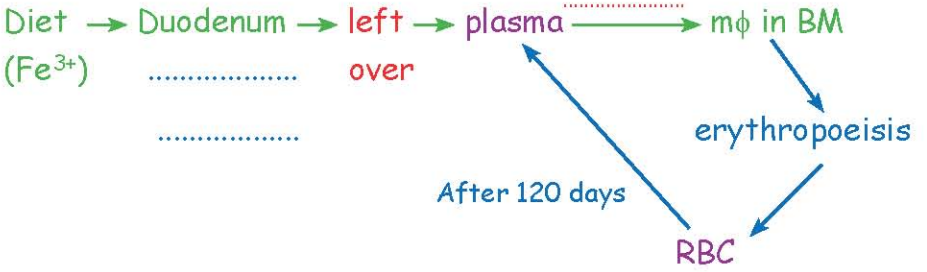
Causes of Iron Def :



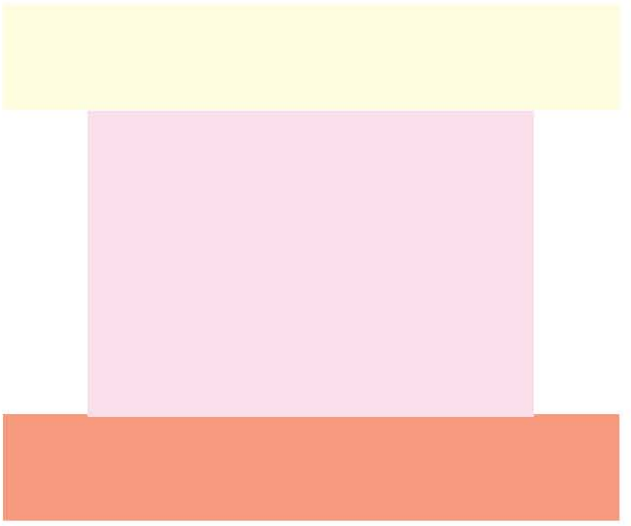
↑ Absorption	↓ Absorption
• Acidic pH	• Tea
• Vit C	• Tannates
• Amino acide	• Phytates

Active space

❑ Metabolism of Iron



❑ Absorption of iron



- Ferritin → Stored form of iron
- S. Ferritin → reflective value of stored iron

Active space

Hepcidin :

- Produced by Liver
- (Acute phase reactant - ↑ during Inflammatⁿ)
- Regulate Iron absorptⁿ : ↑ Hepcidin → ↓ S. Fe

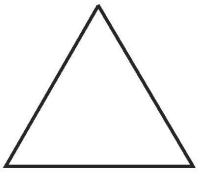
C/F :

- Pallor
- Fatigue
- Dyspnea
- Palpitations.
- Angular stomatitis
- Cheilitis
- PICA
  (unusual craving for clay / chalk)

Koilonychia

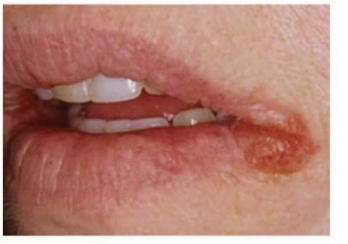
Plummer-vinson synd.

Microcytic Anemia



Atrophic glossitis

Esophageal Web.



Investigatⁿ :

CBC				Indices			
Hb	RBC	WBC	PLC	MCH	MCV	MCHC	RDW
.....	.....	.....	.....	.....	.....	.....	.....

Iron Profile :

Iron	.....
Ferritin	.....
TIBC	.....
T%	.....

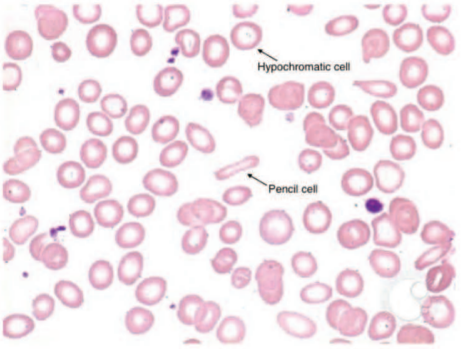
TIBC : Total Iron binding capacity :

T% Transferrin Saturatn

Amt of Sites on transferrin bound to Fe3+

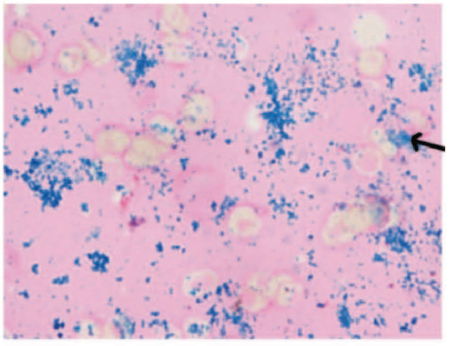
Active space

PBS :



✦ Microcytic Hypochromic Anemia
- Anisopoikilocytosis
- Pencil Cells

✦ BM Iron Stores : (Gold standard Investⁿ)

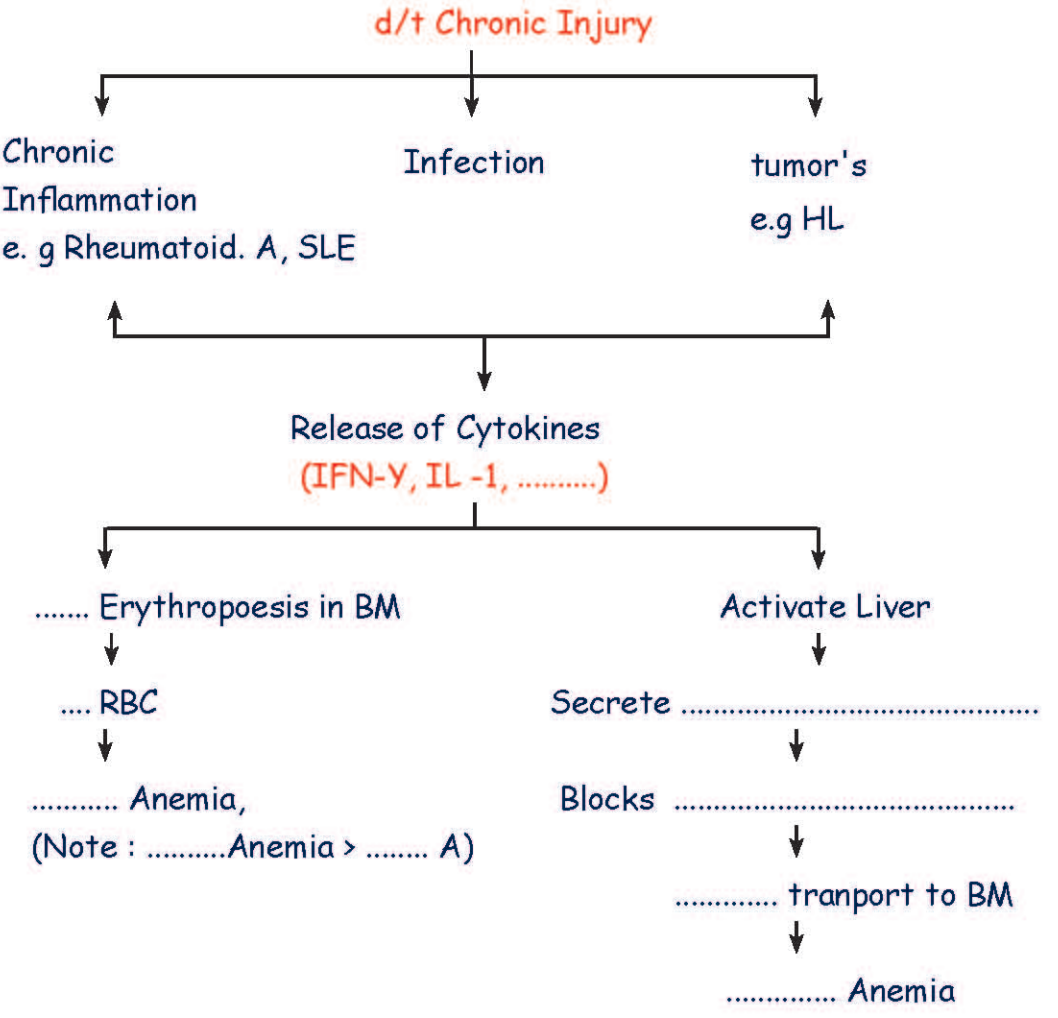


← Iron → Blue
Prussian blue/perl stain

MENTZER INDEX : MCV / RBC Count → ↑↑ (>13) → IDA
↓↓ (>13) → Thal. minor

Active space

ANEMIA OF CHRONIC DISEASE



Investigation :

CBC	
Hb	
WBC	
PLC	

RBC Indices :

MCV	
MCH	
MCHC	

Iron profile:

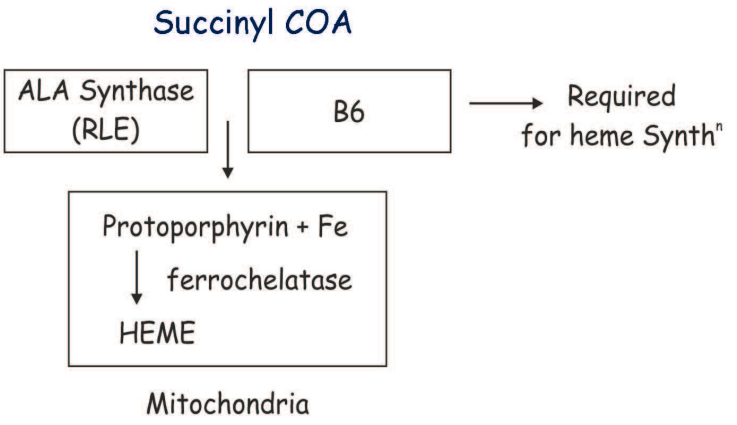
Iron	Ferritin	TIBC	T%

Treatment : Treat the underlying condition

Active space

SIDEROBLASTIC ANEMIA

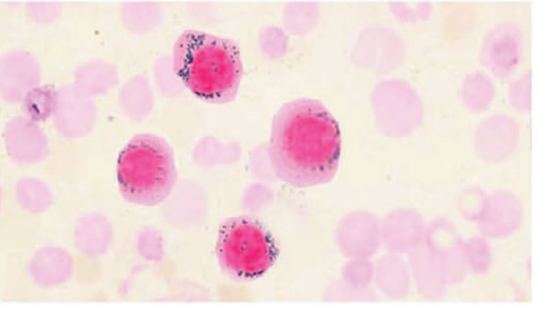
- Excessive Iron deposition In erythroid precursor cells.



Etiology :

Acquired
- Alcohol
- TB drugs (Isoniazid)
- Vit B6 def. *

genetic/congenital
- X-linked
- Enzyme def*



Active space

CBC	
Hb	
WBC	
PLC	

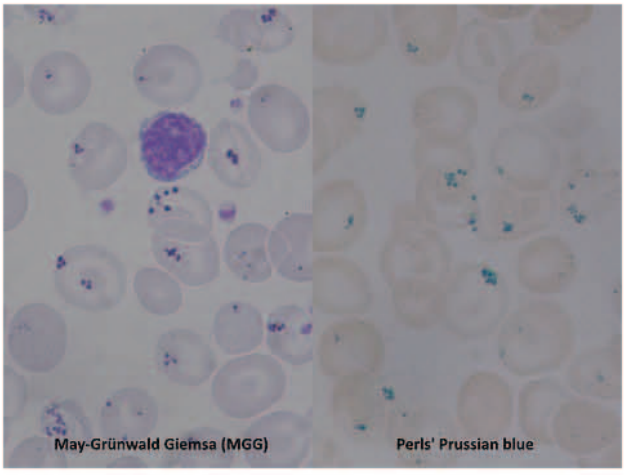
MCV	
MCH	
MCHC	

Iron	
Ferritin	
TIBC	
T%	

PBS :

Papenheimer Bodies
(Accumulation of Iron
In matured RBC)

- MCHC. Anemia
- Coarse basophilic stippling.



Investigation :

BMA : Ringed Sideroblasts.
accumation of Iron In
Mitochondria
Prussian blue Stain (perinuclear)

Treatment :

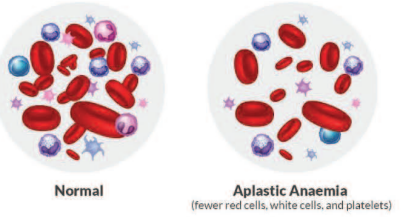
Phelebotomy +
Iron chelators.

Iron Profile	IDA	SA	ACD
Iron			
Ferritin			
TIBC			
T %			

Active space

APLASTIC ANEMIA

D/t Complete BM Suppression



M/C cause → Idiopathic

Inherited/congenital
- Fanconi's anemia
  - → AR
  - → Defect in DNA Repair Gene

Acquired
- Drugs-e .g
- Viruses - (EBV, CMV, PV-B19 , HIV)
- Radiation

CBC — **C/F**

Pancytopenia
- Hb/RBC ↓ → Anemia Pallor, Fatigure
- WBC ↓ → Leucopenia ↑ risk of Infection
- PLC ↓ → thrombocytopenia 3P's → Petechiae, Purpura, Prolonged bleedthg.

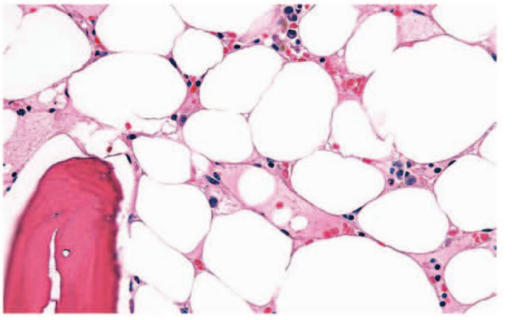
↓ Retic count

PBS :
- NCNC anemia

BMA : Dry tap

BMB : → ↑ Fat

(10C) → ↓ Cellularity



Rx : Bone Marrow transplantion
- Erythropeitin (TOC)
- GM CSF

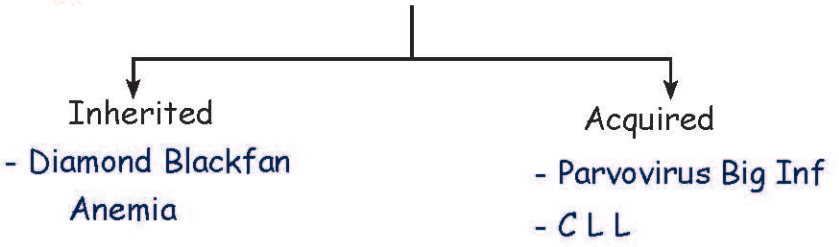
Active space

PURE RED CELL APLASIA

Only one cell lineage is compromised (RBC)

Anemia , reticulocytopenia , ↓↓ erythroid precursors.

Etiology :



Diagnosis :

CBC	
Hb/RBC ↓	PL, WBC } (N)
HCT ↓	
Retic count ↓	

Bone Marrow Biopsy

→ Red cellaplasia +

Giant pronormoblast

Active space

DNA Testing :

PCR for DNA of Parvo-B19

Rx :

-Treat the underlying cause

- Supportive
  - RBC transfusion
  - Iron chelators
- NIG (for parvo-Big)

- Steriods & Immunosuppressive (for idiopathic)

Active space

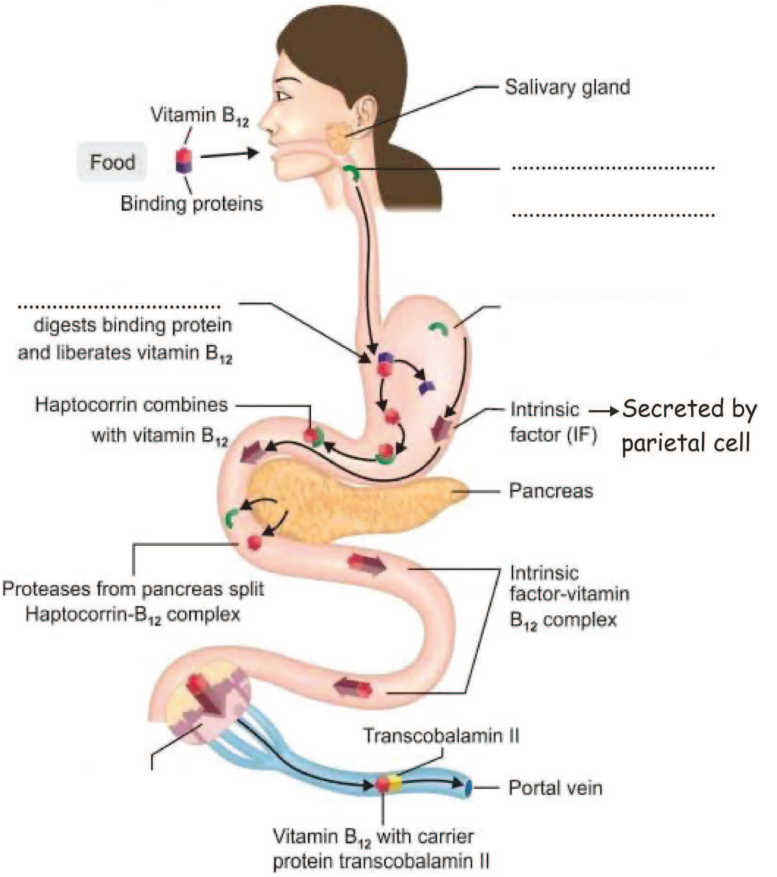
MEGALOBLASTIC ANEMIA

D/t deficiency of Vit ..... & Vit .....:

- Needed for DNA Synthesis & Nuclear maturation

	Vitamin B12	Vitamin B9
Aka	Cyanocobalamine	folate or Folic Acid .
Daily req	2-3 µg	50-200 mg.
Source	Dairy products, egg. fish, meat	Green leafy, vegetable
Site of absorption	.....	.....

Mechanism of Absorption :



Active space

Etiology :

↑ Demand
- Adolescence
- Pregnancy
- Lactation

Impaired absorption
- Pernicious Anemia
- Gastrectomy; ileal resection
- Malabsorption Synd.
- Drugs (methotraxate)
- Alcohol .

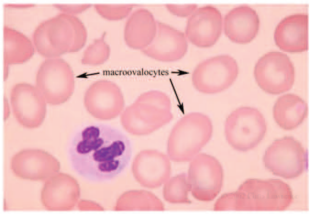
↓ Dietary Intake
- Socio-economic
- Vegetarian diet

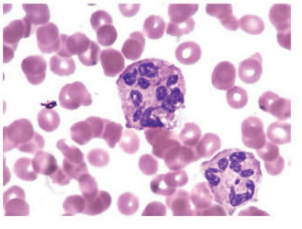
Investigation :

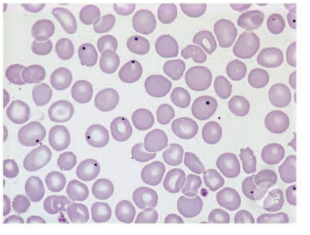
CBC	
Hb	
WBC	
PLC	
Retic	

RBC Indices	
MCV	
MCH	
MCHC	

PBS :







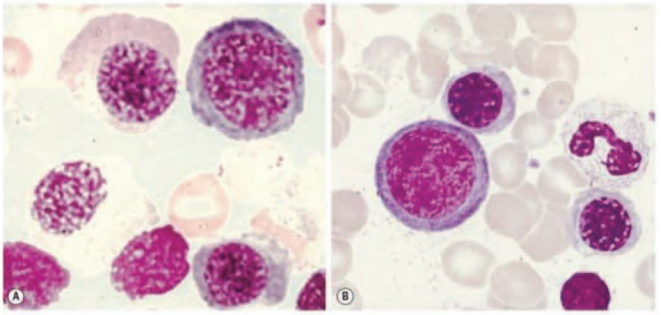
Classical Triad :

Macroovalocytes (Large oval RBCS)
- Earliest finding.

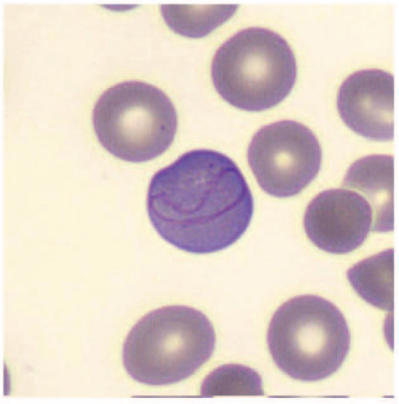
Hypersegmented Neutrophils.

- Howell-Jolly bodies (Precipitated DNA)

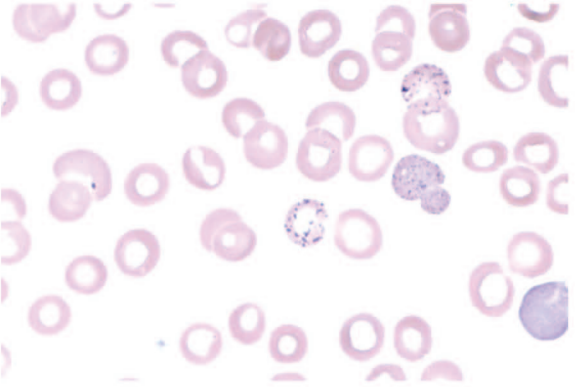
Active space



BMA : ..... (Large cell with Sieve like chromatin)



Cabot ring
(Made up of .....)



Basophilic stippling.

- S. homocysteine levels

- S. Methylmalonyl co-A levels

Succinyl COA (Component of myelin sheath)

Neurological complications. Subacute combined degeneration of Spind cord (SACD)
(M. Sev complication of MA)

Other complications:
Hyperpigmented knuckles
Beefy tongue

Active space

Biochemical parameters :	For folate :
..... Vit B12 assay	S. Folate .....
..... S. Homocysteine levels.	RBC Folate .....
..... S. Methylmalonyl - CoA	FIGL_U Test → ..... Figlu
..... S. LDH	

Histidine → ..... Figlu → ..... glutamic acid.

↓ B9

Forminoglutamic acid

Treatment :

- If ↓ Vit B9 → .....
  If ↓ Vit B12 → .....

Active space

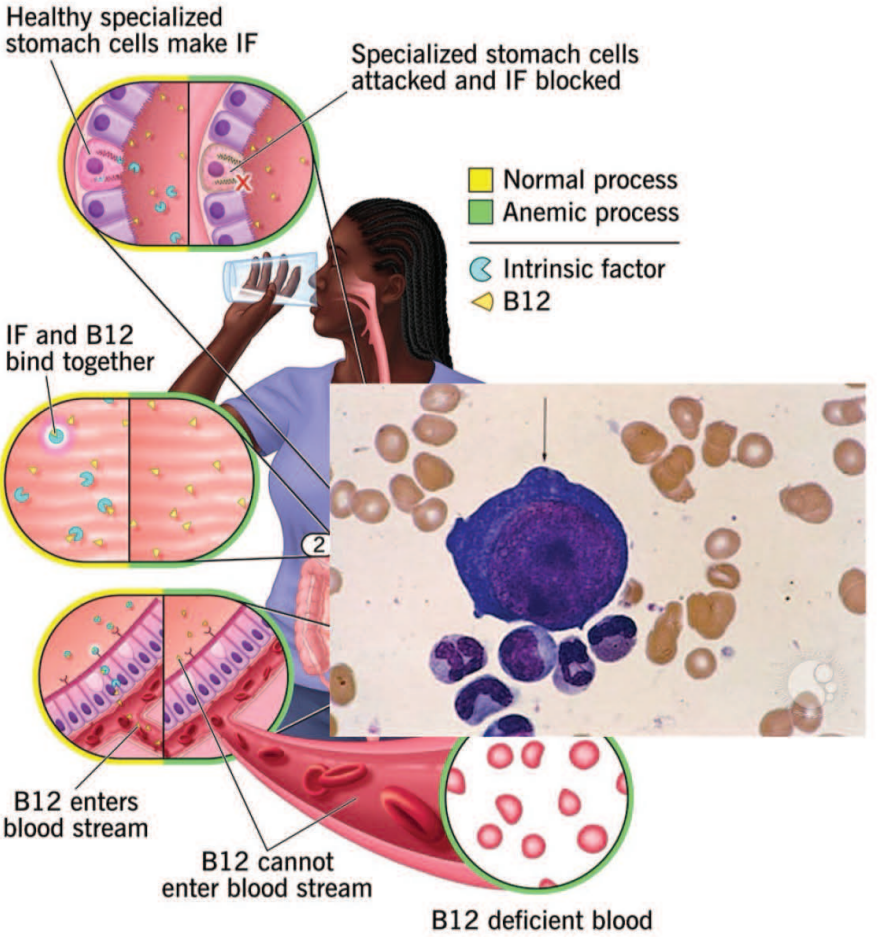
PERNICIOUS ANEMIA

Autoimmune d/s

→ Type-..... HS Rxn (..... mediated)

Type I	Binding of .....
Type II	Binding of Vit B12- If complex to .....
Type III	..... →↓ Vit B12 absorption

Pathogenesis :



Active space

C/F :

- Pallor
- Fatigue
- Beefy tongue
- Atrophic glossitis

- ↑ Risk of :

- Autolmmune d/s
- gastric adenocarcinoma

Beefy Tongue

Atrophic glossitis.

Schilling Test :

Use : to R/O cause of Vit B12 def Anemia.

Active space

EXERCISE

1. A 18 yr old boy came to the dermatology OPD with complains of skin color change over the knuckles. He seems to be worried and also states that there is some changes happening to his tongue as shown in the image. On CBC, Hb is 9.1 gm/dl, MCV is 111 fl. Diagnosis of the following condition is

   A. Strawberry tongue revealing kawasaki disease

   B. Macroovalocytic anemia

   C. Pelger huet anamoly

   D. Scarlet Fever

2. A 16 yr old female came to the medicine OPD with complains of generalized weakness and irritability. She is pale and gives a history of heavy menstrual bleed. On CBC, her Hb is 8.7gm/dl and MCV is 63fl. True regarding the above condition is

   A. Tear drop cells seen on peripheral blood smear

   B. Hemolysis of RBC at 0.3% NaCl

   C. High Transferrin saturation

   D. High Transferrin binding capacity

3. The integrity of rbc maintained by g6pd is:

   A. Controlling oxidative stress on rbc

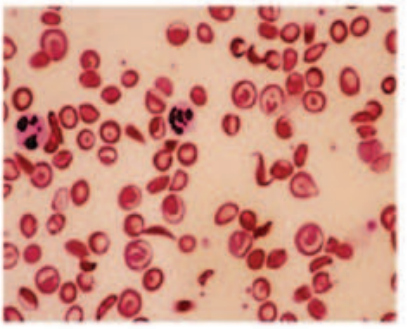
   B. Controlling reduction stress on rbc

   C. Maintaining flexibility of cell membrane

   D. Component of electron transport chain

Active space

4. 25/f came to opd with anemia, jaundice and joint pain.peripheral smear shows the following. What is the pathogenesis associated with the following smear finding



A. Autosomal dominant mutation

B. Abnormality of RBC memebrane

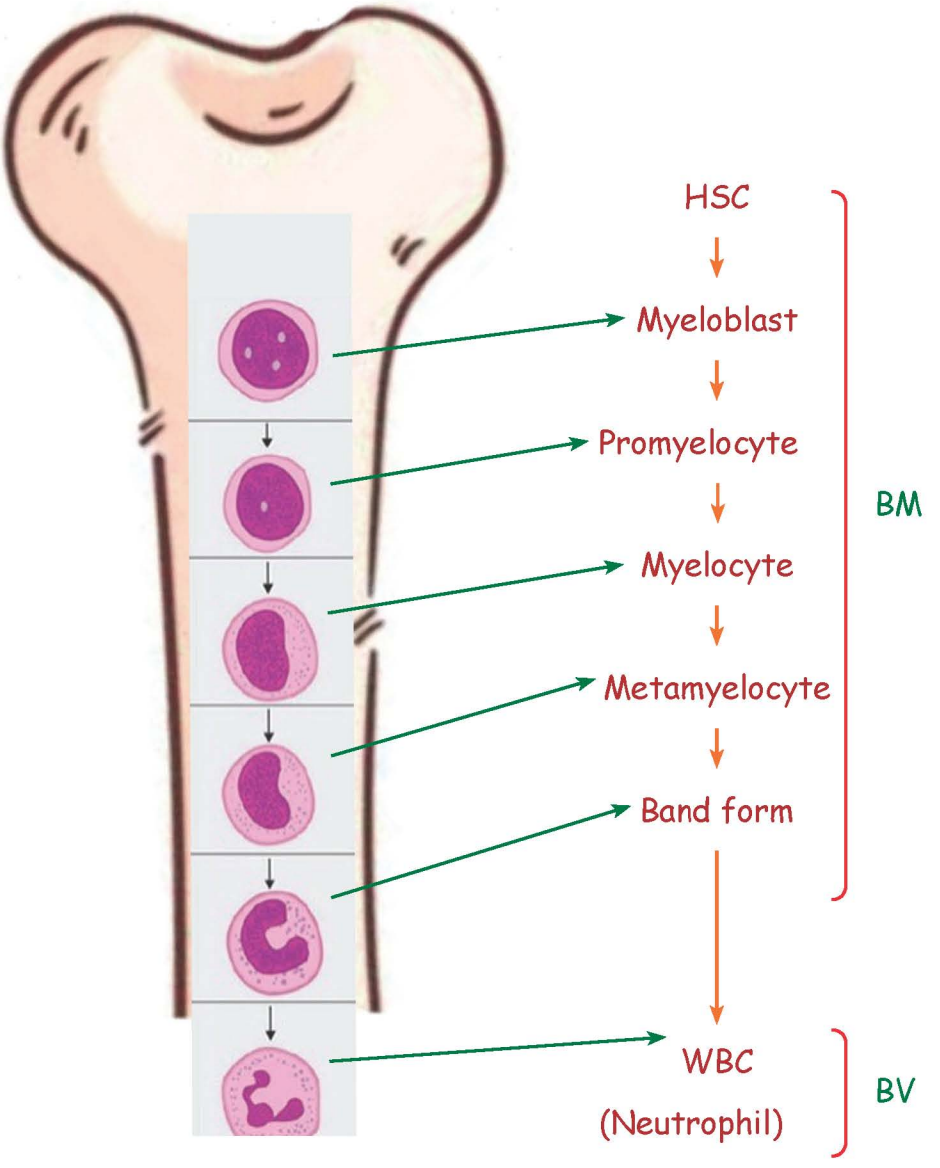
C. Missense point muttion

D. Severe iron absoprtion deficiency leading to pancytopenia

Active space

WBC-DISODERS

WBC DEVELOPMENT



(i) ↓ Size

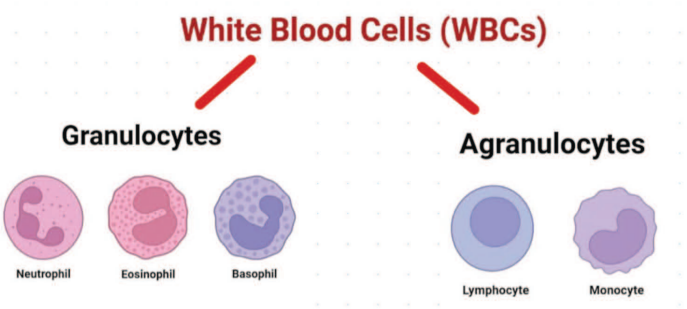
(ii) ↑ Maturation

(iii) ↑ No. of cells

Active space

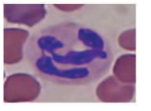
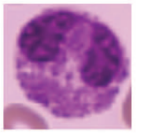
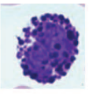
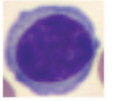
✦ In Myeloproliferative disorders .... RBC's, ... WBCs .... PLT (Since all 3 cell lines have a common precursor)	✦ Lymphoproliferative disorders ↓ will have ..... Lymphocytes

- Normal Total leukocyte Count (TLC) : 4000 - 11,000 cells / Cu. mm



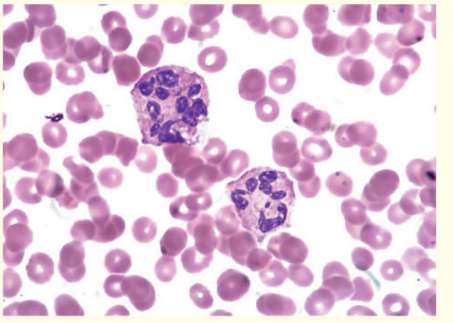
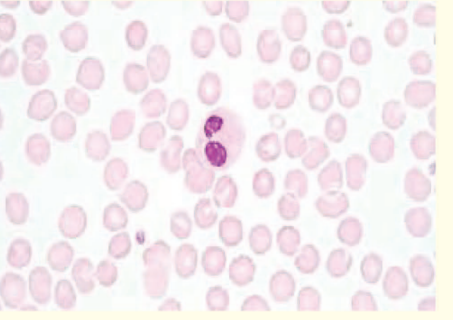
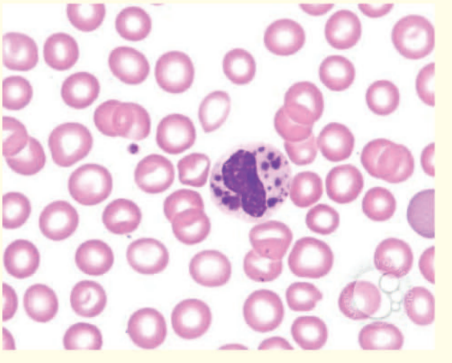
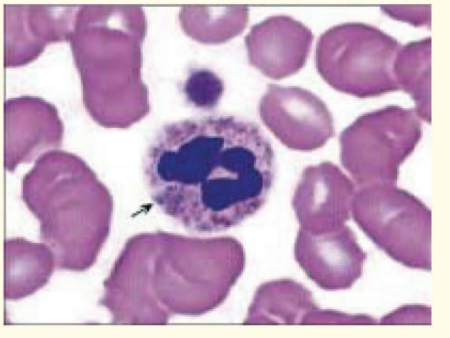
- Granules in cytoplasm

- No granules.

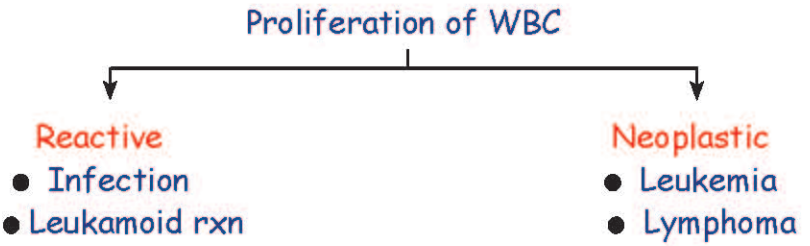
WBS	Description	Image
Neutrophils	3-5 lobed nucleus, small bluish granules in the cytoplasm	
Eosinophils	Bilobed/Spectales Shaped (2 lobes) Nucleus, Brick-red / pinkish granules	
Basophils	Bluish Granules obscuring the Nucleus (2 lobes)	
Lymphocyte	Smallest WBC, big nucleus, Minimal cytoplasm, no granules.	
Monocyte	Largest WBC, Kidney Shaped Nucleus.	

Active space

✦ Neutrophils based on no. of lobes :

(1) Hypersegmented Neutrophils • ..... lobes. • Seen in	
(2) Hyposegmented Neutrophils • ..... lobes • aka .... Seen • In Myelodysplastic syndrome	
3. • Chediak Higashi Syndrome : • d/t abnormal gene ..... • Giant granules In the Neutrophils.	
4. • May Hegglin Anomaly • d/t ..... mutation • HEG Gaint platelets Low platelets count. I → Inclusion In the N → Neutrophils.	

Active space



- ↓sed WBC Count : Leucopenia
  - CMT drugs
  - Radiation
  - BM disturbances.

➢ ↑ WBC count : Leucocytosis

WBC cells	(N)	Causes of ↑ WBC
a) Neutrophils	40-70 %	• Neutrophilia : • Bacterial Infection : [except TB Lymphocytosis.] • Leukemoid reaction
b) Lymphocytes	20-40%	• Lymphocytosis : - Chr. Infection - Viral Infectⁿ - T. B - C L L
c) Monocytes	2-9 %	• Monocytosis - Chr. Infectⁿ - Malaria - Rickettsiosis
d) Eosinophils	1-6 %	• Eosinophilia : - Allergic reactⁿ - Type I HS - BA - HodgKin Lymphoma.
e) Basophils	< 1 %	• Basophilia : - Allergic reactions - CML - Polycythaemia vera, - Myelofibrosis.

Active space

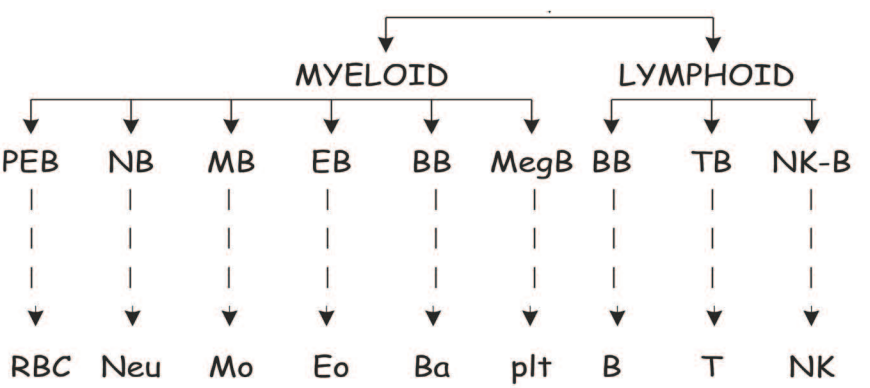
LEUKEMIA AND LYMPHOMA

Leukemia	Lymphoma
•	• .....
•	• .....

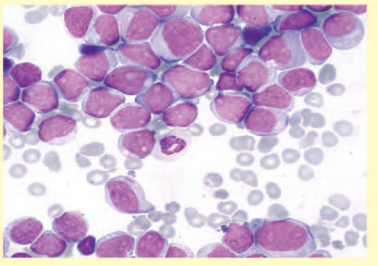
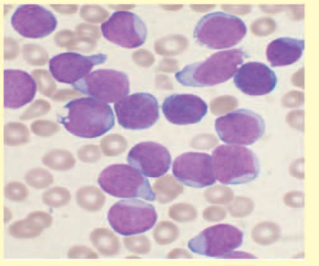
LEUKEMIA

- WBC Blasts in Blood
- Pleuripotency
- Asymm. Division
- Trans differentiation

→ HSC



Active space

	Myeloblast	Lymphoblast
Size	Large	Small
Nucleoli	Nuclei / Nucleoli (prominent)	+/-
Chromatin	Coarse (dark)	Homogenous (Light)
Vacuoles	+	-
Cytoplasm	Abundant cytoplasm	Scanty cytoplasm.
Marker	.....	.....
Stain	.... : terminal-deoxy nucleotidyl transferase	....
		

Active space

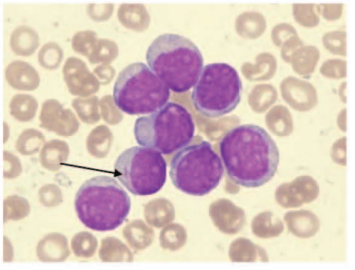
ACUTE LEUKEMIA

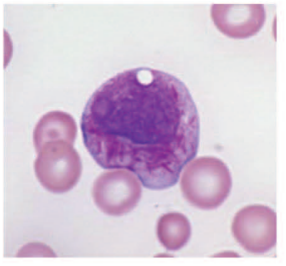
≥ 20% Blasts in Blood/BM

Acute Lymphoid Leukemia		Acute Myeloid Leukemia
.....	RBC	.....
.....	WBC	.....
.....	Platelet	.....
.....	Blast	.....
.....	CNS involn	.....
.....	Testicular involⁿ	.....
.....	Mediastinal involⁿ	.....
.....	Gum hyperplasia	.....
.....	Auer Rods	.....

.....

.....





ACUTE LYMPHOID LEUKEMIA

MC leukemia seen in children .....

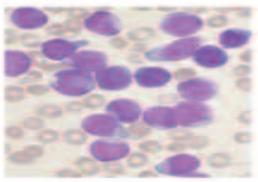
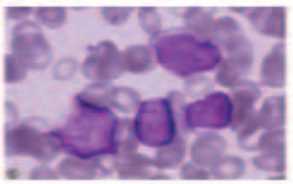
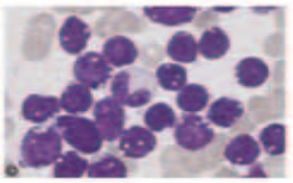
- Risk factors:
  - B • Bloom synd
  - R • Radiations
  - A • Anemia (Fanconi), Ataxia Telengectasia
  - D • Down's synd
  - D • Drugs (Chemo)
  - V • Virus [HTLV-1, EBV]
  - T • Translocations [t (12:21), t (8.14), t (9:22)]

Active space

- Classification: → FAB, WHO

(i) FAB - French American British

- ≥ 30% Blasts
- Based on → Morphology, Stain

L_1	L_2	L_3
- Small Homogenous	Heterogenous	c vacuoles
- Child	- Adult	- Old
- Good	- Intermediate	- Poor
- Pas ⊕	- PAS ⊕	- PAS ⊖
		- SBB ⊕ → Sudan Black B.

ii) WHO: ..... → .....

PRE-B ALL	PRE T-ALL
•	• .....
•	• .....
•	• .....
•	• .....

Good	Bad
•	• .....
•	
•	• .....
•	• .....
•	• .....

Active space

ACUTE MYELOID LEUKEMIA

≥ 20% Myeloblasts in Blood/BM.

- Classificatⁿ: → FAB / → WHO

(i) FAB: • ≥ 30% Blasts

↓ • Based on → Differentiation / → Lineage

8 types

M_0 → AML c̄/out differentiation → .....

M_1 → AML c̄ differentiation.

M_2 → AML c̄ maturation → .....

M_3 → Acute promyelocytic leukemia → .....

M_4 → Acute myelomonoblastic leukemia

M_5 → Acute monocytic leukemia

→ .....

M_6 → Acute Eythroid leukemia

M_7 → Acute megakaryocytic leukemia → .....

↓

.....

.....

.....

✦ Chloroma: aka: - granulation sarcoma
- gp. of immature cells
- t(8:21)

Active space

ii) WHO classifⁿ: ≥ 20% Blasts in BM.

4 types:

a) AML c̄ translocations

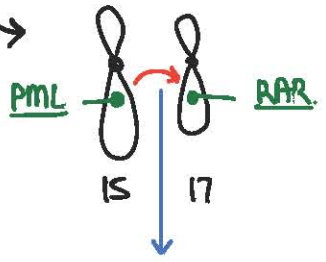
- .....
- .....
- .....

→ Good prog

.....

↓

.....



t(15:17)

PML → Promyelocytic leukemia gene

RAR → Retinoic Acid receptor

b) AML therapy related

c) AML c̄ myelodysplastic features

d) AML (NOS)

or

PML-RARA gene → ↓vit A acivity → ↓differentiation

↓

↑↑promyelocytes → ↑↑mucin → ↑↑Breakdown of Auer rods

↓

multiple thrambi in BV → DIC (Disseminated I/V coagulation)

✦ CAUSES OF AML: .....

S → Schwann Diamond syndrome

I → Idiopathic

R → Radiation

A → Ataxia Telengectasia, Anemia (Fanconi)

B → Bloom synd

C → Chemical (Benzene), Chloramphenicol

D → Down's synd, Drugs (anticancer), Diamond black fan synd

Active space

❑ △: (i) Blood investigatn

PBS & CBS:
- RBC
- WBC
- Platelet
- Blasts

Aleukemic Leukemia :
.....
↓
.....

if Blasts ↓

(ii) BMA/B → Sites
- Klima
- Jamshedi
- Salah

Adults → .....
Child → .....

>20% ↓ Acute
<20% ↓ Chromic

(iii) CD-Markers:

(iv) Stain

CD - Markers	Stain
CD 34 : Blasts CD45 : (N) Fⁿ WBC CD 16, 56 : NK cell CD 19, 20, 21 : B-cell CD 3, 5, 7 : T-cell CD 41, 61 : Platelets CD 14, 64 : Monocytes	PAS : Lymphoblast MPO : Myeloblast tdT : Lymphoblast

☞ Q: (i)
- CD34 ⊕
- MPO ⊕
- CD41 ⊕
- Blast ≥20%

→ .....

(ii)
- CD34 ⊕
- PAS ⊕
- CD3 ⊕
- Blast ≥20%

→ .....

Active space

(iii) FISH:
- t(8:21)
- t (12:21)
- t(15:17)
- t (8:14)
- t(16:16)
- t (9:22)

✦ Rx: Purpose: <0.1% of cancer cells

D → Daunorubicin
A → L-Asparginase
V → Vincristine
I → Intrathecal methotraxate
(D) P → Prednisolone

- t(9:22) → Tyrosine Kinase inhibitor → Imatinib
- t(15:17) → AML-M_3: Rx: ATRA

↓

All transretinoic acid

⊕

Arsenic trioxide

(Breakdown cancer cells)

- RXOC: BMT

ALL	AML
(IOC) B-ALL : CD 19, 20, 21 T-ALL : CD 2, 3, 5, 7 → + CD 10 C ALL A ↓ (Common ALL Antigen.)	CD 13, CD 33, CD 117, MPO + Ve

Active space

SUMMARY

	ALL	AML
Age	2-9 yrs.	15- 39 yrs.
Clinically	CNS, testis, LN Involvement	Gurm bleeding, Chloroma, DIC
P/S	..... Lymphoblast	..... Myeloblasts
Stain .	PASt +	MPO, Sudan black- B, NSA.
Markers	B-ALL → ..... T-ALL ..... Both ..... Hyperdiploidy	CD13 33 17 MPO
Cytogenetics.	....	....

Note : ..... positivity is not seen in AML.

Active space

CHRONIC LEUKEMIA

Chronic Lymphoid Leukemia

✦ 10-19% Blasts (Lymphoblasts) in Blood/BM

MC → ..... Type → ↓ ..... Abn Plasma cells

.... Abn Ab's

RBC .....

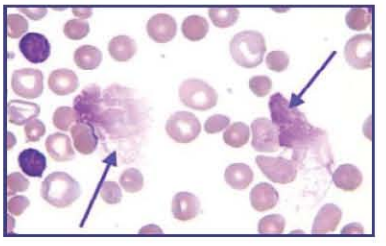
WBC .....

P/t.....

- Never arises .....
- Mutatⁿs:
  - ..... - mutat c̄ good prog
  - 17p-
  - Bcl 2↑↑
  - ..... (Zeta associated protein)
  - .....

✦ C/F:
- Fatigue, pallor, (AIHA-Spherocytes)
- Infections
- Bleeding
- Cervical LNadenopathy
- Hepatosplenomegaly → d/t infiltratⁿ by cancer cells
- .....

✦ Δ: Blood:
- ↑TLC → ALC > 5000/ul



- PBS: .....

  (Blasts c̄ improper cytoskeletal → ↓vimentin)
- Spherocyte
- Thrombocytopenia

Active space

BM: ↓Ey, ↓Myeloid, ↑↑Lymphoid

IHC: B → .....

T → .....

NK → .....

✦ RX: • Milder Pts: Symptomatic Treatment → followup

• Severe: C → Cyclophasphamide

R → Rituximab

F → Fludarabine

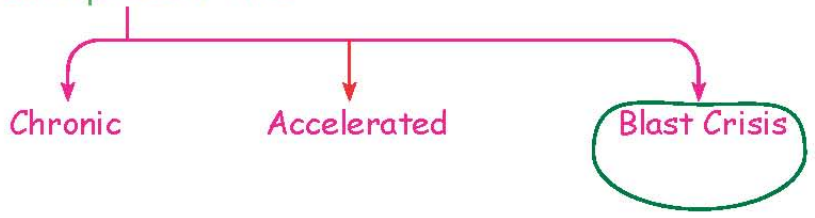
✦ RxOC: BMT

CHRONIC MYELOID LEUKEMIA

- ..... in Blood/Bm
- 50-60 yrs
- M > F

✦ Pathogenesis: .....

✦ C/F: Acc. to phase of CML



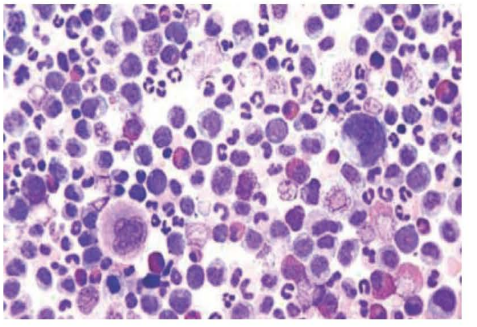
Blasts

Basophils

Active space

- Fever, malaise, wt loss
- Itching (↑mast cells/Histamine)
- Splenomegaly → Massive → compress stomach → Fullness
- Hepatomegaly intestine → constipation

Δ: (i) PBS & CBC:



- ↓ Hb
- ↓ WBC
- ↓ Plat
- Blasts (10-19%)
- .....
- .....

ii) LAP Score: Leucocyte Alkaline phosphate

CML	Leukamoid rxn
.....	.....
.....	.....
	.....

- .....
- .....

iii) FISH: t(9:22)

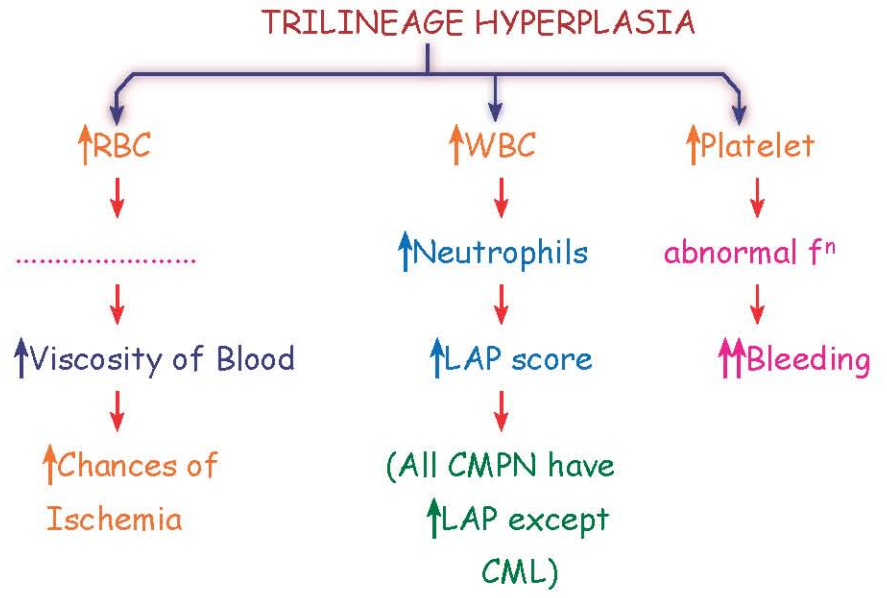
iv) BMA/B:
- Hypercellular marrow → M:Eratio: 2:1-3:1 - (N)
- Pseudo Gaucher Cells 15:1-20:1 CML

✦ Rx: TK ⊖ → Imatinib

✦ RxOC: BMT

Active space

POLYCYTHEMIA VERA

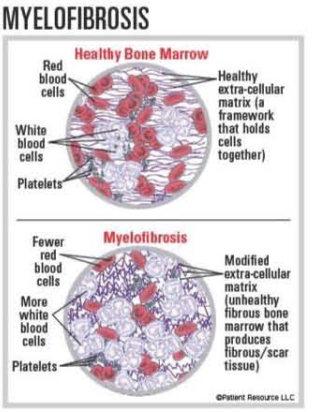


- Pathogenesis: .....
  - Valine → Phenyl alanine
- Criteria
  - Minor → .....
  - Major → • .....
    - > .....
    - • .....
- Rx: Phlebotomy

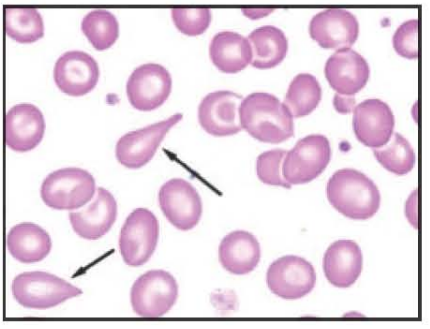
Active space

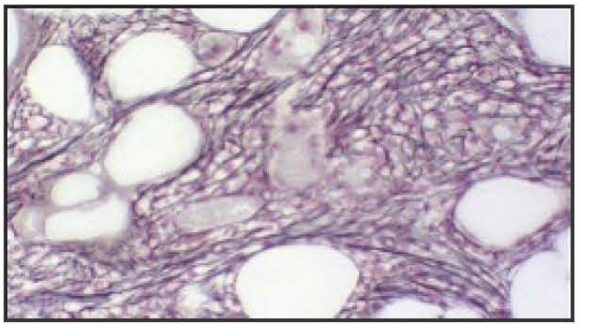
1° MYELOFIBROSIS

- Patho: .....
  - ↑abn/cancer megakaryocytes
  - ↑PDGF
  - ↑Collagen deposition inside BM



- BMA/B:
  - .....
  - .....
  - Splenomegaly → ↑extramedullary Hematopoeisis
- Δ: .....





ESSENTIAL THROMBOCYTHEMIA

- ↑↑Platelets → Abn →
  - ↑Thrombotic episodes
  - Bleeding
  - Parasthesia
  - TIA
- Patho: .....

Active space

LYMPHOMA

WBC Blasts in Lymphoid organs Lymphoblast → B → (MC) T NK

LYMPHOMA

HODGKIN	NON HODGKIN
• 15%	85%
•	.....
.....	
• Good prog	Bad prog

CONTIGOUS SPREAD

NON-CONTIGOUS SPREAD

HODGKIN LYMPHOMA

- Mainly involves B cells
- Strong a/w →
- Pathogenesis

↓

↓ expression of LMP 1 (Virulence factor for EBV)

↓

↑ Activation of NF k ß growth signaling pathway.

↓

↑ B-cell proliferatn

↓

Hodgkin Lymphoma (HL) HL is a disorder of B-cell arising from ..... center B-cell

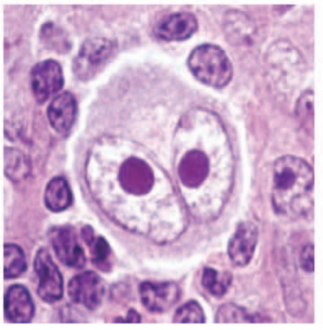
Active space

❑ Classificatⁿ

HL

NON-CLASSICAL	CLASSICAL
NLPHL	• NSHL → Nodular sclerosis HL
Nodular Lymphocytic	• MCHL → Mixed cellularity HL
Predominant HL	• LRHL → Lymphocytic rich HL
	• LPHL → " " Poor HL

❑ M/Characteristic finding: .....



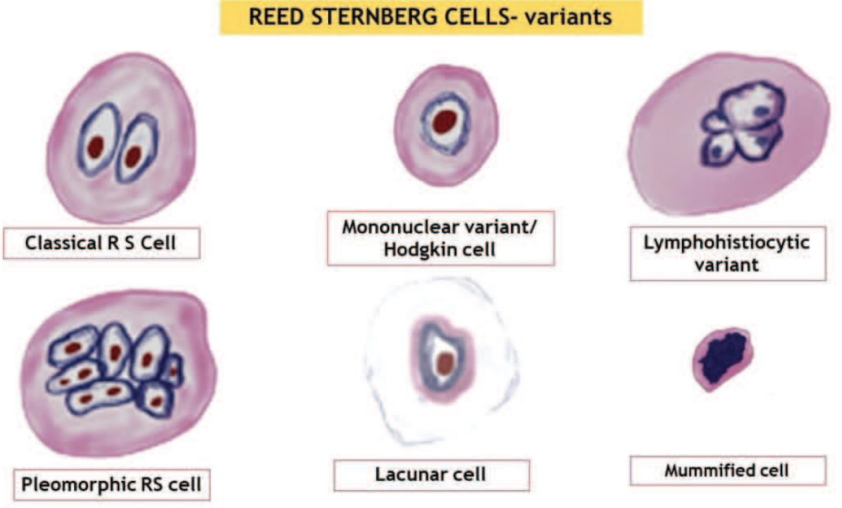
→ .....

.....

.....

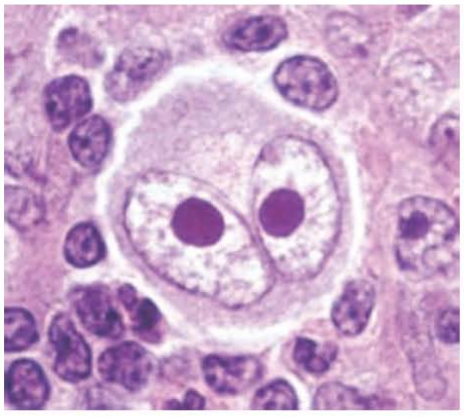
.....

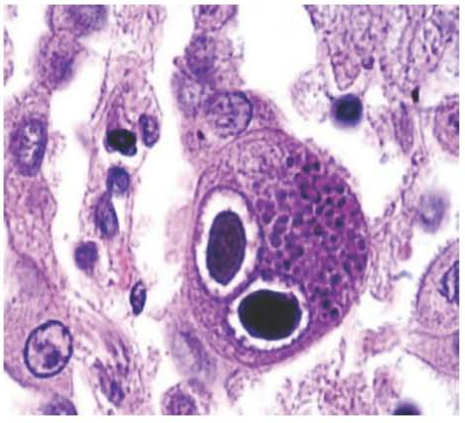
❑ Types of RS cells:



'Pleomorphic & mummified variants' of RS cells are seen in Lymphocyte depleted. HL.

Active space





- IL-13 → RS cells → Never present alone
  - → Secrete 1L 6, 8, 5 Eotaxin
    - ↓
    - Infiltratn of inflammatory cells

Non Classical | Classical

Features	NLPHL	NSHL	MCHL	LRHL	LPHL
i) %		←	←	→	→
ii) Age gp					
iii) RS cell					
iv) EBV	0%	0%	70%	40%	95%
v) IHC CD 15					
20		←	←	→	→
30					
45					

Active space

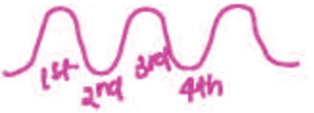
❑ C/F:

A → Specific → i) LN Enlargement → MC-Cevical

B → Non-specific

- Fever →
- Nausea
- Vomiting
- Night sweats

Cylical Trend → PEL EBSTEIN



Paraneoplastic Syndᵐ : • Amyloidosis : AA type • Pain a/c. Ingestⁿ
• Intense Pruritis

❑ Δ:
- Blood
- FNAC
- Bx - IOC → RS cells against background of inflⁿ cells
- IHC

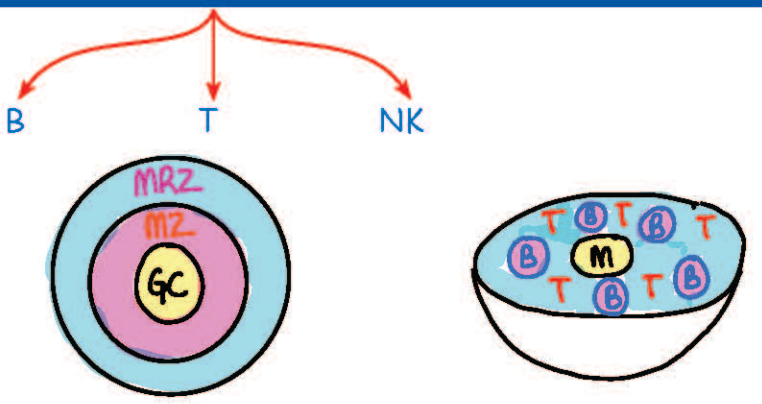
❑ Rx: A → Adriamycin B → Bleomycin
V → Vincristine D → Dacarbazine

	AN ARBOR STAYING SYSTEM :
Stage 1 :	Single IN region or single extralymphatic organ Involvem.
Stage 2:	≤ 2 LN on same side of diaphragm.
Stage 3 :	Both sides of diaphragm Involved.
Stage 4 :	Multiple/disseminated foci Involved with ≥ extralymphatic organs.

Prognostic factor : Staging > tumor type.

Active space

NON HODGKINS LYMPHOMA



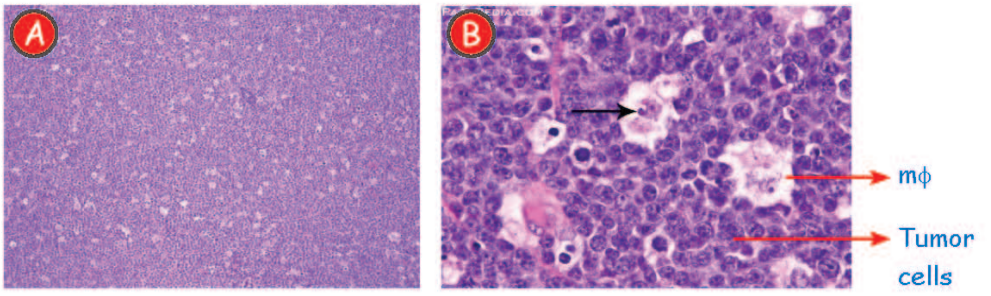
Type	Marginal Zone	Mantecell	Follicular	DLBCL	Burkitt's
MC	-	-	M/C Indolent NHL in world	MC NHL in India	-
Mutation	.....	.....	..... MLL gene		....
M/E			- Follicular arrangement - centrocytes - centroblasts		Starry sky appearnce
IHC	CD 43 (+)	Cyclin D, ... CD 5 ..... CD 23 ..... SOX -	BCL -	CD 19 (+), CD 20 (+) BCL	

Active space

• Burkitt's Lymphoma

- ..... prognosis
- ..... mutation
- .....
- 3 variants → Endemic (MC) -- Africans
  - → HIV
  - → Sporadic

❑ M/E: ..... → ..... → .....
  → ..... → .....



❑ C/F:

A
- Lymphadenopathy
- BM involⁿ → Cytopenia

B
- Fever, nausea etc

❑ Δ:
- Blood
- FNAC
- Bx → IOC
- FISH
- IHC

❑ Rx:
- R → Rituximab
- C → Cyclophosphamide
- H → Hydroxyduanomycin
- O → Oncovin
- P → Predrisolone

Active space

HAIRY CELL LEUKEMIA

A → .....

B → .....

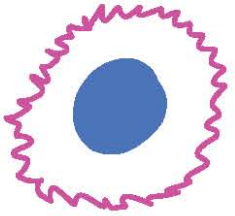
C → .....

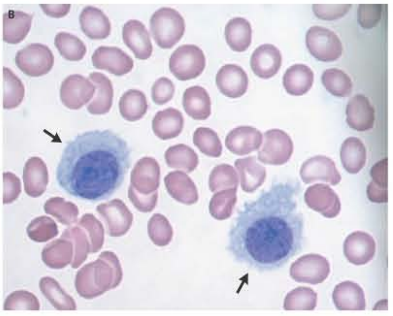
T → .....

❑ M/E: Hairy cell

↓

Fried egg app

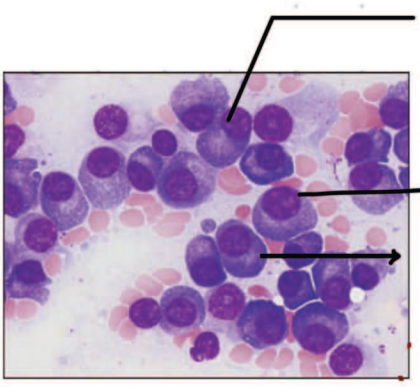




Active space

PLASMA CELL DISORDER

Multiple Myeloma :



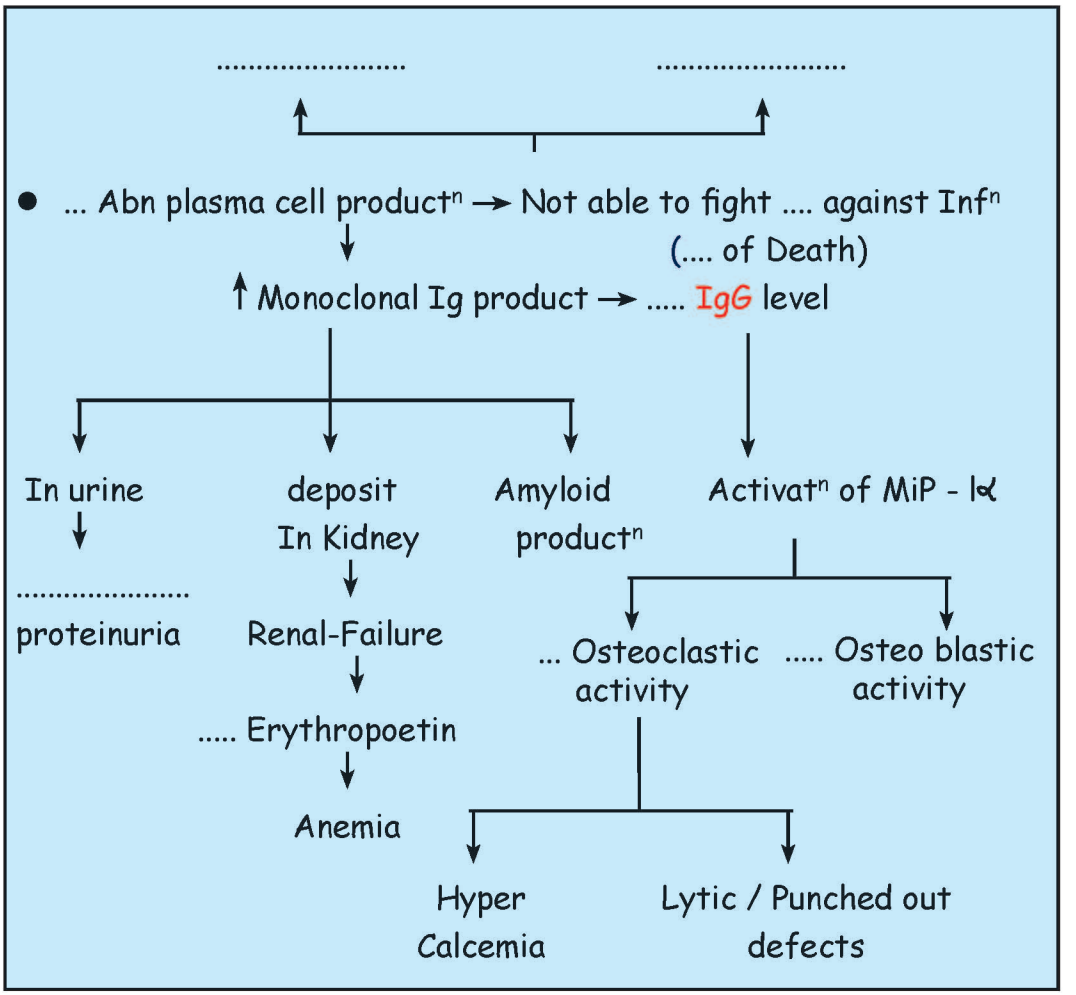
Eccentric Nucleus

Oval Shaped

Peri-nuclear hoff / halo
(Empty Space around the Nucleus
d/t presence of golgi apparatus

Plasma Cells

Etiopathogenesis :



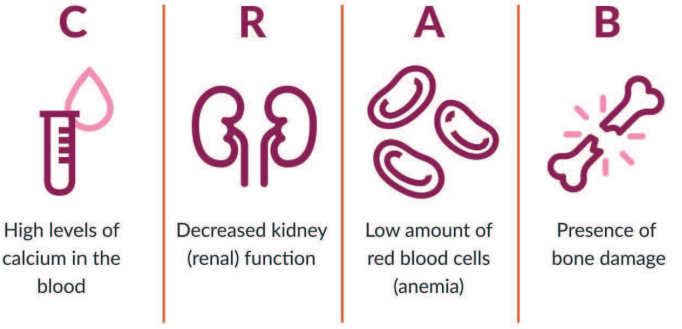
Active space

Criteria :

- ✦ BM plasma cells > .....
- ✦ M Protein .....

+

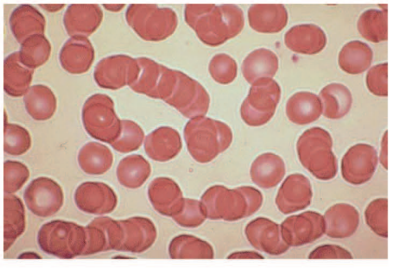
- ✦ Any ..... Myeloma defining event (MDE)

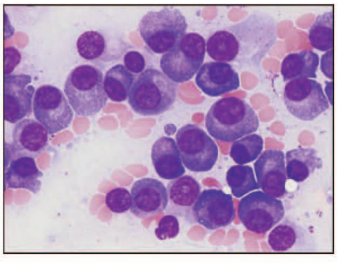


S	Sixty percent ≤ BM Plasma cell
Li	Light Chain ratio > 100
M	MRI with > 1 focal lesion
C	Hyper Calcemia
R	Renal Failure
A	Anemia
B	Bone lytic lesion

In :

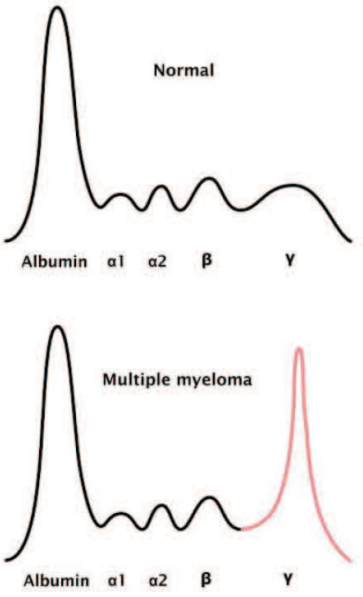
i) PBS and CBC :
- RBC (Rouleux formatⁿ)
- Abn plasma cells.
- Spherocytes





Active space

ii) Electrophoresis :

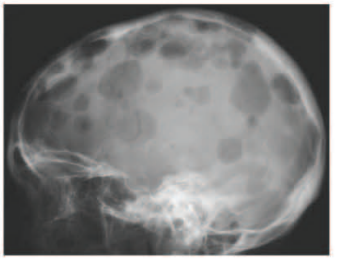


M Protein

Serum	Urine
.....-Spike	↓
..... Region	Proteinuria
..... chain	↓
.....	.....

iii) Biochemical Tests :

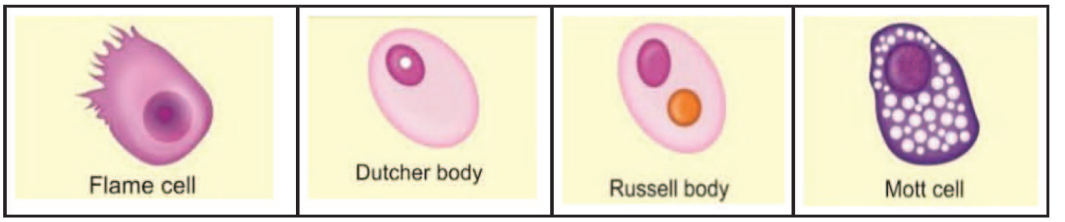
- .... S. Ca2+ (Metastatic cancer)
- .... Hb
- .... ESR
- .... S. B-2 Microglobulin
  (Best prognostic factor)
- RFT → S. Creatinine .....
- S . Alb/Glb : .....
- X-ray, CT Scan, PET Scan
- BMB → Abn plasma cells
  deposit^{n} of M- Protein

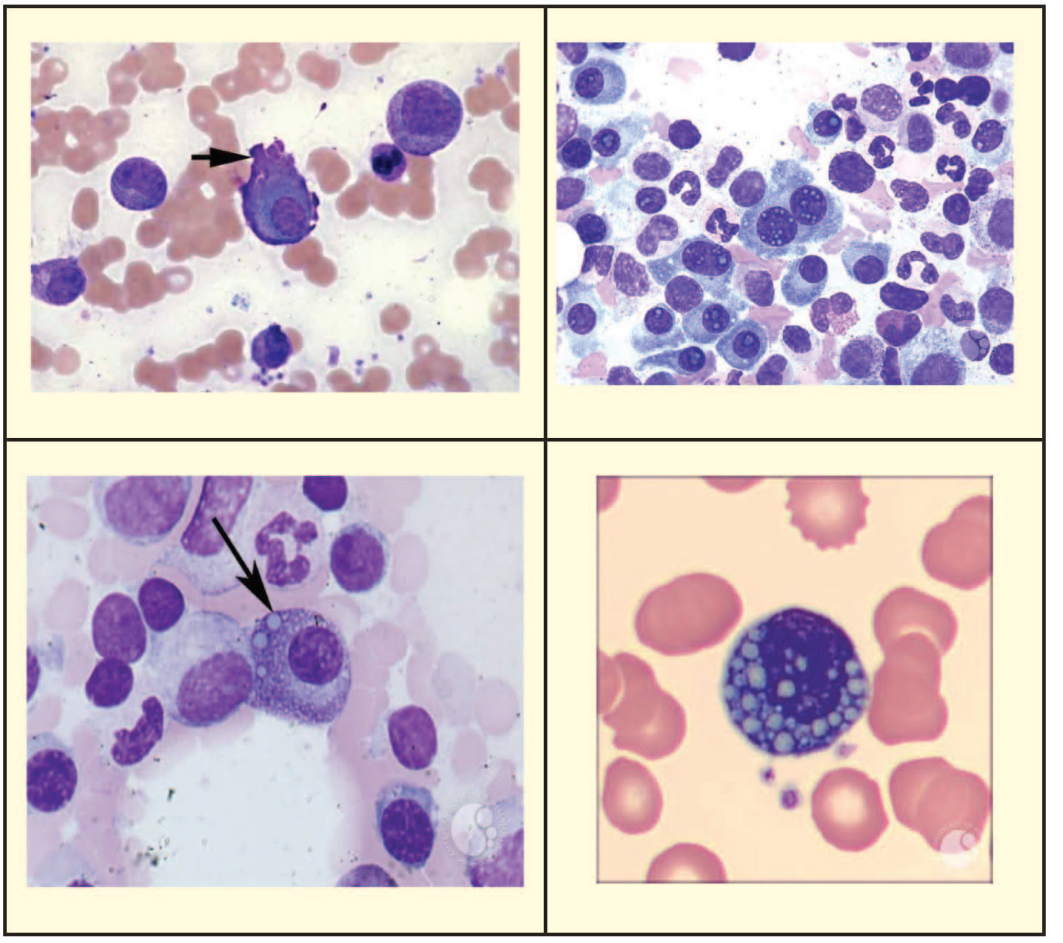


X-ray

PET Scan

Active space





Rx : Stem Cell Transplantat^{n} : Most effective :

Chemotx : B → Bortezomib

L → Lenalidomide

D → Dexamethasone

Active space

Otheplasma cell d/s :

MGUS (Monoclonal Gammopthy of undermined Significance)	Smoldering Myeloma	Waldenstorm's Macroglobulinemia.
- ..... Plasma cells BM - CRAB ..... - S . Ig... : <3g/dl	- ..... PC'S In BM - CRAB ..... M protein >3g /dl	- .... IgM (Hyper viscosity Syndⁿ) ↓ ↑ Viscosity of blood → Head ache → Plethora → Dizziness

Active space

EXERCISE

1. Parents came with their 2 yr old child to the pediatric OPD with complains of fever, lethargy, loss of appetite. Child is a known case of down's syndrome. On CBC , Hb is very low with very high TLC. On PBS, >30% immature cells are seen in blood which are MPO positive. Clinical examination reveals hepatosplenomegaly. Most likely diagnosis of this child would be

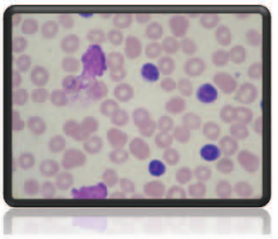
   A. T-ALL

   B. B-ALL

   C. Acute megakaryocytic leukemia

   D. Acute promyelocytic leukemia

2. A middle aged man came to medicine dept with repeated episodes of fever and throat infection. Routine blood investigations revealed a case of pancytopenia but the absolute lymphocyte count was raised. Coomb's test was positive. PBS image is given below demonstrating Smudge cells. True statement regarding the same is



   A. Most common genetic abnormality with the associated dieases is 13p-

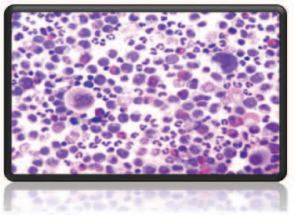
   B. Myeloid to lymphoid ratio is deranged leading to mutiple fold rise in lymphoid origin cells in the blood and bone marrow

   C. NOTCH gene mutation signifies B cell mutation

   D. All of the above

3. 58 yr old female came to the hospital with complains of heaviness in the abdominal area & constipation. She also complains of bleeding from the gums sometimes. CBC reveals marked eosinophilia and basophilia with TLC 63000/cmm. Metamyelocytes (33%), myelocytes (19%) and band form (23%) are seen. PBS finding is attached below. All are true regarding the same except:

Active space



A. No maturation arrest

B. Low LAP score

C. Hepatospenomegaly seen

D. No risk of acute lymphoid leukemia

4. On performing biopsy of a patient with cervical lymphadenopathy with a history of weekly on-off fever, inflammatory infiltrate is seen with presence of giant cells which are CD 15 and 30 positive. image is attached below. Spot the diagnosis.

A. Smudge cells on the Lymph node biopsy

B. Glycogen filled mesangial cells

C. RS cells with clearing of cytoplasm

D. Target cells on muscle biopsy

5. A 5 yr old boy came to hospital with his parents having complains of fever, weakness, fatigue, pallor and purpuric spots on skin. His CBC shows Hb-8.1gm/dl, Tlc-2.2lakhs/cmm. BM evaluation reveals >36% blasts which are tdT positive. Clinical examination shows mediastinal infiltration. True atatement regarding the same.

A. PML-RAR gene mutation

B. NOTCH gene mutation is positive in this patient leading to mediastinal infiltartion

C. EBF gene mutation is positive in this patient leading to mediastinal infiltartion

D. None of the above

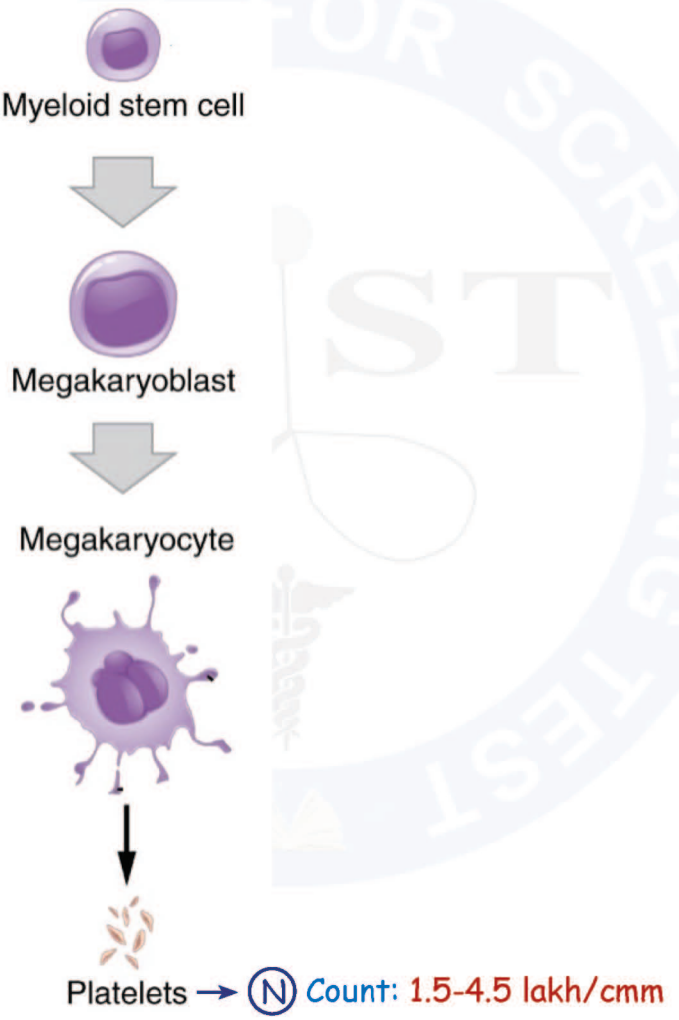
6. 53 yr old male is diagnosed with secondary hyperparathyroidism. Further workup revealed high S. Calcium, high S. creatinine. On Xray osteolytic lesions were seen on skull. BM examination revealed abnormally high levels of plasma cells with very high levels of monoclonal protein in serum. Most common translocation associated is:

A. t(14:18), B. t(11:18) C. t(12:21) D. t(11:14)

Active space

PLATELET & COAGULATION DISORDERS

PLATELET DEVELOPMENT



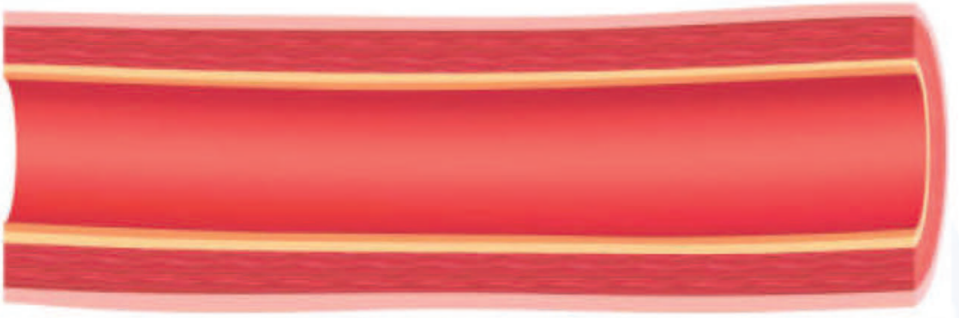
- (N) **fn:** Clot formatn

Active space

HEMOSTASIS

- Physiological process
- Formation of clot at the site of Injury.
- Prevent formation at the site of No Injury

❑ Steps of Hemostasis - 4



i) Vasoconstriction - Endothelin

ii) Formation of Platelet Plug - 1° Hemostasis

a) .....

b) .....

c) .....

d) .....

iii) Deposition of fibrin - 2° Hemostasis

iv) Clot stabilization and resorption

↓ (F XIII)

↓ Endothelium secretes

1) Thrombomodulin → Modulate Thrombin

↓

↑Fibrinogen into ↓Fibrin

2) Tissue plasminogen activator (t-PA)

↓

Plasminogen → (Plasmin) → Cutter for fibrin

Active space

CLOTTING FACTORS

F I	-	Foolish
F II	-	People
F III	-	Try
F IV	-	Climbing
F V	-	Long
F VI	-	
F VII	-	Slopes
F VIII	-	After
F IX	-	Christmas
F X	-	Some
F XI	-	People
F XII	-	Have
F XIII	-	Fallen

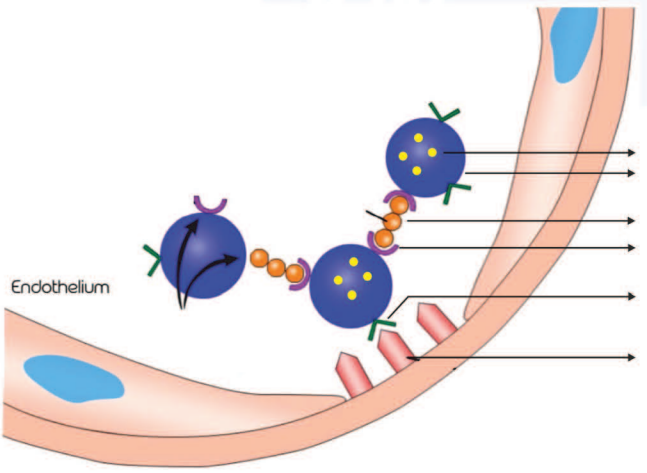
Vit. K Dependent factors : .....

COAGULATION CASCADE

Active space

TESTS FOR HEMOSTASIS

Defects	Platelet Defects		Coagulation Defect	
Tests	Platelet count (PC)	Bleeding time (BT)	Pro Thrombin time (PT)	Activated partial thrombo-plastin time. (aPTT)
(N) Range	.....	.....	.....	.....
	Thrombo--cytosis Essential thrombocytosis. Myelo proliferative diseases Thrombocytopenia d/t....destruction ITP d/t .... production Aplastic anemia	Quantitative defect ....PC Quantitative defect Plt function	Ext & common pathway ● .....PT : ..... of F-..... ● Warfarin Rx	Int & common pathway ● .....aPTT : .....of F-..... ● Heparin Rx



Active space

Bleeding

Platelet Defects		Coagulation Defects
Qualitative	Quantitative	• Hemophilia • vWD • DIC
PC	PC	
• GP Ib/IX ↓ • GP IIb/IIIa ↓ • Granules, ↓	• ITP • HUS • TTP • HIT	

	Platelet Disorder	Coagulation Disorder
Inheritance	Autosomal	X-linked
Gender	M = F	M > F
Clinically	• Petechiae • Purpura • Mucosal Bleedings • Prolonged Bleeding	• Ecchymosis • Hemarthrosis
PC	.....	.....
BT	.....	.....
PT	.....	.....
aPTT	.....	.....

Active space

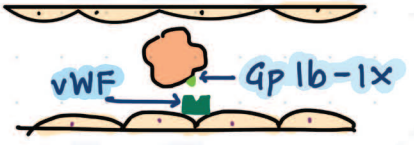
MIST

VON WILLEBRAND DISEASE

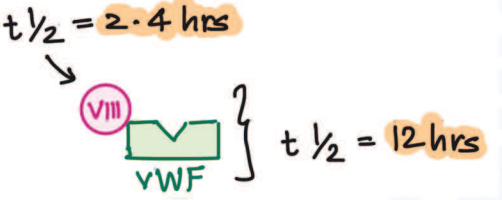
- Gene for vWF is on ch.....
- Quantitative & Qualitative defect of VWF
- ..... Inherited bleeding disorder
- Aka .....hemophilia (Defect in VWF → 2° ..... of .....)
- VWF Stored In
  - Endothelial cells
  - Liver
  - Weibel Palade Bodie
  - Megaka Ryocytes

Ⓝ Function :

a) Platelet .....



b) Activation of .....



Quantitative Defect		Qualitative Defect
↓ VWF		Dysfunctional VWF
Type 1 (M/C) - AD - Mild form	Type 3 - AR - Severe form	Type 2 - AD

Active space

C/F :

Gum Bleeding

Bruises

Menorrhagia

Nose bleeding

Diagnosis :

PC	BT	PT	aPTT
.....	.....	.....	.....

- Ristocetin Induced platelet aggregation (RIPA)
- Ristocetin cofactor assay
- IOC : VWF Assay

Note : Ristocetin helps in higher interaction b/w Gplb-IX & vWF

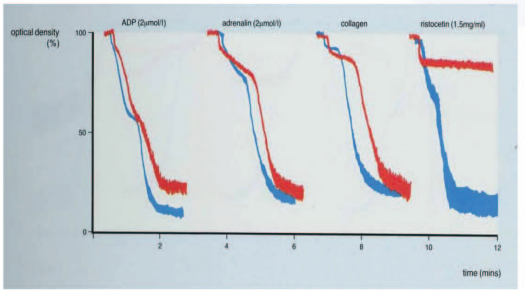
Rx :

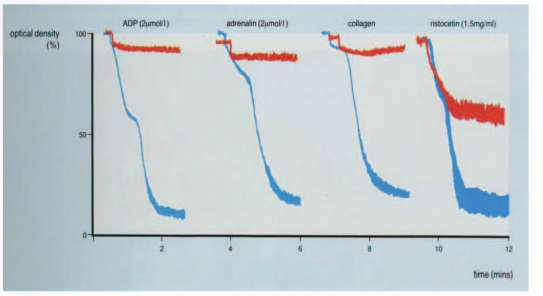
- Desmopressin (Enhances Secretion of VWF from Storage Sites)
- Cryoprecipitate
- FFP (Fresh Frozen Plasma)

Active space

MIST

Features		Bernard Soulier Syndrome	Glanzmann Thrombasthenia
Inheritance		.....	.....
Defect		..... (Platelet ..... defect)	..... (Platelet ..... defect)
C/F		• Petechiae • Purpura • Prolonged Bleeding	
			Bleeding From Umbilical Stump
PC		.....	
BT		.....	
PT		.....	
aPTT		.....	
Platelet Aggregation Study	Ristocetin	.....	.....
	ADP Collagen Epinephrine	.....	.....



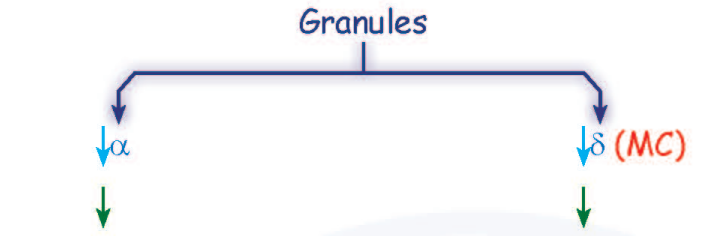


....

Active space

MIST

STORAGE POOL DEFECT



- .....
- TGF- B
- Fibrinogen
- Fibronectin
- PDGF

- .....
- ADP
- Ca2+
- Epinephrine
- Serotonin

ITP : IMMUNE THROMBOCYTOPENIC PURPURA

(Ab's → .... Platelets → Superficial bleeding)

Features	Acute ITP	Chronic ITP
Onset	Sudden	Insidious
Duration	..... months	..... months
Ab	.....	.....
Age gp	Children	Adult
Gender	M= F	F: M (3 : 1)
Trigger	Viral Infection	Idiopathic
Treatment	Self-Limiting	Immunosuppressive drugs / Steroids

Lab test :

PC	.....	PT	.....
BT	.....	aPTT	.....

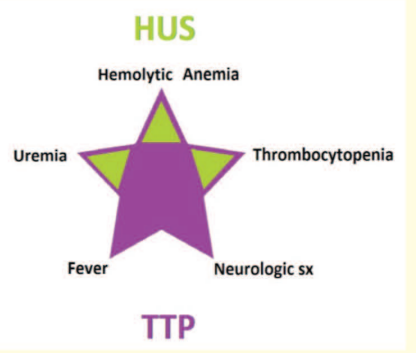
➢ Coomb's Test .....

- Ab's Directed Against ↓
  - .....
  - .....

Active space

HEMOLYTIC UREMIC SYNDROME (HUS)

THROMBOTIC THROMBOCYTOPENIC PURPURA (TTP)

Features	HUS	TTP
Etiology	Typical → E. coli 0157 : H7 Shigella → Release toxin → injury to vessel → Thrombus formation Atypical → • Factor H, 1 • Drugs • A1I Problems → Thrombus formation	Mutation of ..... → Accumulation abnormal VWF → Platelet activation → Thrombus formation
C/F		
PC	.....	
BT	.....	
PT	.....	
aPTT	.....	
Rx	Plasmapheresis	Hemodialysis

Active space

MIST

HEPARIN INDUCED THROMBOCYTOPENIA (HIT)

Features	Type I	Type II
Pathogenesis	M/C d/t platelet aggregation effect of heparin	d/t Ab's against PF - 4
Postexposure heparin	Immediately	5 - 15 days later
Fate	Self - resolving	May lead to A/V thrombosis

MICROANGIOPATHIC HEMOLYTIC ANEMIA

- AKA Thrombotic Microangiopathies

Hemolytic uremic Syndrome	Thrombotic Thrombocytopenic purpura	Disseminated Intravascular coagulation

Active space

DISEMINATED INTRAVASCULAR COAGULATION (DIC)

- AKA Consumptive Coagulopathy
- State of Hyper & Hypo Coagulability
- Combination of 'Thrombosis' & Hemorrhage

➢ Triggering Factors

a) Endothelial Injury
   - Obstretic complication
   - AF Embolism
   - Dead fetus

b) Infectn

c) Severe burns

d) AML-M_3

e) Sx ,Trauma

→ Multi-organ Ischemia.

Activation of I/V Coagulation

↓

Endothelial damage + Plt + Coag. Consumption

↓

Thrombus formation in the micro circulation

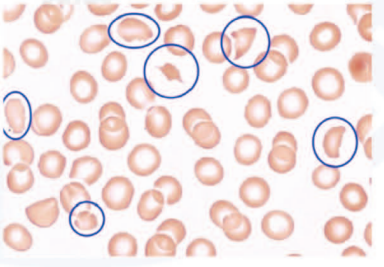
↓

↑ Fibrinolysis → Hemorrhagic Synd

➢ Poor Prognosis

Lab test :

PS : RBC : Schistiocytes (Fragmented)



PC	
BT	
PT	
aPTT	
S. Fibrinogen	
FDP	
D-dimer	

Rx : Fresh Frozen plasma. (FFP)

Active space

HEMOPHILIA

Features	Hemophilia A	B	C
Inheritance	.....		.....
Factor	.....	.....	.....
M : F	.....		.....
BT	.....	.....	.....
PC	.....	.....	.....
PT			
APTT			
IOC	..... assay	..... assay	..... assay
Rx	- Cryoppt - Desmopressin	- FFP	

✦ C/F :

- Joints Bleeding (Hemarthrosis)
- Gum bleeding
- Nose bleeding
- Bruises
- Prolonged bleeding from Injuries.

Active space

SUMMARY

Features	Defect	PC	BT	PT	aPTT
Von-willebrand disease	.....				
Bernard Soulier Syndrome	.....				
Glanzman Thrombasthenia	.....				
ITP	.....				
TTP	....				
HUS	....				
Hemophilia A	.....				
Hemophilia B	.....				
Hemophilia C	.....				
DIC	....				

Active space

MIST

EXERCISE

1. Which of the following is involved when there is excessive bleeding after labor, and the laboratory studies demonstrate decreased platelets, prolonged prothrombin time and partial thromboplastin time, and increased fibrin split products?

   Disseminated intravascular coagulation

   Hemophilia A

   Severe liver disease

   Vitamin K deficiency

2. In TTP, all of the following are seen except:

   Renal failure — Hemolyic anemia

   Thrombocytosis — Neurological features

3. To test hepatic function to assess bleeding disorder clinical test done is

   CT, BT — PT

   Factor 8 — None of the above

Active space

CENTRAL NERVOUS SYSTEM

CNS TUMOR

- MC Brain tumor → .....
- MC 1° Brain tumor → .....
- MC (Benign) Brain tumor in children → .....
- 1° malignant Brain tumor

  Children → .....

  Adult → .....

WHO Grading :

A → Atypia

M → Mitosis

E → Endothelial Proliferation

N → Necrosis.

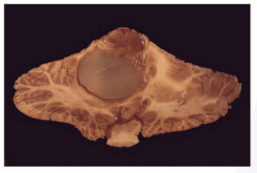
Grading	Features	Prognosis.
I	.....	Good
II	.....	↓
III	.....	
IV	.....	Worst

Active space

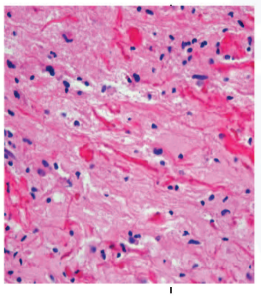
➢ Glial Tumors : mϕ

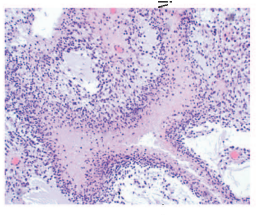
Arise from mϕ of Brain (Microglial cells)

Astrocytoma	Oligodendroglioma	Ependymoma
↓	- ....., & ..... gene mutation - ..... appearance.	- Child - 4th ventricle, SC - ..... gene mutation - .... rosette



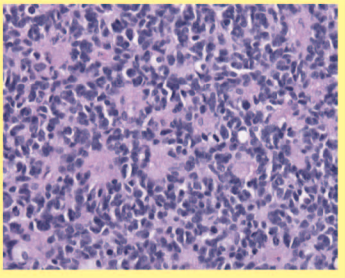
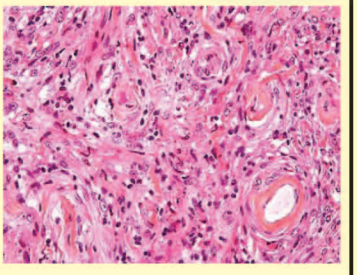
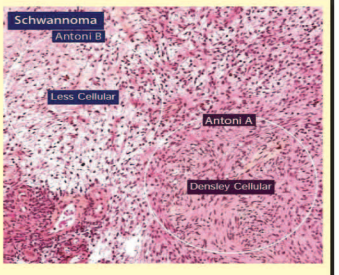
Grade	Types	Prognosis.
I	Pilocytic (Childhood)	Good
II	Anaplastic	↓
III	Atypical	
IV	Glioblastoma Multiformae	Worst





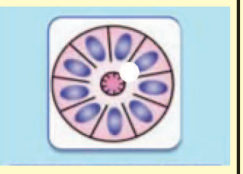
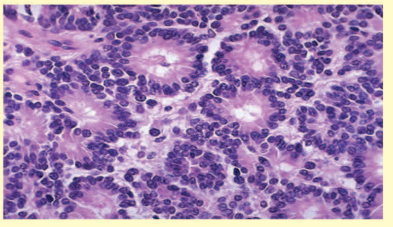
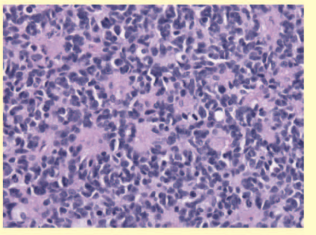
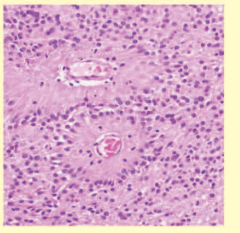
	Pilocytic Astrocytoma	Glioblastoma
• WHO grade	I	IV
• Age	Children	.....
• Location	Cerebellum ; 4th ventricle	- Frontal lobe
• Pathogenesis	..... Mutation	..... gene mutation
• Gross features	Cystic Nodule	Butterfly tumors
• Microscopy:	No AMEN features	All AMEN features.
	-	- .....
	-	..... necrosis.
	.....	- ..... bodies

MIST

Medulloblastoma	Meningioma	Schwannoma
• MC 1° Malignant Brain tumor in children • M.C Site : ..... (Early Mets → .....)	• M.C 1° Brain Tumor • F>M • Arise from meninges (Arachnoid Cap cells)	- aka Acoustic Neuroma - PNST : .....
• ..... 2 mutation • ..... mutation	• ..... 2 mutation	• ..... 2 mutation
• Grade IV WHO Tumor : Poor Prognosis	Benign tumor : .... Radiatⁿ exposure →	
• M/E : .... (..... - rosette) Small round blue cells with Scanty Cytoplasm.	• M/E :	• M/E : Antoni - ..... (hypercellular area) Antoni - ..... (Hypo cellular area) - ..... bodies
		

Active space

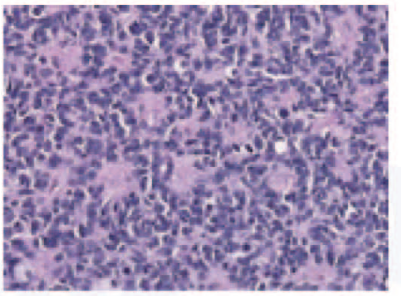
ROSETTE

Types	True Rosette	Pseudo Rosette	Perivascular rosette
			
Examples	....	....	
Tumor	.....	M : ..... E : ..... N :	.....
 True Rosette			 Perivascular rosett

Active space

EXERCISE

1. Which of the following are histological features of medulloblastoma as shown in the diagram?



A. Polygonal cells

B. Abundant cytoplasm and hyperchromatic nuclei

C. Homer-wright rosettes

D. Pseudorossette

2. Most common primary site for the mets in the brain comes from

A. Stomach

B. Liver

C. Eyes

D. Lung

3. The histopathological examination of a neural tumor in a patient with neurofibromatosis reveals hypercellular, hypocellular and acellular areas. Which of the following cranial nerves does this tumor arise from in such patients?

A. Cranial nerve V

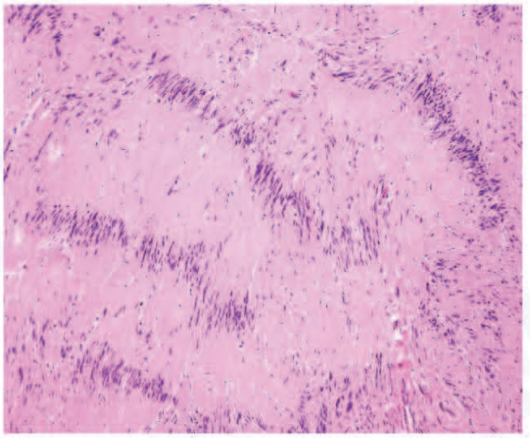
B. Cranial nerve VIII

C. Cranial nerve XII

D. Cranial nerve X

Active space

4. 34yrs old male suffering from Neurofibromatosis 2 came to OPD and revealed that it began with ringing sensation in ears which Progresses to balance problems and hearing loss. Which of the following microscopic finding is seen in given disease?



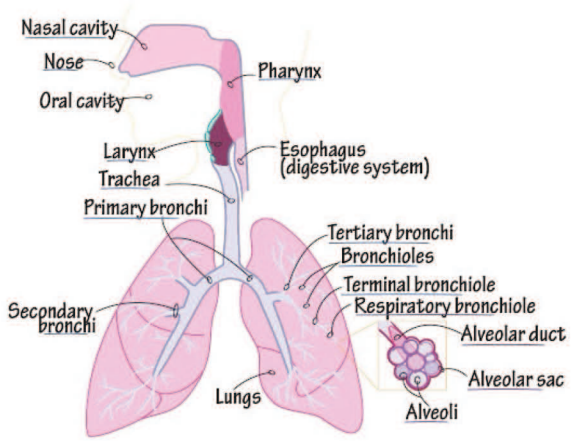
A. Spindle cells
B. Desmoplasia
C. Perineuronal invasion
D. Antoni A and B pattern

5. Most common primary benign brain tumor in children is
A. Pilocytic astrocytoma
B. Glioblastoma
C. Medulloblastoma
D. Ependymoma

Active space

RESPIRATORY SYSTEM

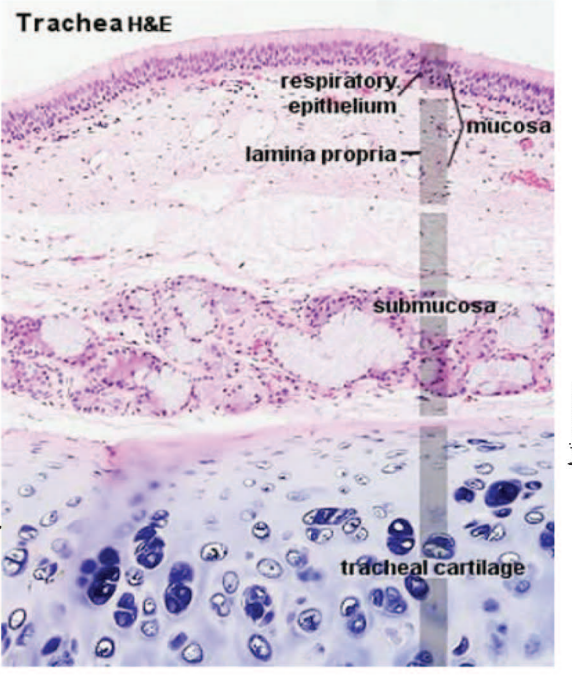
Functional unit of lungs :-



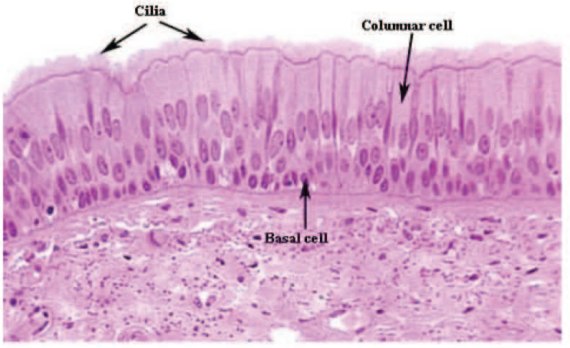
LAYERS OF RESPIRATORY TRACT

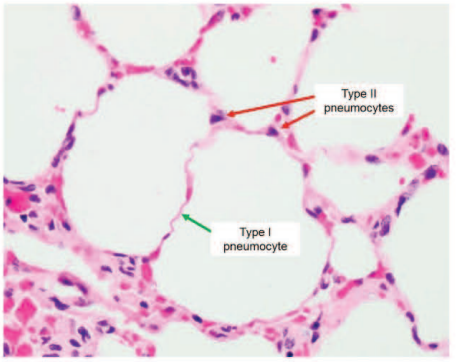
Lined by : .....

- Mucosa :
  - Epithelium
  - Lamina propria
  - Muscularis mucosa
- Submucosa :
  - .....
    - .....
- Muscularis
- Cartilage.



Active space





Alveoli Lined by :

Type I Pneumocytes	Type II Pneumocytes
(95%)	(5%)
• Flat cells	• Globular Cells
•	• ..... • .....

❑ Disease of Resp. system

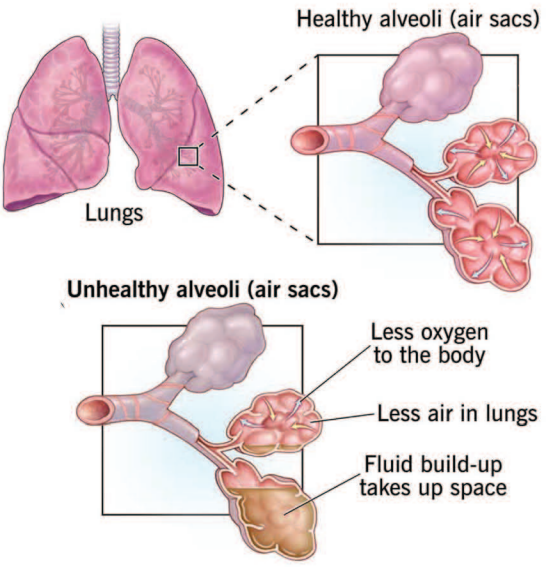
- Acute.
  - ARDS
- Chronic
  - Obstructive
    - Emphysema
    - Ch. Bronchitis
    - Bronchial Asthma
    - Bronchiectasis
  - Restrictive
    - Pneumoconiosis

Active space

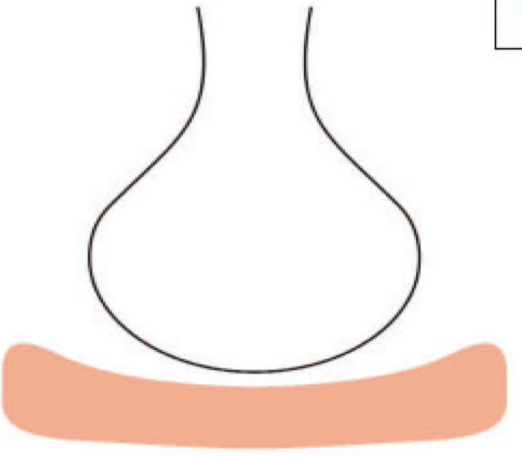
ARDS : ACUTE RESPIRATORY DISTRESS SYNDROME

- Diffuse Alveolar damage
- Stiff lung
- Hyaline memb. disease

❑ **Etiology:** ....



❑ **Pathophysiology:**



Ag entry → Mϕ Activatⁿ → Secrete IL-8 → Neutrophil recruitment inside alveoli

↓

Damage alveolar walls. ← Secrete granules/ enzymes

⊕

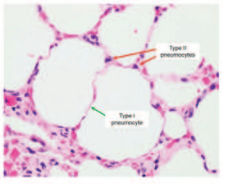
fibrin deposition

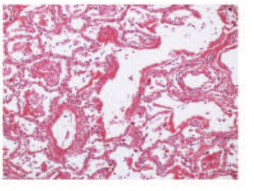
Type I ~~>>>~~ Type II

↓ pneumocytes ↓

~~gaseous exchange~~ ~~repair surfact prodⁿ.~~

- Diffuse alveolar Damage (DAD)
- Hyaline memb. D/S.
- Stiff Lung.





Active space

EMPHYSEMA :

- Permanent destructⁿ & dilatation of .....

- AR

PIMM gene - .....
PIMZ - .....
PIZZ - .....
M - (N) allele
Z - Mutated allele.

Mechanism :-

Destructive	Protective.
..... Enzymes ↓ Neutrophils granules	..... (a1-AT) ↓ Produced by Liver (PIMM gene) (Chr. 14)

MCC : A) ..... & B) .....
↓
..... Elastase activity
↓
Destructⁿ of Lung tissue/parenchyma.
↓
Emphysema

C/F :

- Dyspnea
- Pursed Lip
- Pink puffer's (Entry of O_2, retention of CO_2) (No Cyanosis)
- Barrel shaped chest
- Flattened diaphragm

Active space

Centriacinar - ..... - ..... - Prox-acinus - Upper Lobe	
Panacinar .... Prox + distal acinus - Lower Lobe.	
Paraseptal Aka distal acinar ↓ Alveolar damage ↓ Spontaneous Pneumothorax.	
Irregular MC microscopic Variant ↓ Patchy Involvement.	

Active space

CHRONIC BRONCHITIS :

- Persistent productive cough for ..... months for 2 consecutive years in the ..... of any other known cause
- MCC-.....

CB is not a/w .....

Chr. Bronchitis

↓ mucin production ↓ ↑ Risk of 2^0 Infection

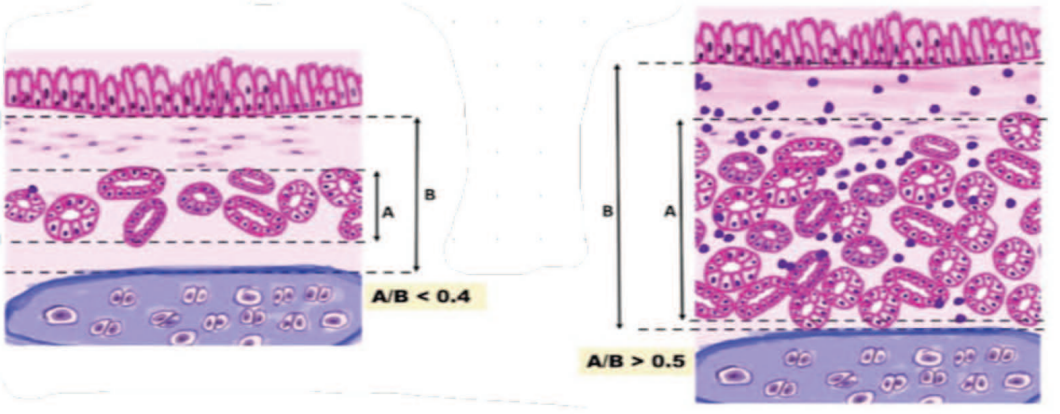
metaplasia ↓ Dysplasia ↓ Anaplasia

CB is a Premalignant cond^n

↓ $Pa^0 2$ ↓ VC ↓ Pulmonary HTN ↓ RHF (Cor pulmonale) (↑ mortality)

(N) : .....
CB : .....

C/F :
- Persistent productive cough.
- Blue bloaters



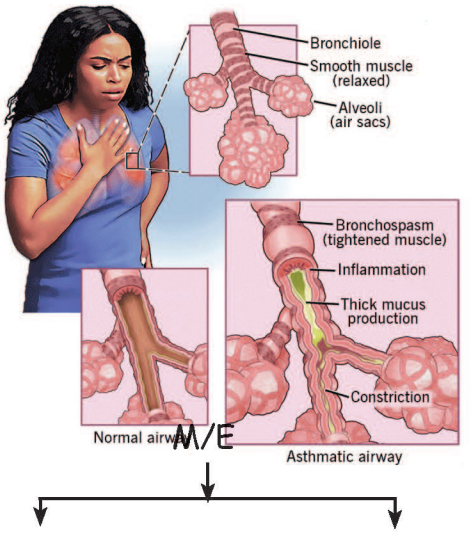
Reid Index : Glandular Layer Thickness / Thickness b/w epithelium & Cartilage

Rx :
- O_2
- Bronchodilators

Active space

BRONCHIAL ASTHMA :

- Type I HS rxn
- a/w : ..... gene mutation
  YKL gene - ↑ Severity
- Gene for atopy. Chr. .....
- Cytokines : IL-...., IL-...., 1L-.....
- Cells Involved :
  Mast cells, Eosinophils-
  ↓
  Histamine (earliest mediator)



M/E

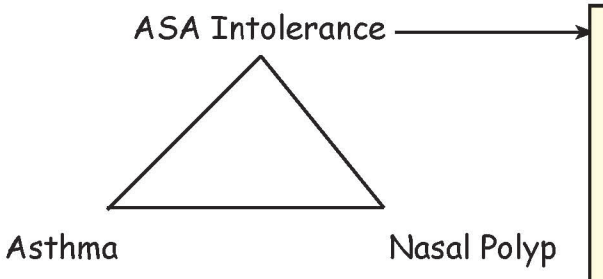
↑ Fibrosis ↑ Vessels

Airway Remodeling.

Note : Reid Index is Normal

Extrinsic	Intrinsic
• IgE	• .....
• Atopic	• Non-atopic
• Type I HSN	• Drug, Virus
• Child	• Adult
• F/H	• .....

Samter's triad :



Drug Induced BA

Inhibits COX Pathwayas

↓

LT'S

↓

Asthma

Active space

Sputum Examinatⁿ :

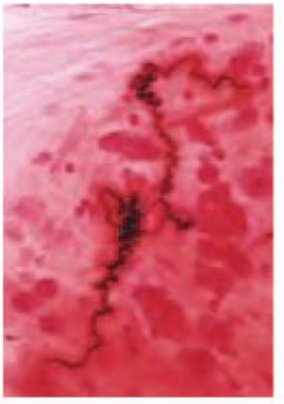
↑ Eosinophils & ↑ mucin
(↑IL-5) (↑IL-13)

↓

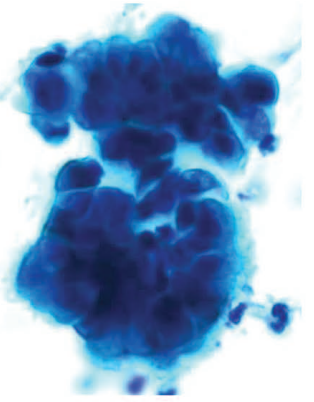
3.C'S

↓ ↓ ↓

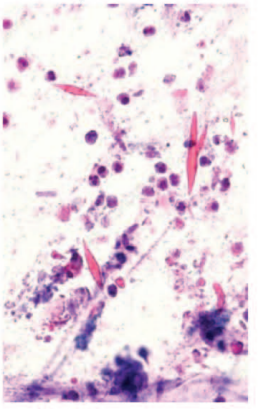
Curschmann's Spiral | Creola bodies | Charcoat leyden Crystal



Mucin Impacted plug



Sloughed out resp. epi cells



Sputum with Eosinophilic deposits

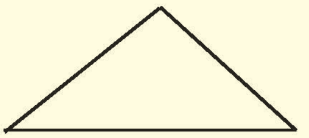
BRONCHIECTASIS :

- Permanent destructⁿ & dilatatioⁿ of .....

Genetic	Acquired
✦ α_1 AT ↓↓	✦ Smoking
✦ Cystic Fibrosis	(.....)
✦ Kartegener Syndrome	✦ Inf./Tumors

* Bronchiectasis is a/w with amyloidosis (AA type)

* Kartagener Synd → (1° ciliary dyskinesia)
Bronchiectasis (d/t - defect in ..... arm)
(Infertility <50%)



Situs Inversus Allergic rhinitis.

Active space

C/F :

- Productive cough
- Dyspnea
- Lower Lobes affected (B/L)

Complication :

- Sepsis
- Lung abscess
- Empyema

Gross

.....

CT Scan

.....

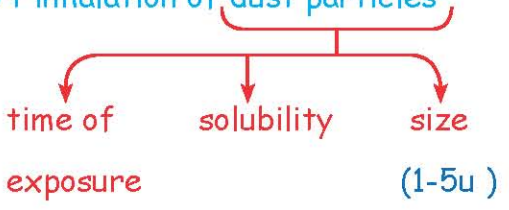
X-ray

.....

RESTRICTIVE LUNG DISEASE :

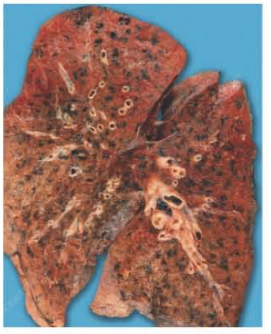
Pneumoconiosis

- .....
- d/t inhalation of dust particles



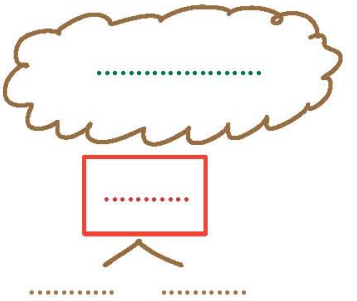
Active space

(1) Anthracosis

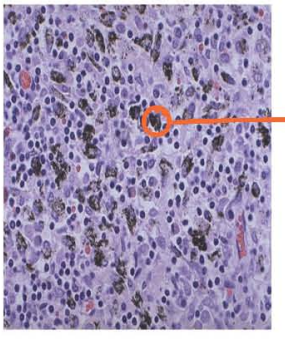


- .....
- .....
- .....

3 types : .....
.....
.....



M/E :- .....

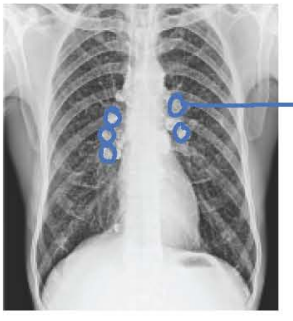


.....
.....
.....

(2) Silicosis → .....

- aka - .....
- d/t Silica dust (quartz) → .....

.....



Hilar LNadenopathy.

.....
.....

Fig. 10.8 Hilar Lymphadenopathy in silicosis

✦ M/E: Non immune Granuloma

Active space

(3) Asbestosis:

✦ d/t Asbestos dust → .....

.....

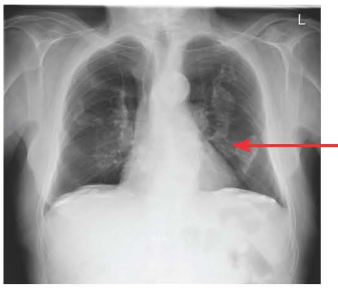
Amphibole — Serpentine

(more dangerous) (LL)

Ferruginous Bodies

OR

Asbestos Bodies



Pleural Plaque (MC)

⊕

Extensive fibrosis

- TB risk → .....
- Lung Ca risk → .....

✦ M/C cancer c̄ Asbestosis: .....

✦ M/specific cancer c̄ Asbestosis: .....

LUNG CA :

- Mc Lung Ca : ..... (m/c Breast)
- M/C Lung Ca :

Benign	Malignant
⇨ Pulmonary hamartoma ⇨ Carcinoid tumor.	⇨ Sq. cell Ca. ⇨ Adenocarcinoma ⇨ Small cell ca ⇨ Large cell ca.

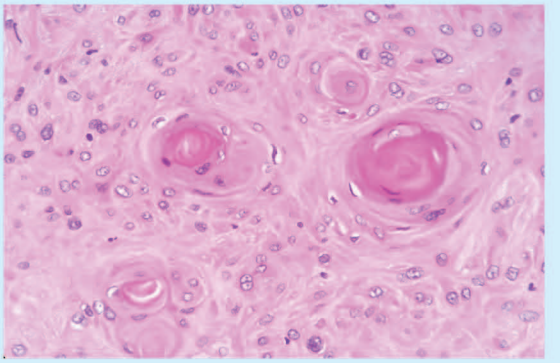
Active space

Features	Sq cell ca	Adeno Ca	Small cell ca	Large cell ca
M : F	M > F	M > F	M > F	
Location	.....	.....	.....	
Smoking	.....	.....	.....	
Mutatⁿ	P53	....	.....	
PNS	....	• Migratory Thrombophl - ebilitis (Trousseu Synd)	• SIADH • Cushing Syndrome [..... a/w PNS]	
IHC	• Cytokeratin • P40, P63	• TTF-1 • Napsin - A	....	-
Prognosis	Good	Good	Poorest	-
M/E	Keratin pearls	Lepidic pattern	Azzopardi-effect (Salt & Pepper Chromation)	

Active space

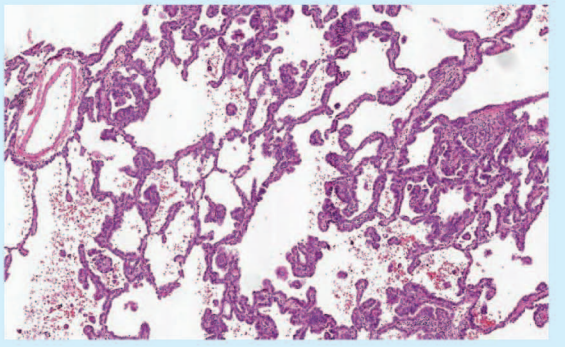
Sq. cell ca

- Keratin pearls



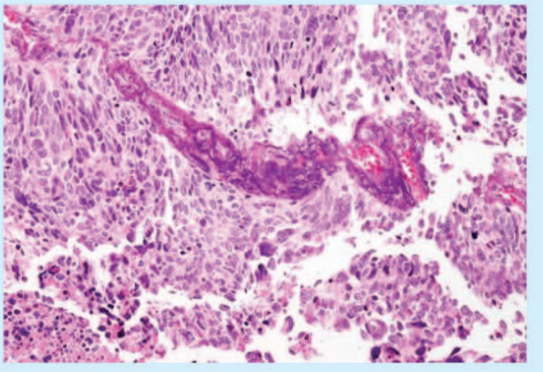
Adeno Ca

Variant - Adeno ca Insitu
(Broncho alveolar ca)
↓
Aerogenous Spread
↓
..... pattern
or
..... pattern



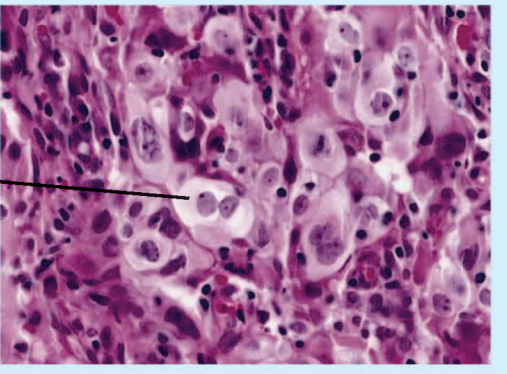
Small cell ca

- ..... effect
  (Basophilic staining
  of BV wall
  d/t broken DNA)
- Early mets → .....



Large cell ca

- Large
- Pleomorphic cells



Active space

PANCOAST TUMOR :

Tumor at the apex (Lung)

↓

- Adenoca ca → Compresses cervical Symp. Chain
- Peripheral ca ↓

.....

E → .....

L → .....

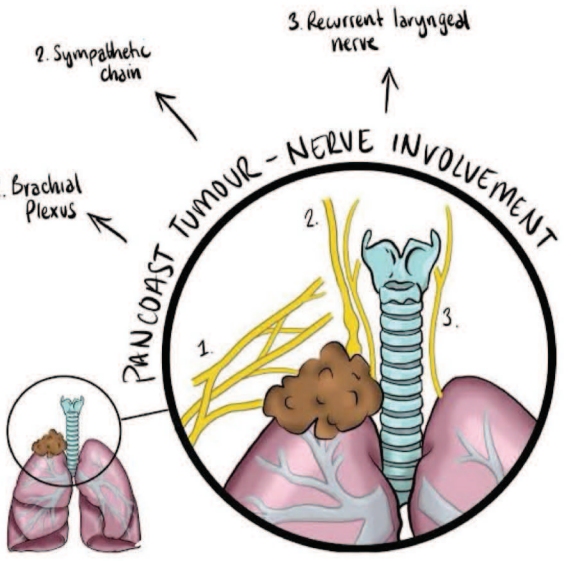
A → .....

M → .....

P → .....

✦ Hoarsness of voice :

d/t compression of RLN



Active space

EXERCISE

1. True statement amongst the following is-

   A. Lepidic pattern is characteristic finding of small cell carcinoma lungs

   B. Coal macule is characteristic of Richter syndrome

   C. Ground glass opacities are seen in lungs with quartz dust prolonged exposure

   D. None of the above

2. One is not true about chronic bronchitis?

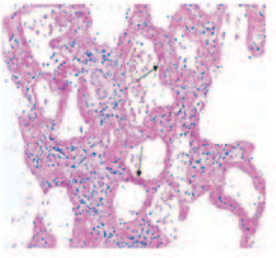
   A. Ratio of total respiratory epithelial thickness to glandular layer thickness is increased.

   B. Smoking leads to increase risk of chronic bronchitis

   C. Increased risk of adenocarcinoma lungs

   D. Not associated with Amyloidosis

3. Autopsy of a person who died of severe respiratory distress following blood transfusion shows the following. He has been diagnosed with ARDS. All are true except



   A. Surfactant production is compromised in ARDS

   B. IL-8 is the cause of inflammatory damage in ARDS

   C. Mast cells are responsible for severe respiratory distress

   D. Can lead to Pulmonary edema

Active space

4. Which of the following histological type lung carcinoma has worst prognosis:

   Squamous cell CA

   Adenocarcinoma

   Alveolar-carcinoma

   Small cell carcinoma

5. Aerogenous spread of lung cancer seen in

   Squamous cell carcinoma

   Oat cell carcinoma

   Carcinoid

   Bronchoalveolar carcinoma

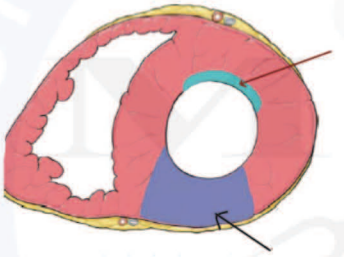
Active space

HEART

MYOCARDIAL INFARCTION

Infarction → area of ischemic necrosis

- MC → .....
- LC → .....



✦ Δ: (i) ECG →
- ST segment elvat^{n}
- Pathological Q waves
- T- wave inversion.

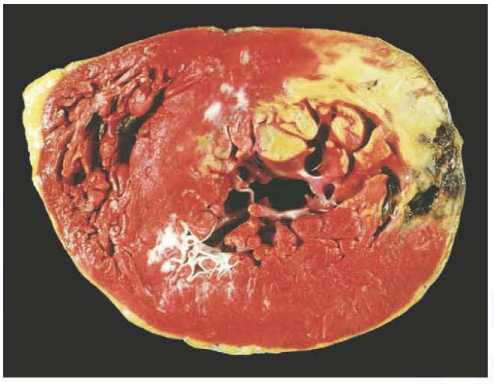
Enzyme	Onset	Peak	Comes back to normal	Use
Myoglobin				
CK-MB				
LDH				
Trop-I				

Active space

(iii) Stain (Dead ♡/ Autopsy) (< 12 hr)

↓

.....



(iv) Morphology.

Time	Gross	LM	EM
☆ < 30 min	⊖	⊖	....
☆ 30 min - 4 hrs	⊖	....	.....
☆ Day 1-3	Red infarct	✦ ..... ✦	⊖
☆ Day 3-7	Light Red	....	⊖
☆ Day 7-14	Yellowish	.....	⊖
☆ Day 28 & beyond	Silvery White infarct	....	⊖

Active space

✦ Complication of MI:

P: - .....

A: - .....

C: - .....

T: - .....

..... → ..... / ..... / .....

Vegetations

- Growth on value.

eg: (a) Rheumatic carditis
(b) NBTE
(c) Libman Sac Endocarditis
(d) Infective Endocarditis
} Mitral value

❑ (a) Rheumatic Carditis → .....

- Type II HSN rxn.
- Immunologically mediated Pancarditis
- Seen in children (5-15 yrs), H/O of throat infⁿ (2-3 wks)

✦ Etio: Gp A β Hemolytic streptococci

↓

Ag

↓

Ab's

↓

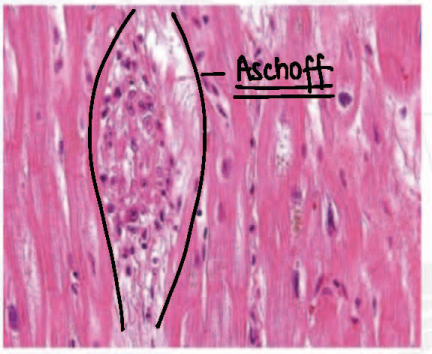
Cross Reactivity / Or / Mol. Mimicry → Kidneys → PSGN.
→ Heart → RC
→ Joints → RA

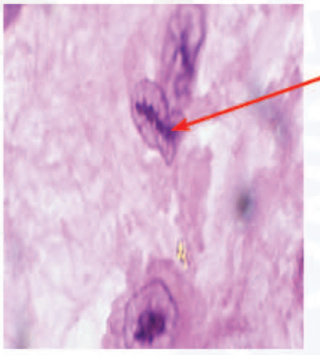
Active space

Pancardits (RC)

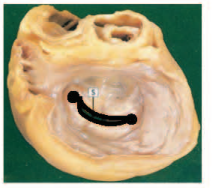
Pericarditis | Myocarditis | Endocarditis

MC value : Mitral valve





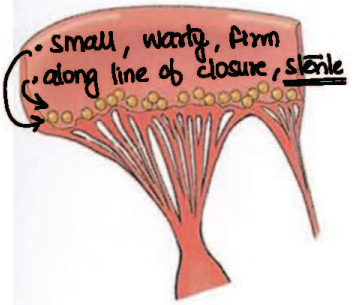
Anitschkow Cell



◀FIG. 13.4 Fish mouth deformity

❑ **Endocarditis: (RC)**

- MC value involved →
  - initially
  - Or



MIST

Active space

✦ Δ: Revised Jones Criteria

Major		Minor	
☆ J	Joint (mig. polyarthritis)	P	Previous Rh. fever
☆ ♡	Heart (carditis, value damage)	E	ECG c PR prolongation
☆ N	Nodules (suncutaneous, ext surface)	A	Arthalgia
☆ E	Erythema marginatum	C	↑↑ CRP
☆ S	Sydenham's chorea.	E	Elevated temp.

✦ For Δ: H/O Gp A β Hemolytic strep

⊕

2 major or 1 major + 2 minor

❑ (b) NBTE - Non Bacterial Thrombotic Endocarditis / Marantic EC

Etiology: Hyper coagulable state

↓

Thrombosis

↓

Embolize

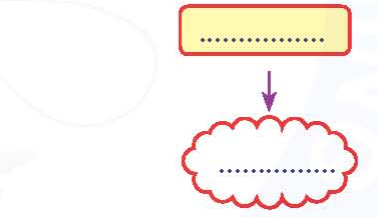
↓

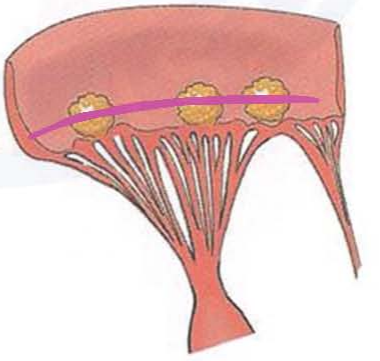
Damage of values

↓

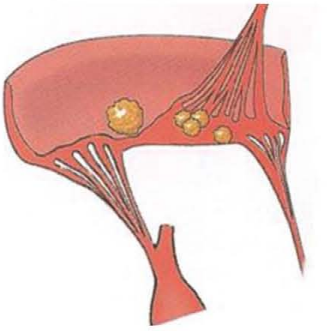
Deposition

- small/medium
- -sterile
- -friable
- .....





Active space



❑ **(c) LSE - Libmann Sachs Endocarditis**

- ✦ a/w SLE
- ✦ MC value → .....
  - small / medium
  - sterile, veruccous
  - .....
  -

❑ **(d) Infective Endocarditis** - MC value : .....

Acute	Subacute
↓	↓
■ Native valve	■ Prosthetic valve
■	■ .....

✦ For Δ: Duke's Criteria: ☆ 2 major
☆ 1 major + 3 minor
☆ 5 minor

❑ **Duke's Criteria**

✦ Major Criteria

- Persistently +ve blood culture for typical organisms → Staph / Strep
- ECHO: vegetation, dehiscence, abscess
- New valvular regurgitation murmur
- Coxiella burnetii infection

✦ MINOR CRITERIA

- Predisposing heart condition or IV drug use
- Fever - 38.0°C
- Vascular: emboli to organs/brain, hemorhages
- Immunologic: glomerulonephritis, Osler's Nodes,
- Roth spots, rheumatoid factor
- Positive blood cultures that do not meet specific criteria MIST

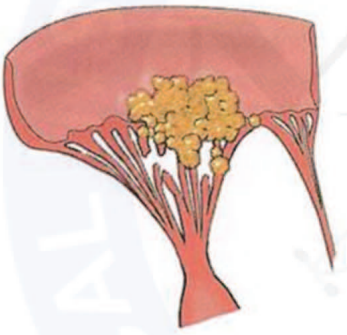
Active space

❖ Roth spots

- Oval Retinal H'ages → DM
  - → Pernicious Anemia
  - → leukemia

❖ Osler's Node

- S/C lesion in pulp
- Painful



- .....
- Friable
- .....
- .....

CARDIAC TUMORS

✦ MC cardiac tumor: .....

✦ MC Benign cardiac tumor:

- Child: .....
- Adult: .....

✦ MC 1° malignant cardiac tumor:

- Child: .....
- Adult: .....

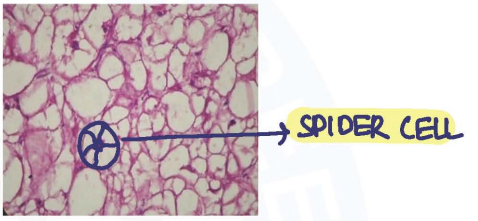
MIST

Active space

Myxoma	Rhabdomyoma/Sarcoma
MC - Adults	Children
Site -	.....
Gene -	.....
a/w Carney Syndrome	a/w Tuberous Sclerosis

Carney Syndrome → Endocrinopathy, Pigmentation, Myxoma

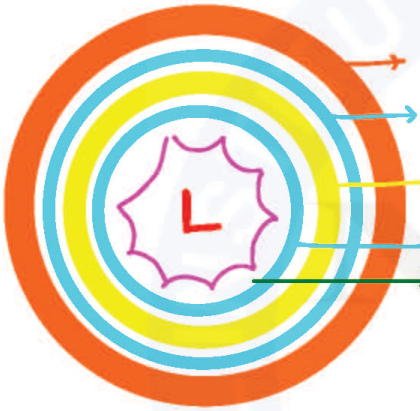
2 types:
- Familial
- Sporadic (90%)



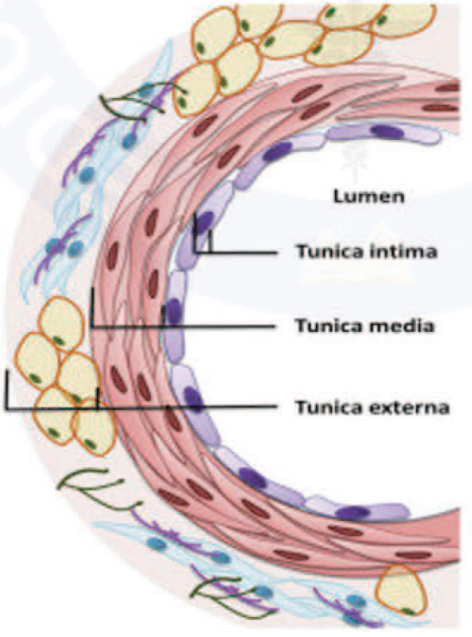
Active space

VASCULAR SYSTEM

- (N) St of vessels: 3 layers
  - T. Adventitia
  - T. Media
  - T. Intima



T. Adventitia → collagen + Elastic fibres
Ext. Elastic lamina
T. media → S|M + Elastic fibres.
Int. Elastic lamina.
T. intima.
(single layer of endothelial cells)



Active space

❑ ARTERIOSCLEROSIS → (3) types

✦ (1) Arteriolosclerosis: 2 types

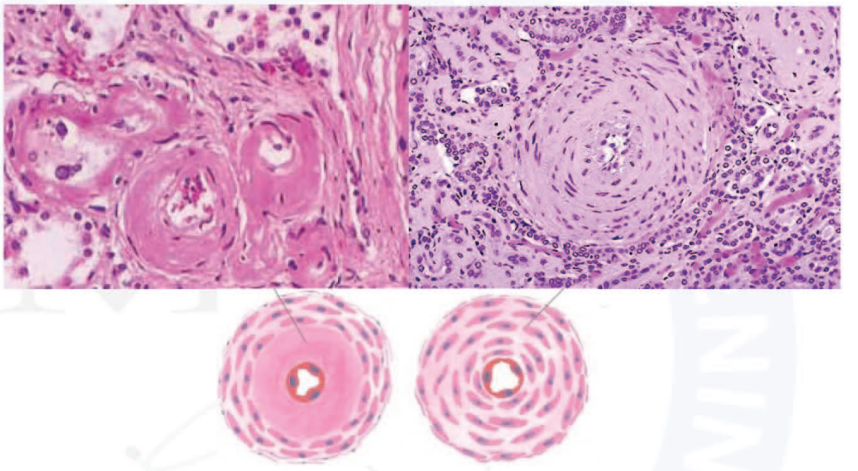
Hyaline

Hyperplastic

.....

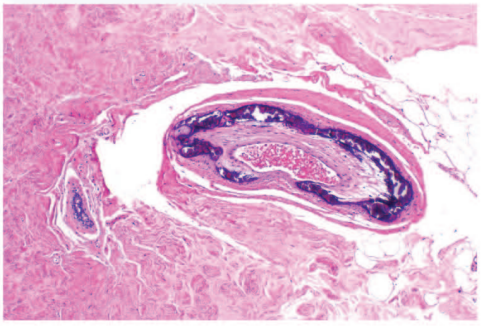
.....

.....



✦ (2) Monckeberg Medial Sclerosis

- Involves T. media
- > 50 yr (not clinically significant)
- .....



Active space

✦ (3) Atherosclerosis (M/I)

- Inflammatory damage to T. Intima

↓

Formation of Plaque

☞ Risk Factors:

Modifiable	Non modifiable
- Alcohol	- Stress (Type A)
- Smoking	- Age
- Hyper lipidemia	- ♂
- Obesity	- F/H
- HTN	
- Diabetes.	

❑ PATHOGENESIS:

Injury

↓

✦ .....

✦ .....

↓

.....

↓

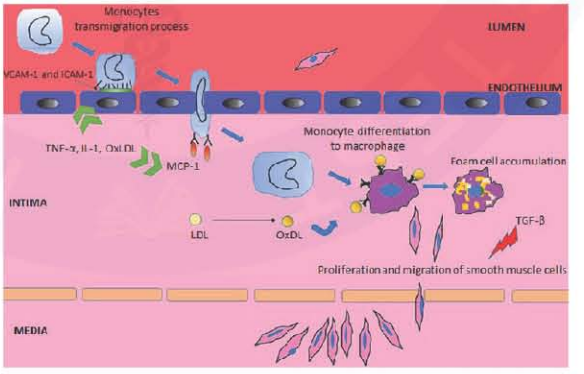
.....

↓

.....

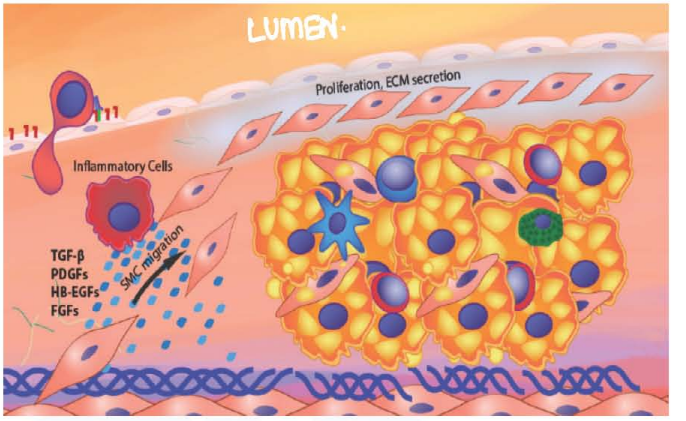
↓

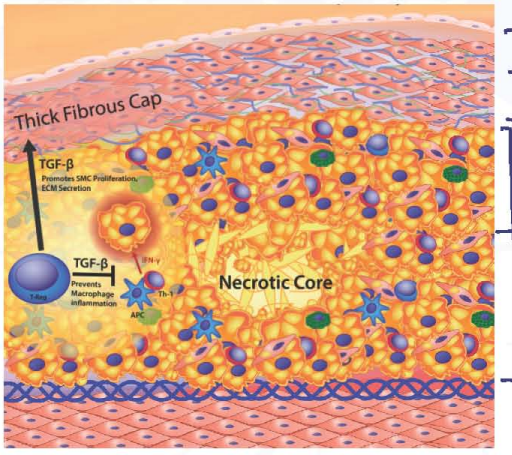
.....



❑ Atheromatous Plaque

Active space

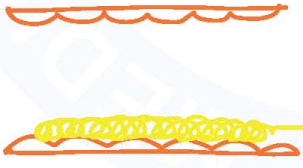




FIBROUS CAP
S/M cells & collagen

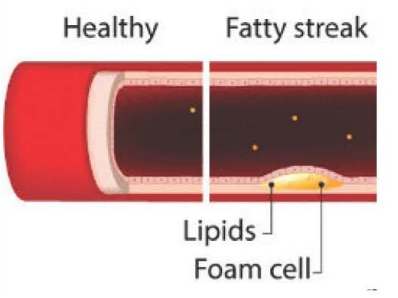
NECROTIC CORE
- Cellular Debris
- Lipid Laden MΦ (FOAM CELLS)

☞ Earliest lesion:



MC vessel ↓ .....

LC vessel ↓ .....



MIST

Active space

☞ Complications:

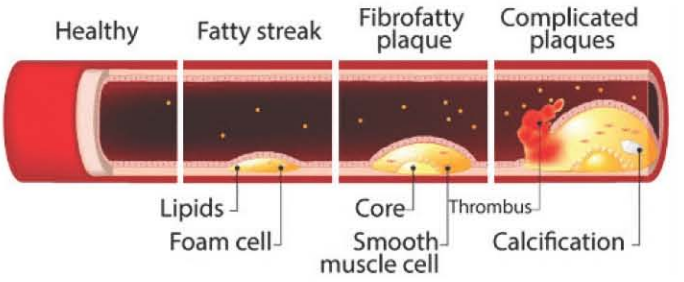
A: .....

C: .....

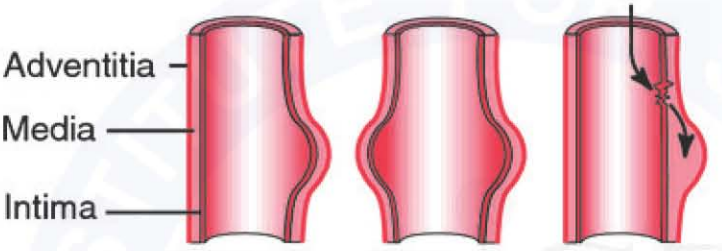
U: .....

T: .....

E: .....



❑ Aneurysm: .....



SACCULAR FUSIFORM FALSE

(i) True
- involⁿ of all 3 layers
- MCC → Atherosclerosis

(ii) Pseudo
- Break in vessel wall
- Post MI rupture

VASCULITIS

infiⁿ of BV size

Large
- Giant cell Arteritis
- Takayasu Arteritis

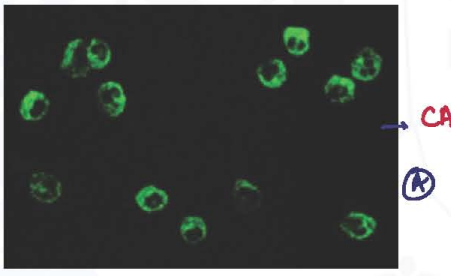
Medium
- PAN
- Kawasaki d/s
- Buerger's d/s

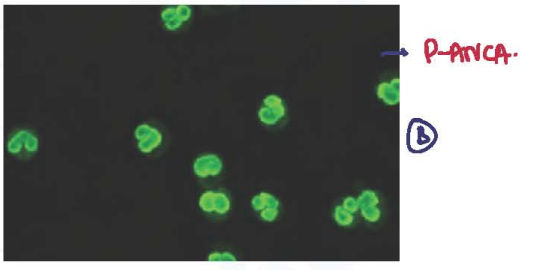
Small
- Immune – HSP
- Pauci Immune
  - ☆ MPA
  - ☆ Churg Strauss d/s
  - ☆ Wegner's Granulomatosis

MIST

Active space

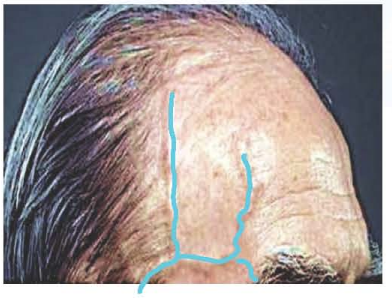
Antibodies in vasculitis – (3)





Large Vessel

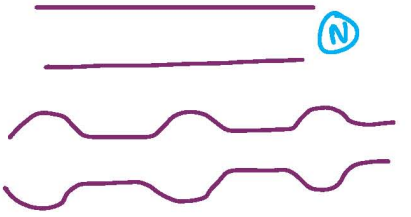
❑ (1) Giant Cell Arteritis



- aka ..... (sup. Temporal A)
- .....
- MC symptom – .....
- M/specific symptom → .....
- M/feared symptom → .....

✦ .....

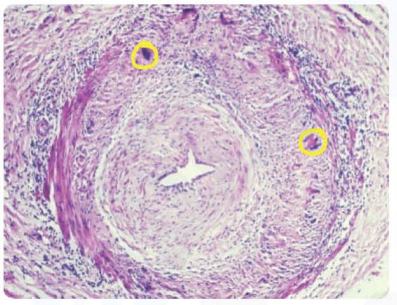
Active space

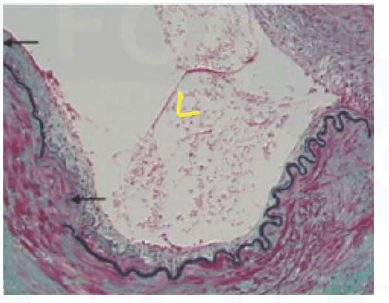


- ✦ Gross: Nodular Segmental thickening
- ✦ M/E: →
  - ● .....
  - ● Granuloma
  - ● Rupture of internal Elastic lamina

↓

(Verhoff Von Gibson Stain)





- ✦ Rx: Steroid

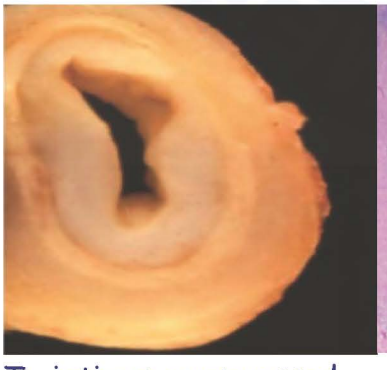
❑ **(2) Takayasu Arteritis** aka - .....

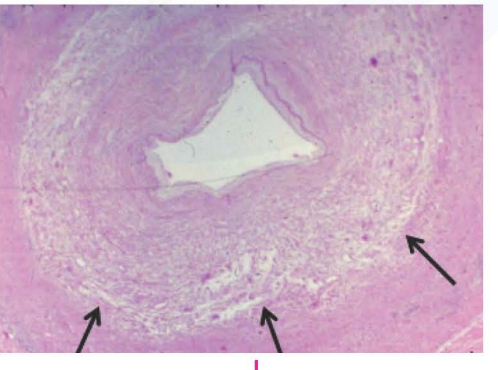
MC vessel: .....

aka: .....

- ● < 50 yr.
- ● MRI spine → (N) but pulse not (N) → TA.

- ✦ Rx: Steroids.





T . intima: compressed

T. Media: Thickened

T. Adventitia: Fibrosis

↓

- ● inflammatory infiltrate
- ● granuloma.

MIST

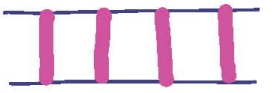
Active space

Medium Vessel

❑ (1) Polyarteritis Nodosa (PAN)

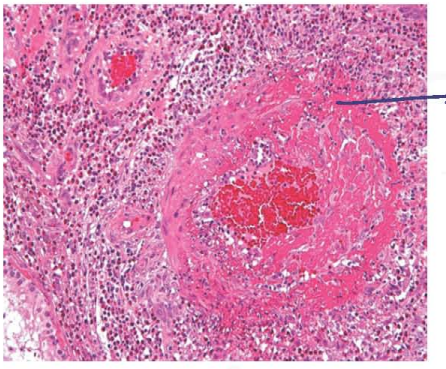
MCC of Death
↓
Renal failure

- .....
- ..... ↘ .....
  .....
- .....
- .....
- .....



✦ M/E: Segmental Transmural involⁿ

✦ Rx: Steroids



❑ (2) Kawasaki Disease:

aka → .....

T → ↑Temp > 5days

E → Erythema, Anti Endothelial Ab

M → .....

P → Pain ..... ← .....

E → Eyes (B/L non.supp. conjunctivitis)

R → Rash

S → Swelling → LN → > 1.5 cm

Active space

- 2nd MC vasculitis in children (MC – HSP)
- .....

↓

Arrhythmias

Rx: IvIg's

❑ (3) Buerger's Disease:

- aka: .....
- smokers □ 35 yrs ♂
- Raynaud phenomenon
- No Lymphatics involⁿ.
- HLA B5 ⊕
- Intermittent claudication

↓ later

Rest pain

Small Vessels

❑ (1) Henoch Schonlein Purpura

⇨ .....

.....

⇨ .....

⇨ involⁿ:

- Skin → .....
- Joints → .....
- GIT → .....
- Abd → .....

⇨ Rx - Steroids

Active space

❑ (2) Microscopic Polyangitis .....

⇨ .....

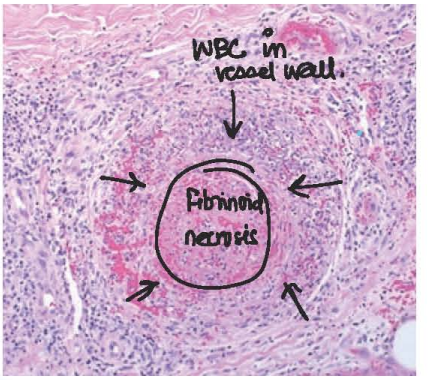
⇨ .....

⇨ .....

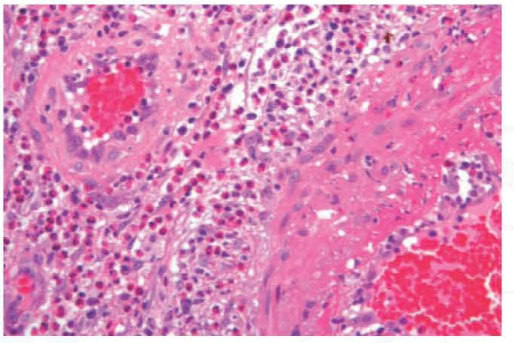
⇨ .....

⇨ ..... → .....

→ .....



❑ (3) Churg Strauss Synd: aka allergic Granulomatosis c̄ Angitis



Eosinophilic infiltration

⊕

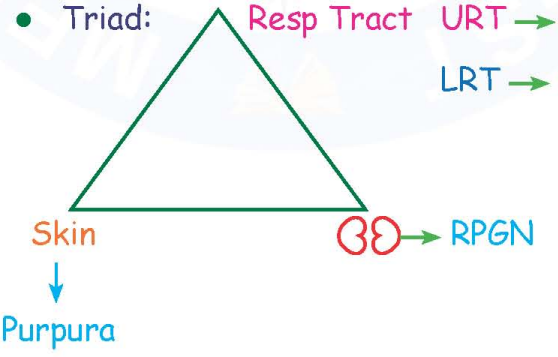
Granuloma.

✦ .....

✦ Rx: Steroids

❑ (4) Wegener's Granulomatosis

- aka: .....
- .....
- Triad:



✦ M/E: Granuloma c̄ giant cells

Active space

VASCULAR TUMORS

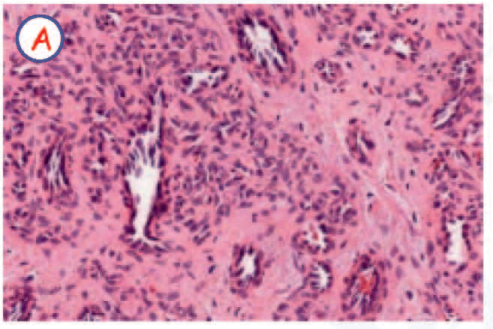
- Benign
  - Hemangioma
  - Lymphangioma
  - Glomus tumor
- Intermediate
  - Kaposi sarcoma
- Malignant
  - Angiosarcoma

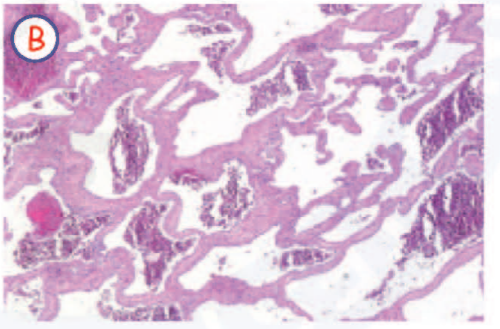
❑ Benign

✦ (1) Hemangioma

Capillary	Cavernous
MC	—
Children	Adults
Skin	Skin, liver, spleen, (CNS)
regress	⊕

VHL gene





✦ (2) Lymphangioma

- involⁿ of Lymphatics
- 2 types → capillary
  - Cavernous → Cystic Hygroma (45 X0) (neck swelling)

MIST

Active space

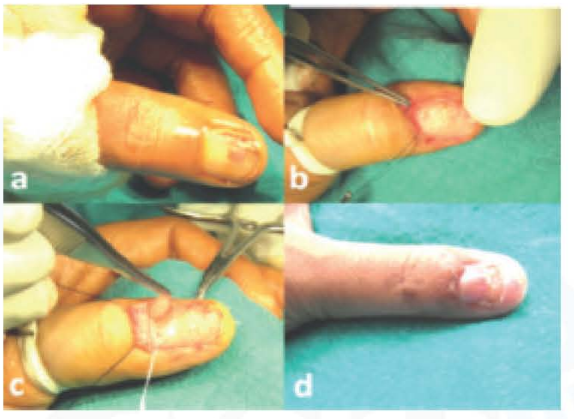
✦ (3) Glomus Tumor

- Tumor of glomus body (Thermoregulatⁿ)

↓

under Nail bed

- extremely pain full tumor



❑ **Intermediate**

✦ KAPOSI SARCOMA

- Caused by .....
- .....

⇨ 4 types:

(a) Epidemic → .....

(b) Endemic → LN

(c) Classical → Skin

(d) Post Transplant → Skin, LN, vis. org.

.....

.....

❑ **Malignant**

✦ ANGIO SARCOMA

- Marker: .....
- V → .....

A → .....

T → .....

Active space

EXERCISE

1. The most likely diagnosis in the case of a patient with multiple pulmonary cavities, hematuria and crescents in kidney is

   A. Anti gbm disease

   B. Churg-strauss allergic granulomatosis

   C. Systemic lupus erythematosus

   D. Wegner's granulomatosis

2. Cytokine responsible for proliferation of smooth muscle in the pateint suffering from atherosclerosis is -

   A. IFN- Gamma

   B. IL-6

   C. TGF-beta

   D. FGF

3. All the following are true regarding PAN except:

   A. Affects large size blood vessels

   B. May affect renal vessels

   C. Necrotizing granuloma may be seen on microscopy

   D. HBsAg positivity is seen with severe patients

4. Gene mutation seen with most common tumor of heart in child is prognosis:

   A. Beta catenin

   B. TSC 1

   C. PRKAR

   D. HBX

Active space

5. In a patient with periorbital edema, oligouria, high serum creatinine and respiratory distress, a biopsy was taken from renal artery. There appears presence of granuloma and biopsy is consistent with presence of Perinuclear-ANCA. These findings are most suggestive of which disease?

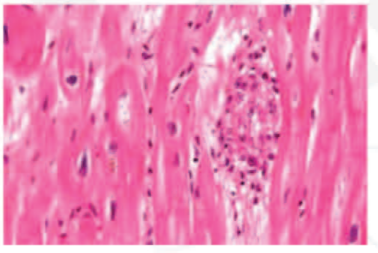
A. Giant cell arteritis

B. Classic polyarteritis nodosa

C. Wegener's granulomatosis

D. Microscopic polyangiitis

6. Heart muscle biopsy of the patient shows following histological finding consistent with Aschoff bodies as shown in figure. All are true about the finding except



A. Mitral valve is the most common valve involved in the following disease

B. Contaminated growth on one side of the mitral valve is seen

C. Catterpillar like chromatin cells are found well immeresed in myocardium

D. Mitral valve stenosis is the valvular damage seen in long standing disease

Active space